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Novartis Research and Development

Clinical Trial Protocol

A phase I open-label, multi-center study to evaluate the safety, tolerability and preliminary activity of [^{225}Ac]Ac-DWJ155 and safety of [^{68}Ga]Ga-DWJ155 in participants with solid tumors

Clinical Trial Protocol Number: CPBQ663A12101

Version Number: v01 (Clean)

Compound: [^{225}Ac]Ac-DWJ155, [^{68}Ga]Ga-DWJ155

Brief Title: A phase I study of [^{225}Ac]Ac-DWJ155 and [^{68}Ga]Ga-DWJ155 in participants with solid tumors

Study Phase: I

Regulatory Agency Identifier Number(s): 2025-522650-39-00

Sponsor name: Novartis Pharma AG or its affiliates outside of the EEA (where applicable)

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Amendment 1 (04-May-2026)

Amendment rationale

The protocol has been amended due to formatting requirements.

Study Update: At the time of this amendment, the study has not been submitted to any Health Authority.

Changes to the protocol

Changes to specific sections of the protocol are shown in the track changes version of the protocol using strike through for deletions and underline for insertions. The changes made to the protocol due to this amendment and the rationale for the changes are described below.

Section 5.1: Corrected the inclusion criteria numbering and bullet points

Section 5.2: Corrected the exclusion criteria numbering and bullet points

IRBs/IECs

A copy of this amended protocol will be sent to the Institutional Review Board (IRBs)/Independent Ethics Committee (IECs) and Health Authorities.

The changes described in this amended protocol require IRB/IEC approval prior to implementation.

The changes herein do NOT affect the Informed Consent.

1 Protocol summary

1.1 Description of the study

Protocol Title:

A phase I open-label, multi-center study to evaluate the safety, tolerability and preliminary activity of [^{225}Ac]Ac-DWJ155 and safety of [^{68}Ga]Ga-DWJ155 in participants with solid tumors

Brief Title:

A phase I study of [^{225}Ac]Ac-DWJ155 and [^{68}Ga]Ga-DWJ155 in participants with solid tumors

Purpose:

The purpose of this phase I study is to evaluate the safety, tolerability and preliminary anti-tumor activity of [^{225}Ac]Ac-DWJ155 and the safety of [^{68}Ga]Ga-DWJ155 in participants with histologically or cytologically confirmed advanced (defined as metastatic or locally advanced, i.e., not amenable to definitive multi-modality therapy including surgery) Human Epidermal Growth Factor Receptor 2 (HER2) +, HR+/HER2-, or triple negative breast cancer (TNBC), non-small cell lung cancer (NSCLC), HER2 3+ or 2+ (in situ hybridization (ISH) + or -) gastroesophageal adenocarcinoma (GEA), and bladder cancer.

Study Indication/Medical Condition:

Advanced HER2+, HR+/HER2-, or TNBC, NSCLC, HER2 3+ or 2+ (ISH + or -) GEA, and bladder cancer

Treatment Type:

Radioligand therapy

Study Type:

Interventional

Objectives, Endpoints, and Estimands:

Table 1-1 Objectives and Endpoints

Objectives	Endpoints
Primary	
Assess the safety and tolerability for [^{225}Ac]Ac-DWJ155 and identify the recommended dosage(s) of [^{225}Ac]Ac-DWJ155 for further clinical evaluation	• Safety: Incidence and severity of adverse events (AEs), including dosage limiting toxicities (DLTs) and serious adverse events (SAEs), changes in laboratory values, vital signs and electrocardiograms (ECGs) • Tolerability: Frequency of dosage interruptions and reductions, dosage intensity

Secondary	
Evaluate the preliminary anti-tumor activity of [^{225}Ac]Ac-DWJ155	Overall response rate (ORR), disease control rate (DCR), duration of response (DOR), and progression-free survival (PFS) based on RECIST v1.1 by local investigator assessment
Characterize the pharmacokinetics (PK) of [^{225}Ac]Ac-DWJ155 in blood post-secular equilibrium	PK parameters (i.e., clearance, maximum concentration (Cmax), time to maximum concentration (Tmax), area under the curve (AUC), half-life, systemic clearance (CL), distribution volume) of [^{225}Ac]Ac-DWJ155 in blood
Characterize the safety of [^{68}Ga]Ga-DWJ155	Incidence and severity of AEs, including SAEs

Trial Design:

This is a first-in-human (FIH), phase I, open-label, multi-center study in participants with locally advanced or metastatic breast cancer, NSCLC, GEA, and bladder cancer. The study consists of a dosage escalation part, followed by a dosage expansion part.

Treatment of Interest:

The investigational drugs are [^{225}Ac]Ac-DWJ155 and [^{68}Ga]Ga-DWJ155.

DWJ155 is a HER2-targeting peptide labeled with Actinium-225 (Ac-225) for therapy or with Gallium-68 (Ga-68) for imaging.

Treatment Groups:

During the imaging period, all participants will receive [^{68}Ga]Ga-DWJ155 and will be evaluated for eligibility for [^{225}Ac]Ac-DWJ155 treatment. Participants who are eligible for treatment are planned to receive one administration of [^{225}Ac]Ac-DWJ155 at the beginning of each cycle. Participants will be followed up for 36 months from the last [^{225}Ac]Ac-DWJ155 administration, as described in Section 1.3.

Number of Participants:

Escalation part: A minimum of 21 participants will be treated with [^{225}Ac]Ac-DWJ155. The exact number of participants will depend on the number of dosage levels.

Expansion part: Approximately 135 participants in total, including HER2+ breast cancer (n~15), HR+/HER2-low breast cancer (n~15), HR+/HER2-0 breast cancer (n~10), non-Actionable Gene Alteration (AGA) NSCLC (n~20), AGA NSCLC (n~20), and HR-/HER2-low breast cancer (n~15), HR-/HER2-0 breast cancer (n~10), HER2 3+ or 2+ (ISH + or -) GEA (n~15), bladder cancer (n~15), will be assigned to receive [^{225}Ac]Ac-DWJ155 treatment.

Key Inclusion Criteria:

- Male or female participants age ≥ 18 years.
- Participants with one of the following histologically or cytologically confirmed and documented malignancies, for whom no standard therapy is available, tolerated, or appropriate in the judgement of the investigator:

- Dosage escalation (first cohort only):
  - Advanced HER2+ breast cancer with disease progression after at least two prior lines of systemic therapy in the advanced setting.
- Dosage escalation (from second cohort onward):
  - Advanced HER2+ breast cancer with disease progression after at least two prior lines of systemic therapy in the advanced setting.
  - Advanced HR+/HER2-low breast cancer with disease progression after at least one line of hormone-based therapy in combination with a CDK4/6i and at least one additional line of systemic therapy administered in the advanced setting. Participants progressing on, or within 12 months of completing, hormone based therapy plus CDK4/6i in the adjuvant setting are only required to have been treated with one line of additional therapy in the advanced setting.
  - Advanced NSCLC without Actionable Genetic Alterations (AGAs) with disease progression after prior therapy in the advanced setting who have received a platinum-based chemotherapy and an immune checkpoint inhibitor therapy either in combination or in sequence, unless participant was ineligible to receive such therapy. Participants must have known HER2 immunohistochemistry (IHC) ≥ 1+ or confirmed tumor positivity based on local assessment of [^{68}Ga]Ga-DWJ155 uptake.
  - Advanced NSCLC with AGAs who have received prior treatment with at least a targeted therapy or ADC, platinum-based chemotherapy, and immune-oncologic (IO) therapy if appropriate, unless participant was ineligible to receive such therapy. Participants must have known HER2 IHC ≥ 1+ or confirmed tumor positivity based on local assessment of [^{68}Ga]Ga-DWJ155 uptake.
- Dosage expansion:
  - Advanced HER2+ breast cancer with disease progression after at least two prior lines of systemic therapy in the advanced setting.
  - Advanced HR+/HER2-low breast cancer with disease progression after at least one line of hormone-based therapy in combination with a CDK4/6i and at least one additional line of systemic therapy administered in the advanced setting. Participants progressing on, or within 12 months of completing, hormone-based therapy plus CDK4/6i in the adjuvant setting are only required to have been treated with one line of additional therapy in the advanced setting.
  - Advanced HR+/HER2 0 breast cancer with disease progression after at least one line of hormone-based therapy in combination with a CDK4/6i and at least one additional line of systemic therapy administered in the advanced setting. Participants progressing on, or within 12 months of completing, hormone-based therapy plus CDK4/6i in the adjuvant setting are only required to have been treated with one line of additional therapy in the advanced setting.
  - Advanced HR-/HER2-low breast cancer with disease progression after prior therapy in the advanced setting after at least two lines of systemic therapy including cytotoxic chemotherapy, antibody-drug conjugate, targeted therapies and IO for metastatic disease.

- Advanced HR-/HER2 0 breast cancer with disease progression after prior therapy in the advanced setting after at least two lines of systemic therapy including cytotoxic chemotherapy, antibody-drug conjugate, targeted therapies and/or IO.
- Advanced NSCLC without AGAs, who have received a platinum-based chemotherapy and an immune checkpoint inhibitor therapy either in combination or in sequence, unless participant was ineligible to receive such therapy. Tumor must be positive for [^{68}Ga]Ga-DWJ155 uptake based on local assessment.
- Advanced NSCLC with AGAs, who have received prior treatment with at least a targeted therapy or ADC and platinum-based chemotherapy, and/or IO therapy if appropriate, unless participant was ineligible to receive such therapy. Tumor must be positive for [^{68}Ga]Ga-DWJ155 uptake based on local assessment.
- Advanced HER2 IHC 3+ or 2+ (ISH + or -) GEA with HER2 IHC 3+ or 2+ (ISH + or -), following disease progression after prior therapy in the advanced setting including all standard of care (SOC) systemic therapies per local guidelines including chemotherapy, targeted agents, IO, and/or ADCs, unless participant was ineligible to receive such therapy.
- Advanced bladder cancer with disease progression after prior therapy in the advanced setting including all SOC systemic therapies per local guidelines, including chemotherapy, ADCs, and/or immunotherapy, unless participant was ineligible to receive such therapies. Tumor must be positive for [^{68}Ga]Ga-DWJ155 uptake based on local assessment.

Key Exclusion Criteria:

- Out-of-range laboratory values defined as:
  - Estimated glomerular filtration rate (eGFR) < 60 mL/min/1.73m^2 measured or calculated by Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formula for adults.
  - Total bilirubin > 1.5 x upper limit of normal (ULN; except for participants with Gilbert's syndrome who are excluded if total bilirubin > 3.0 x ULN) or direct bilirubin > 1.5 x ULN; Alanine aminotransferase (ALT)/Aspartate aminotransferase (AST) > 3 x ULN, except for participants with tumor involvement of the liver who are excluded if ALT > 5 x ULN; lipase > 1.5 x ULN
  - Absolute neutrophil count (ANC) < 1.5 x 10^9/L, hemoglobin < 9 g/dL, platelet count < 100 x 10^9/L
- Ongoing treatment with hematopoietic colony stimulating factors, thrombopoietin mimetics, or erythroid stimulating agents.
- Use of transfusion support ≤ 4 weeks prior to the first administration of [^{225}Ac]Ac-DWJ155
- Impaired cardiac function or clinically significant cardiac disease.
- Unmanageable urinary tract obstruction or urinary incontinence.
- Prior treatment with any therapeutic radiopharmaceutical
- Treatment with any of the following anti-cancer therapies prior to the first administration of [^{225}Ac]Ac-DWJ155 within the stated timeframes:

- < 10 half-lives for any imaging radiopharmaceutical
- ≤ 4 weeks for external beam radiation therapy (EBRT) or brachytherapy (not lung-directed)
- ≤ 6 months for lung-directed external beam radiotherapy

- Participants with uptake of [^{68}Ga]Ga-DWJ155 in non-tumor tissues or organs that, in the opinion of the investigator, increase the risk associated with [^{225}Ac]Ac-DWJ155 treatment. This may include, but is not limited to, bone marrow or other normal tissue/organ uptake.

Data Monitoring/Other Committee:

Yes; data monitoring committee for the Dosage expansion part only

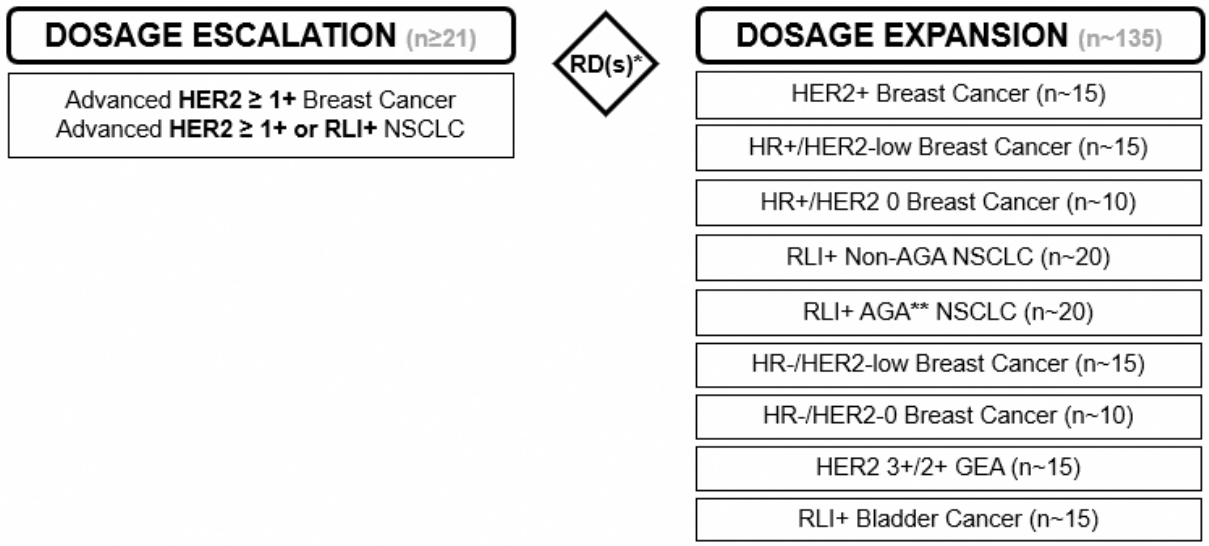
Key Words:

Breast cancer; non-small cell lung cancer (NSCLC); bladder cancer; gastroesophageal adenocarcinoma (GEA); radioligand therapy (RLT); radioligand imaging (RLI); [^{225}Ac]Ac-DWJ155; [^{68}Ga]Ga-DWJ155; Human Epidermal Growth Factor Receptor 2 (HER2); PBQ663; FKL480

[^{225}Ac]Ac-DWJ155 (PBQ663) and [^{68}Ga]Ga-DWJ155 (FKL480) will hereafter be referred to as ^{225}Ac-DWJ155 and ^{68}Ga-DWJ155, respectively.

1.2 Schema

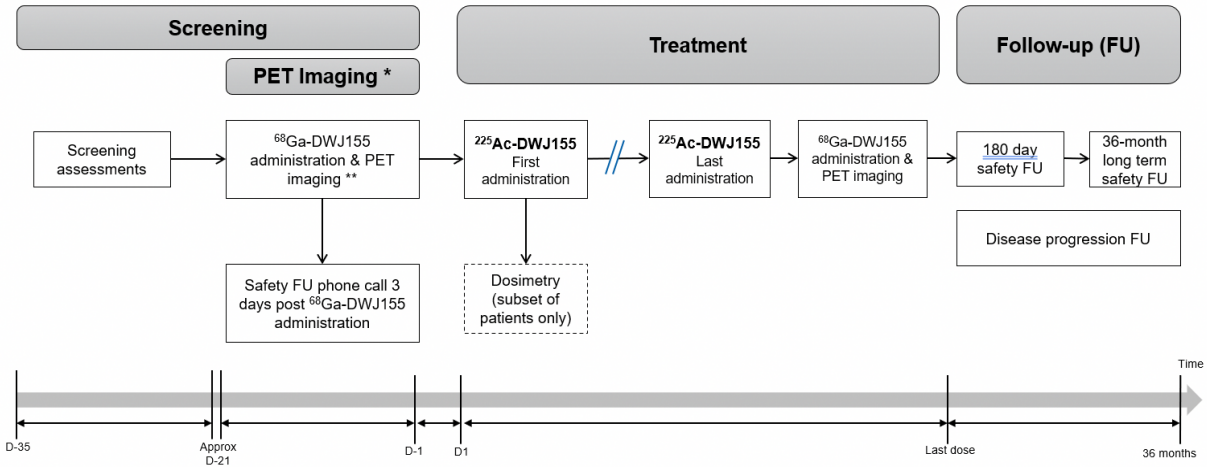
Figure 1-1 Study design



*RD(s) = recommended dosage(s) (where dosage is defined as administered activity per cycle, total number of cycles and schedule) of ^{225}Ac-DWJ155

**AGA are non-Actionable Gene Alterations any genetic alterations for which a targeted therapy is approved. Please refer to Section 5.1 for details on inclusion criteria

Figure 1-2 Study flow



* PET imaging will be performed in all participants; eligibility for treatment with ^{225}Ac-DWJ155 in NSCLC and bladder cancer will be confirmed based on positive ^{68}Ga-DWJ155 tumor uptake. Please refer to Section 5.1 for details on inclusion criteria.

** ^{68}Ga-DWJ155 administration and PET imaging should be performed as soon as possible after completion of screening assessments for indications with known HER2 status (i.e., breast cancer and GEA). Please refer to Section 5.1 for details on inclusion criteria.

1.3 Schedule of activities (SoA)

The SoA lists all the assessments and when they are performed according to the study period. Please refer to Section 4.1 for study design details.

- Table 1-3 details the assessments to be performed during screening and imaging periods
- Table 1-4 details the visit evaluation schedule during the treatment and follow-up periods for a 6-week administration schedule
- Table 1-5 details the visit evaluation schedule during the treatment and follow-up periods for a 4-week administration schedule
- Table 1-6 details the visit evaluation schedule during the treatment and follow-up periods for a 8-week administration schedule
- Table 1-7 details the additional activities to be performed for the dosimetry and/or PK subgroup

Table 1-2 Allowed visit windows

Visit name	Visit window
Cycle 1 all visits	±3 days
Cycle 2 onwards all visits (6- and 8-week schedules)	±7 days
Cycle 2 onwards all visits (4-week schedule)	-2/+7 days
Conventional imaging	-7 days after first administration of ^{225}Ac-DWJ155 until EoT, then ±15 days
^{68}Ga-DWJ155 PET/CT or PET/MRI	At specified time points
Post ^{68}Ga-DWJ155 safety follow-up	-1/+2 days
^{225}Ac-DWJ155 SPECT/CT	At specified time points

Visit name	Visit window
PK and ECG	At specified time points
Echocardiogram	-7 days
End of treatment (EoT)	±7 days
Safety follow-up	±15 days
Disease progression follow-up	±15 days

All data obtained from these assessments must be supported in the participant's source documentation.

Rescreening of participants is allowed (refer to Section 5.4 for details).

Participants should be seen for all visits/assessments as outlined in the SoA or as close to the designated day/time as possible, as specified in Table 1-2. Missed or rescheduled visits should not lead to automatic discontinuation.

Participants who complete or discontinue from study treatment are to complete the end of treatment visit and attend the follow-up visits as indicated in the SoA. At this visit, the AEs and concomitant medications not previously reported must be recorded on the electronic Case Report Form (eCRF). All follow-up visits should also be done as per the SoA and the required data entered in the CRF.

In some instances, related to participant safety and/or efficacy, the Novartis Medical Monitor may request copies of de-identified source documents including, but not limited to, photographs, radiographic images, and relevant medical records as allowed per local regulations. Based on local radioprotection agency requirements, additional assessments or safety procedures may be performed (e.g., overnight hospital stays for participant surveillance).

In extraordinary situations (e.g., a potential supply disruption and/or a quality incident) the investigator may immediately be:

- informed to not administer a dose from an impacted batch to the participant, and/or
- asked to closely monitor participants who have received an administration from an impacted batch for AEs, and
- advised how to manage participants in screening, imaging or on treatment in case of supply disruption.

For visits during treatment and at the 180 day safety follow-up visits that only have chemistry, hematology and urinalysis assessments, these assessments may be conducted at an off-site facility. All other visits with these lab assessments and other planned assessments are not eligible for off-site delegation.

During a public health emergency as defined in Section 4.6, alternative methods of providing continuing care may be implemented by the Investigator. If allowable by a local health authority, national and local regulations and depending on operational capabilities, phone calls, virtual contacts (e.g., tele consultation) or visits by site staff/ off-site healthcare professional(s) to the participant´s home, can replace in person visits for certain protocol assessments such as safety monitoring, lab assessments and urine pregnancy test for the duration of the disruption until it is safe for the participant to visit the site again.

If the Investigator delegates tasks to an off-site healthcare professional, the Investigator must ensure the individual(s) is/are qualified and appropriately trained to perform assigned duties. The Investigator must oversee their conduct and remain responsible for the evaluation of the data collected.

Table 1-3 Schedule of activities, Screening and PET imaging periods

Period	**Screening**	**Screening/Imaging**	**Follow-up**
Visit Name	**Screening**	**68Ga-DWJ155 imaging**	**Post 68Ga-DWJ155 imaging**
Days	**-35 to 68Ga-DWJ155 imaging (min. -21)**	**min. -21**	**3 days post 68Ga-DWJ155 imaging (phone call or visit)**
Obtain informed consent	X		
Demography	X		
Inclusion/exclusion criteria	X	X[1]	
Diagnosis and extent of cancer	X		
Relevant medical history/current medical history	X		
Prior antineoplastic therapy	X		
Prior/concomitant medications, surgery, medical procedures and non-drug therapies	X		
ECOG Performance Status	X		
Smoking history	X		
Adverse events		X	
Physical examination	S		
Height	X		
Weight	X		
Vital signs	X	Before and 0-30 min after ^{68}Ga-DWJ155 administration	
SpO2 (for Japan only)	S	S	
Hematology	X		
Chemistry	X		
Coagulation	X		

Period	**Screening**	**Screening/Imaging**	**Follow-up**
Visit Name	**Screening**	**68Ga-DWJ155 imaging**	**Post 68Ga-DWJ155 imaging**
Days	**-35 to 68Ga-DWJ155 imaging (min. -21)**	**min. -21**	**3 days post 68Ga-DWJ155 imaging (phone call or visit)**
Serum pregnancy test (within 72h prior to ^{68}Ga-DWJ155 administration)	S		
Urinalysis	X		
ECG	X		
Cardiac imaging (TTE or MUGA - TTE preferred)	X		
Chest X ray (for Japan only)	X		
Conventional imaging for tumor response as per RECIST v1.1 (CT and/or MRI, and other imaging assessments as clinically indicated)	X[2]		
^{68}Ga-DWJ155 administration		X	
^{68}Ga-DWJ155 PET/CT or PET/MRI		X[3]	
Determination of tumor ^{68}Ga-DWJ155 positivity (after ^{68}Ga-DWJ155 administration)		X	
Archival tumor sample (if available) or newly obtained tumor sample (for all participants in dosage escalation)			X
Newly obtained tumor sample (for participants in dosage expansion)			X[4]
IRT registration	X		

Period	**Screening**	**Screening/Imaging**	**Follow-up**
Visit Name	**Screening**	**68Ga-DWJ155 imaging**	**Post 68Ga-DWJ155 imaging**
Days	**-35 to 68Ga-DWJ155 imaging (min. -21)**	**min. -21**	**3 days post 68Ga-DWJ155 imaging (phone call or visit)**

[X] Assessment to be recorded in the clinical database or received electronically from a vendor
[S] Assessment to be recorded in the source documentation only
[1] For indications with unknown HER2 status (i.e. NSCLC and bladder cancer), eligibility for treatment of ^{225}Ac-DWJ155 to be confirmed based on positive ^{68}Ga-DWJ155 tumor uptake. For details, please refer to Section 5.1 Inclusion criteria.
[2] Collections to take place at least 1 day prior to ^{68}Ga-DWJ155 administration. The same imaging modality as at baseline to be used throughout the study.
[3] All participants will undergo ^{68}Ga-DWJ155 imaging at baseline. Please refer to Section 8.3.2 for additional details on imaging assessments.
[4] For participants in Expansion part who will receive ^{225}Ac-DWJ155 treatment, this collection may be substituted with an archival tumor sample collected up to 6 months prior to Screening, with no intervening treatment. In cases where it is not safe or medically feasible to provide this sample, enrollment may be possible following discussion between investigator and Novartis. Biopsies can be collected separately from the post ^{68}Ga-DWJ155 imaging visit, up to 5 days prior to ^{225}Ac-DWJ155 administration. Specifications for these samples are described in Section 8.8.

Table 1-4 Schedule of activities, Treatment and Follow-up period (all participants) - 6-week schedule

Period	**Treatment**													**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**						**Cycle 2 (and subsequent cycles)**						**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**1**	**8**	**15**	**22**	**29**	**36**	**Last dose +42d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Physical examination	S				S		S						S			
Weight	X						X						X			
SpO2 (for Japan only)	S	S	S	S	S	S	S						S			
ECOG Performance Status	X						X						X			
Chemistry	X	X	X	X	X	X	X		X[1]		X[1]		X	X[1]		
Hematology	X	X	X	X	X	X	X	X[1,2]	X[1]	X[1,2]	X[1]	X[1,2]	X	X[1]		
Urinalysis	X	X	X	X	X	X	X		X[1]		X[1]		X	X[1]		
Thyroid function	X						X									
Vital signs	X[3]	X	X	X	X	X	X[3]						X[3]			
Pregnancy test	S[4]						S[4]						S[4]	S[4]		
^{225}Ac-DWJ155 administration	X						X									
^{68}Ga-DWJ155 administration													X			

Period	**Treatment**													**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**						**Cycle 2 (and subsequent cycles)**						**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**1**	**8**	**15**	**22**	**29**	**36**	**Last dose +42d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
^{68}Ga-DWJ155 PET/CT or PET/MRI													X			
ECG[5]	X						X						X			
Cardiac imaging (TTE or MUGA - TTE preferred)	Every 12 weeks (-7 days) after the first dose of ^{225}Ac-DWJ155 treatment until EoT. Final reports must be locally reviewed prior to ^{225}Ac-DWJ155 administration.												X[6]			
Conventional imaging for tumor response as per RECIST v1.1 (CT and/or MRI, and other imaging assessments as clinically indicated)	Every 6 weeks (-7 days) after first dose of ^{225}Ac-DWJ155 treatment. Radiological report(s) must be locally reviewed prior to ^{225}Ac-DWJ155 administration.														Every 12 weeks after the last dose of ^{225}Ac-DWJ155 treatment during Disease progression FU	

Period	**Treatment**													**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**						**Cycle 2 (and subsequent cycles)**						**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**1**	**8**	**15**	**22**	**29**	**36**	**Last dose +42d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Prior/concomitant medications, surgery, medical procedures and non-drug therapies	Assess continuously													X		
Adverse events	Assess continuously													X[7]		X[8]
Chest X ray (for Japan only)			X													
Subsequent anti-neoplastic therapies															X	
Long term safety follow-up (phone call or visit)																X

Period	**Treatment**												**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**						**Cycle 2 (and subsequent cycles)**					**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**1**	**8**	**15**	**22**	**29**	**36**	**Last dose +42d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Blood sample for soluble factors	pre-dose		X		X		pre-dose, only C2 D1 and C3 D1						X			
Blood sample for cfDNA	pre-dose		X		X		pre-dose, only C2 D1 and C3 D1						X		Every 12 weeks after the last dose of ^{225}Ac-DWJ155 treatment during Disease progression FU	
Urine sample for renal biomarkers	pre-dose						pre-dose									

[X] Assessment to be recorded in the clinical database or received electronically from a vendor
[S] Assessment to be recorded in the source documentation only
[1] Eligible for off-site delegation.
[2] Required only if most recent prior hematologic assessment shows CTCAE grade 2 or higher neutropenia or thrombocytopenia, or CTCAE grade 3 or higher anemia.
[3] Before and 0-30 min after ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155 administration
[4] Serum pregnancy test to be collected within 72h prior to ^{225}Ac-DWJ155 administration except at EoT when it will be collected on the day of study visit. A monthly urine pregnancy test is required for 8 months after the last ^{225}Ac-DWJ155 administration.

Period	**Treatment**												**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**						**Cycle 2 (and subsequent cycles)**						**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**1**	**8**	**15**	**22**	**29**	**36**	**Last dose +42d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**

[5] Refer to Section 8.4.3 for ECG collection timepoints
[6] Collections to take place at least 1 day prior to ^{68}Ga-DWJ155 administration
[7] Only non serious AEs related to ^{225}Ac-DWJ155 treatment and SAEs regardless of causality are to be reported to Novartis. If a new post-treatment antineoplastic therapy has been administered, then only SAEs related to ^{225}Ac-DWJ155 treatment are to be reported to Novartis. Follow local law/regulations if requirements differ. Please refer to Section 8.10.1 and Section 8.10.3 for details.
[8] Only SAEs related to ^{225}Ac-DWJ155 treatment and SAEs of second primary malignancies regardless of causality are to be reported to Novartis, unless otherwise specified by local law/regulations. Please refer to Section 8.10.3.

Table 1-5 Schedule of activities, Treatment and Follow-up period (all participants) - 4-week schedule

Period	**Treatment**													**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**				**Cycle 2**				**Cycle 3 (and subsequent cycles)**				**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**Last dose + 28d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Physical examination	S		S		S				S				S			
Weight	X				X				X				X			
SpO2 (for Japan only)	S	S	S	S	S				S				S			
ECOG Performance Status	X				X				X				X			
Chemistry	X	X	X	X	X	X[1]	X[1]	X[1]	X		X[1]		X	X[1]		
Hematology	X	X	X	X	X	X[1]	X[1]	X[1]	X	X[1,2]	X[1]	X[1,2]	X	X[1]		
Urinalysis	X	X	X	X	X	X[1]	X[1]	X[1]	X		X[1]		X	X[1]		
Thyroid function	X				X				X							
Vital signs	X[3]	X	X	X	X[3]				X[3]				X[3]			
Pregnancy test	S[4]				S[4]				S[4]				S[4]	S[4]		
^{225}Ac-DWJ155 administration	X				X				X							
^{68}Ga-DWJ155 administration													X			

Period	**Treatment**													**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**				**Cycle 2**				**Cycle 3 (and subsequent cycles)**				**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**Last dose + 28d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
^{68}Ga-DWJ155 PET/CT or PET/MRI													X			
ECG[5]	X				X				X				X			
Cardiac imaging (TTE or MUGA - TTE preferred)	Every 12 weeks (-7 days) after first dose of ^{225}Ac-DWJ155 treatment until EoT. Final reports must be locally reviewed prior to ^{225}Ac-DWJ155 administration.												X[6]			
Conventional imaging for tumor response as per RECIST v1.1 (CT and/or MRI, and other imaging assessments as clinically indicated)	Every 8 weeks (-7 days) after first dose of ^{225}Ac-DWJ155 treatment. Radiological report(s) must be locally reviewed prior to ^{225}Ac-DWJ155 administration.														Every 12 weeks after the last dose of ^{225}Ac-DWJ155 treatment during Disease progression FU	

Period	**Treatment**													**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**				**Cycle 2**				**Cycle 3 (and subsequent cycles)**				**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**Last dose + 28d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Prior/concomitant medications, surgery, medical procedures and non-drug therapies	Assess continuously													X		
Adverse events	Assess continuously													X[7]		X[8]
Chest X ray (for Japan only)			X													
Subsequent anti-neoplastic therapies															X	
Long term safety follow-up (phone call or visit)																X
Blood sample for soluble factors	pre-dose		X		pre-dose				pre-dose, C3D1 only				X			

Period	**Treatment**												**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**				**Cycle 2**				**Cycle 3 (and subsequent cycles)**				**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**1**	**8**	**15**	**22**	**Last dose + 28d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Blood sample for cfDNA	pre-dose		X		pre-dose				pre-dose, C3 D1 only				X		Every 12 weeks after the last dose of ^{225}Ac-DWJ155 treatment during Disease progression FU	
Urine sample for renal biomarkers	pre-dose				pre-dose				pre-dose							

[X] Assessment to be recorded in the clinical database or received electronically from a vendor
[S] Assessment to be recorded in the source documentation only
[1] Eligible for off-site delegation.
[2] Required only if most recent prior hematologic assessment shows CTCAE grade 2 or higher neutropenia or thrombocytopenia, or CTCAE grade 3 or higher anemia.
[3] Before and 0-30 min after ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155 administration
[4] Serum pregnancy test to be collected within 72h prior to ^{225}Ac-DWJ155 administration except at EoT when it will be collected on the day of study visit. A monthly urine pregnancy test is required for 8 months after the last ^{225}Ac-DWJ155 administration
[5] Refer to Section 8.4.3 for ECG collection timepoints
[6] Collections to take place at least 1 day prior to ^{68}Ga-DWJ155 administration
[7] Only non serious AEs related to ^{225}Ac-DWJ155 treatment and SAEs regardless of causality are to be reported to Novartis. If a new post-treatment antineoplastic therapy has been administered, then only SAEs related to ^{225}Ac-DWJ155 treatment are to be reported to Novartis. Follow local law/regulations if requirements differ. Please refer to Section 8.10.1 and Section 8.10.3 for details.
[8] Only SAEs related to ^{225}Ac-DWJ155 treatment and SAEs of second primary malignancies regardless of causality are to be reported to Novartis, unless otherwise specified by local law/regulations.. Please refer to Section 8.10.3.

Table 1-6 Schedule of activities, Treatment and Follow-up period (all participants) - 8-week schedule

Period	**Treatment**																**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**								**Cycle 2 (and subsequent cycles)**								**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**Last dose + 56d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Physical examination	S		S						S								S			
Weight	X								X								X			
SpO2 (for Japan only)	S	S	S	S	S	S	S	S	S								S			
ECOG Performance Status	X								X								X			
Chemistry	X	X	X	X	X	X	X	X	X		X[1]		X[1]		X[1]		X	X[1]		
Hematology	X	X	X	X	X	X	X	X	X	X[1,2]	X[1]	X[1,2]	X[1]	X[1,2]	X[1]	X[1,2]	X	X[1]		
Urinalysis	X	X	X	X	X	X	X	X	X		X[1]		X[1]		X[1]		X	X[1]		
Thyroid function	X								X											
Vital signs	X[3]	X	X	X	X	X	X	X	X[3]								X[3]			
Pregnancy test	S[4]								S[4]								S[4]	S[4]		
^{225}Ac-DWJ155 administration	X								X											

Period	**Treatment**																**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**								**Cycle 2 (and subsequent cycles)**								**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**Last dose + 56d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
^{68}Ga-DWJ155 administration																	X			
^{68}Ga-DWJ155 PET/CT or PET/MRI																	X			
ECG[5]	X								X								X			
Cardiac imaging (TTE or MUGA - TTE preferred)	Every 12 weeks (-7 days) after the first dose of ^{225}Ac-DWJ155 treatment until EoT. Final reports must be locally reviewed prior to ^{225}Ac-DWJ155 administration, if coinciding with the next administration.																X[6]			
Conventional imaging for tumor response as per RECIST v1.1 (CT and/or MRI, and other imaging assessments as clinically indicated)	Every 8 weeks (-7 days) after first dose of ^{225}Ac-DWJ155 treatment. Radiological report(s) must be locally reviewed prior to ^{225}Ac-DWJ155 administration.																		Every 12 weeks after the last dose of ^{225}Ac-DWJ155 treatment during Disease progression FU	

Period	**Treatment**																**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**								**Cycle 2 (and subsequent cycles)**								**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**Last dose + 56d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Prior/concomitant medications, surgery, medical procedures and non-drug therapies	Assess continuously																	X		
Adverse events	Assess continuously																	X[7]		X[8]
Chest X ray (for Japan only)			X																	
Subsequent anti-neoplastic therapies																			X	
Long term safety follow-up (phone call or visit)																				X

Period	**Treatment**																**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**								**Cycle 2 (and subsequent cycles)**								**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**Last dose + 56d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Blood sample for soluble factors	pre-dose		X		X				pre-dose, only C2D1 and C3D1								X			
Blood sample for cfDNA	pre-dose		X		X				pre-dose, only C2D1 and C3D1								X		Every 12 weeks after the last dose of ^{225}Ac-DWJ155 treatment during Disease progression FU	

Period	**Treatment**															**36 month Long Term Follow-up**		
Visit Name	**Cycle 1**								**Cycle 2 (and subsequent cycles)**							**EoT**	**180 day Safety FU**	**Disease progression FU**	**Long Term FU**
Days	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**1**	**8**	**15**	**22**	**29**	**36**	**43**	**50**	**Last dose + 56d**	**At 90 and 180 days from last dose**	**Every 90 days from last dose, in parallel to safety FU**	**Every 180 days starting after 180 day safety FU**
Urine sample for renal biomarkers	pre-dose								pre-dose											

[X] Assessment to be recorded in the clinical database or received electronically from a vendor
[S] Assessment to be recorded in the source documentation only
[1] Eligible for off-site delegation.
[2] Required only if most recent prior hematologic assessment shows CTCAE grade 2 or higher neutropenia or thrombocytopenia, or CTCAE grade 3 or higher anemia.
[3] Before and 0-30 min after ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155 administration
[4] Serum pregnancy test to be collected within 72h prior to ^{225}Ac-DWJ155 administration except at EoT when it will be collected on the day of study visit. A monthly urine pregnancy test is required for 8 months after the last ^{225}Ac-DWJ155 administration.
[5] Refer to Section 8.4.3 for ECG collection timepoints
[6] Collections to take place at least 1 day prior to ^{68}Ga-DWJ155 administration
[7] Only non serious AEs related to ^{225}Ac-DWJ155 treatment and SAEs regardless of causality are to be reported to Novartis. If a new post-treatment antineoplastic therapy has been administered, then only SAEs related to ^{225}Ac-DWJ155 treatment are to be reported to Novartis. Follow local law/regulations if requirements differ. Please refer to Section 8.10.1 and Section 8.10.3 for details.
[8] Only SAEs related to ^{225}Ac-DWJ155 treatment and SAEs of second primary malignancies regardless of causality are to be reported to Novartis, unless otherwise specified by local law/regulations. Please refer to Section 8.10.3.

Table 1-7 Schedule of activities, Dosimetry and/or PK subgroup

Period	**Treatment Phase**													
Visit Name	**Cycle 1**													
Days	**1**								**2**	**3**	**4**	**5**	**8**	**15**
Time (post-dose)	**Pre ^{225}Ac-DWJ155 admin**	**T0s**[1]	**T0e**[2]	**T0e + 30min**	**T0e + 1h**	**T0e + 2h**	**T0e + 4h**	**T0e + 6h**	**T0e + 24h**	**T0e + 48h**	**T0e + 72h**	**T0e + 96h**	**T0e + 168h**	**T0e + 312h**
^{225}Ac-DWJ155 administration		X												
ECG[3]	X		+10min		±10min		±15min							
SPECT/CT[4]						±1h			±4h	±4h	±6h[5]		±6h[6]	
Blood for PK and HPLC[7]			+10min	±10min	±10min	±15min	±30min	±30min	±2h	±4h	±6h[5]			
Urine for PK and HPLC[8]			0h - 6h(±30min)						6h(±30min) - 24h(±2h)	24h(±2h) - 48h(±4h)	48h(±4h) - 72h(±4h)[9]			

[X] Assessment to be recorded in the clinical database or received electronically from a vendor
[1] T0s: start of ^{225}Ac-DWJ155 administration
[2] T0e: end of ^{225}Ac-DWJ155 administration
[3] Refer to Section 8.4.3 for additional information on ECG assessments. If clinically indicated as per the investigator's judgment (e.g., if significant ECG abnormalities are observed at the end of the ^{225}Ac-DWJ155 administration), the collection of ECG assessments at additional post-dose time points coinciding with PK sampling time points per Table 8-4 should be considered.
[4] Each participant should void his/her bladder immediately before each SPECT scan. A precise time window of each acquisition will be informed by emerging data and will be communicated to investigators at time of individual participant enrollment. Refer to Section 8.7.4 and Section 8.7.5 for details on SPECT/CT assessments and for additional SPECT/CT scans in subsequent cycles, where required by local regulations.
[5] Not applicable for dosage level 1 participants.
[6] Not applicable for dosage level 1 and dosage level 2 participants.
[7] Refer to Section 8.7.1 for detailed information on blood PK assessments.
[8] Up to 6 participants in selected cohorts at relevant dosage levels and at dedicated sites. The assignment of a participant will be coordinated by Novartis. Refer to Section 8.7.2 for additional information on urine PK assessments.
[9] If the first 3 participants who contribute to urine PK show negligible radioactivity (≤ 5% of IA) detected in the 48-72h collection interval, this collection interval may be omitted in the remaining participants and it will be communicated to investigators at time of individual participant enrollment.

2 Introduction

2.1 Study rationale

DWJ155 is a peptide that binds to the HER2 extracellular domain with high affinity and includes a metal chelator for radiolabeling with either gallium-68 (Ga-68) for PET imaging or actinium-225 (Ac-225) for therapy.

The purpose of this phase I study is to evaluate the safety, tolerability and preliminary anti-tumor activity of ^{225}Ac-DWJ155 and the safety of ^{68}Ga-DWJ155 in participants with locally advanced unresectable or metastatic HER2+ breast cancer, HR+/HER2-negative breast cancer, TNBC, NSCLC, HER2 IHC 3+ or IHC 2+ (ISH + or -) GEA, and bladder cancer.

Data and samples will be collected as described to support the study objectives and may also be used to learn more about the study indications and/or treatments.

2.2 Background

HER2 is a member of the EGFR/ErbB family of receptor tyrosine kinases, which comprises HER1/EGFR, HER2, HER3 and HER4 (Loibl and Gianni 2017 and Drago et al 2022). The oncogenic potential of HER2 is well characterized, which most notably occurs via amplification of the *HER2* genetic locus leading to protein over-expression on the cell membrane (Galogre et al 2023) with resultant poor clinical outcomes (Slamon et al 1987). Immunohistochemistry (IHC) and ISH are the standard assays used to characterize HER2 amplification status in tumor tissue (Wolff et al 2023). However, the accuracy of IHC in the setting of lower levels of HER2 expression, particularly the ability to accurately delineate HER2 0, HER2-ultralow (IHC 0 with weak or incomplete staining in up to 10% of cells), and HER2-low (IHC 1+ and 2+/ISH-), may be limited (Rüschoff et al 2022, Baez-Navarro et al 2023, Zaakouk et al 2023, and Lu et al 2022).

Data from a Novartis-developed ultrasensitive HER2 IHC assay indicate that nearly all breast cancers, non-small cell lung cancers, GEAs, and bladder cancers express detectable levels of HER2 protein, including those scored as 0 by the standard clinical assay using American Society of Clinical Oncology/College of American Pathologists (ASCO/CAP) scoring. For details, please refer to the ^{225}Ac-DWJ155 Investigator's Brochure (IB).

All currently approved HER2-targeting agents inhibit intracellular receptor signaling (Drago et al 2022 and Stern 2000). Historically, HER2-targeted agents have leveraged the protein expression differential between tumor and normal tissue to maximize their therapeutic index. While this mechanism has been successful in the setting of HER2 amplification and genetic dependency, it is less efficacious in tumor types that are either HER2 non-amplified or are not dependent on HER2 (Krug et al 2005, Gatzemeier et al 2004, and Smit et al 2024). More recently, trastuzumab-based antibody drug conjugates (ADCs) have emerged that also inhibit HER2 receptor signaling, but function primarily by delivery of cytotoxic payloads. These agents have been shown to have superior efficacy in participants with HER2-positive disease (Modi et al 2020, Verma et al 2012, and Modi et al 2022). One ADC, trastuzumab-deruxtecan, has recently gained Food and Drug Administration (FDA) tumor agnostic accelerated approval for the treatment of adult participants with unresectable or metastatic HER2 IHC 3+ solid tumors who have received prior systemic treatment and have no

satisfactory alternative treatment options. This same drug has additionally demonstrated efficacy in breast cancer participants with lower levels of HER2 expression, including HER2-low (Modi et al 2022) and HER2-ultralow (Bardia et al 2024). These findings provide proof-of-concept that targeting HER2 expressing, but non-amplified tumors, is a viable therapeutic strategy. The side effect profiles of HER2-targeting ADCs are substantial and may limit their widespread use, particularly in early disease settings (Abuhelwa et al 2022).

2.2.1 Overview of DWJ155

Radioligands are systemically delivered, radioactively labeled molecules that bind specific cell surface receptors and deliver radiation in a targeted manner (Bodei et al 2022 and Sgouros et al 2020). Radioligand use in oncology is often through a theranostic approach whereby the same targeting molecule is labeled with a radioisotope for either diagnostic imaging (e.g., Ga-68) or therapy (e.g., Ac-225). Radioligand imaging (RLI) can be used to determine the biodistribution of the molecule as a surrogate for target protein expression in tumor and non-tumor tissue and can be used in participant selection. On the other hand, radioligand therapies (RLTs) deploy higher energy radioisotopes to damage target cell DNA and induce cell death.

The RLT approach is a platform with the potential to provide a novel therapeutic option for participants with tumors expressing HER2. Advantages of Ac-225 RLTs include targeted local delivery of radiation to tumor cells while sparing nearby normal tissues due to the short range of alpha particles (energy deposition range ~100 μm), high linear energy transfer (LET) of alpha particles, and the potential to directly image the biodistribution of RLT molecules using Single-Photon Emission Computed Tomography (SPECT).

2.2.1.1 Non-clinical experience

DWJ155 comprises a peptide molecule, which binds with high affinity (K_D in sub-nanomolar range) to the extracellular domain of HER2. It is conjugated to a metal chelator for radiolabeling with Ga-68 for Positron Emission Tomography (PET) imaging or Ac-225 for therapy. For a comprehensive overview of the pre-clinical data, please refer to the ^{225}Ac-DWJ155 IB for further details.

Using imaging to identify tumors with lower levels of target expression (such as HER2-low breast cancer and NSCLC) requires an imaging agent with high affinity and specificity. Imaging with ^{68}Ga-DWJ155 may improve the accuracy of HER2 detection and enable more precise participant selection and stratification by HER2 levels, even in the setting of HER2-low disease.

Low normal tissue uptake and high tumor uptake and retention is anticipated based on biodistribution studies of ^{225}Ac-DWJ155 in both non-tumor bearing mice and mice bearing breast cancer xenografts. Efficacy of ^{225}Ac-DWJ155 was evaluated in HER2-low breast cancer using patient-derived xenograft (PDX) models with ASCO/CAP HER2 IHC scores ranging from 1 to 2+, ISH negative. ^{225}Ac-DWJ155 demonstrated compelling single agent activity in all models. In estrogen receptor (ER) positive models, ^{225}Ac-DWJ155 demonstrated superiority to a standard-of-care therapy, fulvestrant and ribociclib. ^{225}Ac-DWJ155 also demonstrated efficacy in HER2 expressing NSCLC xenograft models that were superior to the platinum-doublet regimens tested (refer to the ^{225}Ac-DWJ155 IB for further details).

2.3 Benefit/Risk assessment

Two investigational drugs will be administered to participants enrolled in this study, ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155, both containing the identical ligand (DWJ155), labeled with different radioisotopes. The aim is to ultimately develop both as a theranostic pair, using ^{68}Ga-DWJ155 PET imaging to identify participants who may benefit from treatment with ^{225}Ac-DWJ155.

The potential benefits of this study are anticipated to be due to radiation delivered by ^{225}Ac-DWJ155 to sites of disease. ^{225}Ac-DWJ155 has not previously been administered to humans, and the therapeutic effect of ^{225}Ac-DWJ155 to treat participants with cancer is unknown. However, the potential for efficacy is supported by preclinical data obtained with translationally relevant breast and lung tumor models, where HER2 expression level was 1+ or 2+/ISH negative. Treatment with ^{225}Ac-DWJ155 resulted in tumor regression in these models (refer to ^{225}Ac-DWJ155 IB for further details).

As with other radioligand therapies and imaging agents, the primary risks from this study relate to radiation exposure. This comes in three forms: radiation from ^{225}Ac-DWJ155, radiation from ^{68}Ga-DWJ155, and radiation from standard of care (SoC) imaging assessments. All of these elements contribute to a participant's cumulative radiation exposure.

^{225}Ac-DWJ155 radiation exposure

The primary risks to participants enrolled in this study are anticipated to be due to radiation delivered by ^{225}Ac-DWJ155 to normal tissues / organs-at-risk. Organ-specific toxicities are dependent on the absorbed radiation dose and the radiosensitivity of the organ, and anticipated radiation-dependent effects are summarized in the IB (refer to IB section 7.4.3). These risks are considered justified given the high unmet need and poor prognosis of participants on this trial; all participants enrolled on this trial will have advanced cancer with limited life expectancy and will have exhausted standard-of-care treatment options. The time to development of certain late toxicities from treatment may also be longer than the anticipated survival of many participants on trial. For example, radiation nephropathy is expected to develop on the order of 6-18 months post-treatment (Parihar et al 2022), while the median overall survival for phase 1 oncology trial participants similar to the anticipated participant population in this trial is less than 12 months (Weiss et al 2020, Goel et al 2024, Menon et al 2022).

^{68}Ga-DWJ155 radiation exposure

^{68}Ga-DWJ155 has not previously been characterized in humans. Based on extrapolation from preclinical data (refer to ^{68}Ga-DWJ155 IB for further details), in the scenario where 250 MBq or 500 MBq of ^{68}Ga-DWJ155 is administered, the estimated effective dose for the total body is in the range of 7.6 – 8.1 mSv and 15.2 – 16.3 mSv, respectively, which is below the radiation dose limit of 30 mSv for adult participants receiving a single administration, as per the Guidance for Industry and Researchers, "The Radioactive Drug Research Committee: Human Research Without an Investigational New Drug Application" (FDA 2010), and is approximately the value of maximum allowed effective dose for radiological workers per year of 20 mSv. The estimated whole body effective dose is also approximately that of other standard diagnostic techniques (e.g., computed tomography (CT) scan: 14 – 20 mSv). Participants may receive up to 2 administrations of ^{68}Ga-DWJ155 (prior to Cycle 1 and at EoT), with estimated and cumulative

total body effective doses of 15.2 – 16.2 mSv and 30.4 – 32.6 mSv, for clinical radioactivities between 250 MBq and up to 500 MBq per administration, respectively.

SoC imaging assessment radiation exposure

As per protocol, participants included in the dosage escalation and dosage expansion will undergo repeated tumor assessments in accordance with the Schedule of Assessments during the study to assess the disease over time by Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1) criteria. When using CT for performing these tumor assessments, the contribution of a diagnostic CT scan is maximum 20 mSv per exam. Participants associated with radiodiagnostic procedures (including both radiological and nuclear medicine) is classified as moderate according to International Commission on Radiological Protection (ICRP) Publication 62. This risk is considered acceptable given the potential benefit to participants with high unmet medical need (e.g. advanced breast and lung cancer), and is mitigated by adherence to As Low as Reasonably Achievable (ALARA) principles, specifically: 1) the conventional diagnostic imaging procedures performed in this study are comparable to those performed in routine clinical practice; 2) the number and time points of PET/CT and SPECT/CT assessments are restricted to the minimum necessary to characterize the biodistribution of ^{225}Ac-DWJ155 and to evaluate changes in target expression; 3) the cumulative ionizing radiation exposure from these radiodiagnostic procedures can be considered negligible compared to the exposure resulting from ^{225}Ac-DWJ155 administration.

The low mass dose of DWJ155, administered with either ^{68}Ga-DWJ155 or ^{225}Ac-DWJ155, is not expected to have pharmacologic activity. In the case of DWJ155, this also includes a lack of impact on HER2 signaling, as demonstrated in HER2-expressing cells (refer to ^{68}Ga-DWJ155 IB and ^{225}Ac-DWJ155 IB for details).

In preclinical safety studies, a non-radioactive mixture of DWJ155 and ^{175}Lu-DWJ155 (used as a cold surrogate for Ac-225) was evaluated as a single administration in rats and dogs. Notably, an immediate hypersensitivity reaction was seen in 3 of 10 dogs in the high dose group (1.2 mg/kg), which was successfully managed with diphenhydramine treatment. There were no manifestations of a similar reaction in the low dose group (0.3 mg/kg), providing a body surface area-based safety margin of 25-fold (assuming an average of 1.7 m^2 participant) at this dosage level relative to a highest foreseeable mass dose of 400 µg. Given the large safety margin (25-fold) for this finding and the absence of anaphylactic/anaphylactoid reactions, no standard pre-medication with an antihistamine (e.g., diphenhydramine) is implemented. This will also inform whether pre-medication with an antihistamine is required for future studies. The protocol recommends following institutional standard of care for the treatment of hypersensitivity (refer to Section 6.6.3 for more information).

Appropriate eligibility criteria, specific dosage-limiting toxicity definitions, dosage modification guidelines, and stopping rules are included in this protocol. Recommended guidelines for the management of adverse events are provided in this protocol and in the ^{68}Ga-DWJ155 IB and ^{225}Ac-DWJ155 IB.

The study's exclusion criteria and study procedures, including clinical monitoring, are set up to minimize risks to participants' safety, and must be adhered to closely. As in any clinical trial, there may be unforeseen risks which could be serious. An independent data monitoring committee (DMC) will be used during dosage expansion.

Women of child-bearing potential (WOCBP) and sexually active males must be informed that taking the study treatment may involve unknown risks to the fetus if pregnancy were to occur during the trial and agree that in order to participate in the trial, they must adhere to the contraception requirements outlined in the exclusion criteria. WOCBP are advised not to have children while in the clinical trial and for 8 months after the last administration of ^{225}Ac-DWJ155, and sexually active men are advised not to have children while in the clinical trial and for 5 months after the last administration of ^{225}Ac-DWJ155. If there is any question that the participant will not reliably comply, they should not be entered or continue in the trial.

2.3.1 Risks and benefits for planned off-site procedures

It is not anticipated that participants will be exposed to greater risks when participating in off-site assessments than in on-site assessments.

3 Objectives, endpoints, and estimands

3.1 Objectives and related endpoints

Table 3-1 Objectives and related endpoints

Objective(s)	Endpoint(s)
Primary objective(s)	**Endpoint(s) for primary objective(s)**
• Assess the safety and tolerability for ^{225}Ac-DWJ155 and identify the recommended dosage(s) (where dosage is defined as administered activity per cycle, total number of cycles and schedule) of ^{225}Ac-DWJ155 for further clinical evaluation	• Safety: Incidence and severity of adverse events (AEs), including dosage limiting toxicities (DLTs) and serious adverse events (SAEs), changes in laboratory values, vital signs and ECGs • Tolerability: Frequency of dosage interruptions and reductions, dosage intensity
Secondary objective(s)	**Endpoint(s) for secondary objective(s)**
• Evaluate the preliminary anti-tumor activity of ^{225}Ac-DWJ155	• Overall response rate (ORR), disease control rate (DCR), duration of response (DOR), and progression-free survival (PFS) based on RECIST v1.1 by local investigator assessment
• Characterize the pharmacokinetics (PK) at Cycle 1 of ^{225}Ac-DWJ155 in blood post-secular equilibrium	• PK parameters (i.e., Cmax, Tmax, AUC, half-life, CL, volume of distribution in the terminal phase (Vz)) of ^{225}Ac-DWJ155 in blood
• Characterize the safety of ^{68}Ga-DWJ155	• Incidence and severity of AEs, including SAEs
Exploratory objective(s)	**Endpoint(s) for exploratory objective(s)**
• Characterize the imaging properties of ^{68}Ga-DWJ155	• Quantitative assessment (expressed as Standardized Uptake Values (SUVs)) of ^{68}Ga-DWJ155 uptake in normal tissues and selected tumor lesions over time
• Characterize the radiation dosimetry of ^{225}Ac-DWJ155 in selected organs and tumor lesions	• Absorbed dose-equivalent of ^{225}Ac-DWJ155 in selected organs and tumor lesions
• Assess the correlation between ^{68}Ga-DWJ155 imaging and ^{225}Ac-DWJ155 dosimetry and efficacy and correlate with HER2 expression	• Expression of HER2 in tumor samples, ^{68}Ga-DWJ155 imaging and ^{225}Ac-DWJ155 biodistribution in selected tumor lesions, anti-tumor activity endpoints per RECIST v1.1

Objective(s)	Endpoint(s)
• Comparison of centrally assessed HER2 expression with locally assessed HER2 status	• Expression of HER2 in tumor samples from local and central data
• Assess the concordance of tumor lesion visualization between different imaging methods	• Number of lesions identified using different imaging methods including conventional imaging (contrast or non-contrast diagnostic CT or magnetic resonance imaging (MRI)), ^{68}Ga-DWJ155 PET imaging, ^{225}Ac-DWJ155 SPECT imaging and ^{18}F-fluorodeoxyglucose (FDG) PET imaging (if available, as part of SOC assessments)
• Evaluate the impact of ^{225}Ac-DWJ155 on soluble markers (e.g., soluble HER2) and correlation with response	• Baseline and post-baseline levels of soluble markers (e.g., HER2) in blood and anti-tumor activity endpoints per RECIST v1.1
• Assess baseline genetic characteristics and whole transcriptomic and key protein profiles as potential predictors of response	• Baseline genetic alterations, gene expression, and protein expression profiles in tumor samples and anti-tumor endpoints per RECIST v1.1
• To assess tumor characteristics for potential prognostic markers and/or predictive markers of efficacy and/or resistance	• Baseline and post-baseline tumor-associated characteristics compared with anti-tumor endpoints per RECIST v1.1
• Evaluate the association between treatment related changes in novel renal biomarker levels and established renal function assessments	• Change from baseline in novel renal biomarker levels and associations with established renal function markers (e.g. serum creatinine increase, urine protein increase and/or estimated glomerular filtration rate decrease)
• Evaluate the metabolic stability of ^{225}Ac-DWJ155 in blood and urine	• Metabolite profiling of blood or plasma and urine samples using high-performance liquid chromatography (HPLC) with radioactivity detection
• Assess ^{68}Ga-DWJ155 PET/CT imaging before and after starting ^{225}Ac-DWJ155 treatment	• Quantitative uptake changes in selected tumor lesions (if available) between ^{68}Ga-DWJ155 PET/CT scans performed at baseline and end of treatment
• Characterize the cumulative urine excretion at Cycle 1 of ^{225}Ac-DWJ155	• Quantification of urinary excretion of ^{225}Ac-DWJ155 (percent of injected activity (%IA), renal clearance)
• Assess the relationship between cell-free DNA (cfDNA) detection and clinical measurements of response and progression	• cfDNA quantification with tumor burden measured by RECIST v1.1 criteria and other measures of clinical efficacy

3.2 Primary estimands

Not applicable.

3.3 Secondary estimands

Not applicable.

4 Study design

4.1 Overall design

This is a FiH, phase I, open-label, multi-center study to evaluate the safety, tolerability, and preliminary activity of single-agent ^{225}Ac-DWJ155 in participants with locally advanced unresectable or metastatic breast cancer, NSCLC, GEA, and bladder cancer. As illustrated in Figure 1-1, the study consists of a dosage escalation part followed by a dosage expansion part. Once the recommended dosage(s) (RD(s), where dosage is defined as administered activity per cycle, total number of cycles and schedule) of ^{225}Ac-DWJ155 as a single agent for future evaluation are determined, the expansion part may commence.

Eligibility confirmation will trigger initiation of ^{225}Ac-DWJ155 central manufacturing and shipment of the dose to the site. ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155 are considered to be Investigational Medical Products (IMPs) in this protocol.

Participating institutions may require that participants stay overnight at the hospital or another facility to adhere to the intensive assessment schedule (e.g., urine sampling for ^{225}Ac-DWJ155 PK assessments). This decision is guided by local regulations, medical judgment, and participant preference. Extension of hospitalization may be considered based on the participants' general status assessed by the investigator.

Dosage escalation part

In the dosage escalation part of the study, several dosage levels of ^{225}Ac-DWJ155 will be evaluated beginning with a 6-week schedule. The provisional dosage levels are shown in Table 6-2.

In the first dosage level (DL1), participants will receive an administered activity of 2 MBq of ^{225}Ac-DWJ155 (refer to Section 4.3.1) per cycle for up to 4 cycles.

Initiation of ^{225}Ac-DWJ155 treatment between the first two participants in the first dosage level will be staggered by at least 6 weeks to cover the first participant's DLT period prior to dosing the second participant. Dosing of subsequent participants in this and subsequent dosage levels may proceed concurrently. The first cohort of participants receiving ^{225}Ac-DWJ155 treatment will be participants with HER2+ (as determined locally by ASCO/CAP scoring) breast cancer. Subsequent participants can be enrolled per broader dosage escalation inclusion criteria (refer to Section 5.1).

Each dosage escalation cohort will consist of 3 to 6 newly treated participants. Additional cohorts at any dosage level at or below the highest administered activity (per cycle and cumulative) previously tested and shown to be safe may be opened (without simultaneously compressing the schedule) to better characterize the safety, tolerability, dosimetry, PK and preliminary anti-tumor activity of ^{225}Ac-DWJ155 (please refer to Section 6.5.3).

Dosage escalation will be guided by a Bayesian Hierarchical Logistic Regression Model (BHLRM) following the escalation with overdose control (EWOC) principle. For further details, please refer to Section 6.5.3 and Section 10.4. Dosage escalation decisions will be based on the totality of data, including but not limited to safety and tolerability and guided by the BHLRM with EWOC. Please refer to Section 6.5.2 for more details.

Dosimetry assessments will be performed in the first cycle, in approximately 3 participants in each cohort (preferably the first three participants) (see Section 8.7 for more details). In the earlier dosage levels (DL1 and DL2), where injected activity per cycle ≤ 4 MBq are planned, image-based dosimetry from first participant may be reviewed to assess feasibility before additional participants undergo dosimetry, to confirm feasibility of the analysis.

Refer to Section 6.1.3 for details about treatment duration and to Section 6.5 for details on regimen.

Dosage expansion part

Once the RD(s) have been determined from the escalation part, additional participants will be enrolled in the expansion part to further characterize the safety, tolerability, PK, dosimetry and preliminary anti-tumor activity of ^{225}Ac-DWJ155. Expansion groups may investigate different activities and schedules depending on the participant population being enrolled, as informed by totality of data and analysis conducted during the escalation part of the study. Dosage expansion may begin only after consideration by the investigators and Novartis of the available safety, preliminary anti-tumor activity data, and the assessment of risk to future participants. The expansion part will include groups based on tumor type. Participants with breast cancer will be in separate groups based on the following baseline molecular characteristics by ASCO/CAP scoring: HER2-positive (IHC 3+ or 2+/ISH+), HR+/HER2-low (IHC 2+/ISH- or IHC 1+), HR+/HER2 IHC 0, HR-/HER2-low (IHC 2+/ISH- or IHC 1+), and HR-/HER2 IHC 0. Separate groups exist for participants with NSCLC with AGAs, NSCLC without AGAs, HER2 IHC 3+ or IHC 2+ (ISH+ or -) GEA, and bladder cancer. To be eligible for treatment with ^{225}Ac-DWJ155, all participants with NSCLC and bladder cancer must have at least one measurable lesion per RECIST v1.1 that shows ^{68}Ga-DWJ155 uptake higher than surrounding tissues on PET/CT or PET/MRI as determined by local investigator. Pending emerging data, not all dosage expansion groups need to be opened or may be opened in a staggered fashion. If cumulative study data (including but not limited to safety, tolerability, PK, dosimetry, and/or preliminary anti-tumor activity) from dosage escalation support more than one ^{225}Ac-DWJ155 dosage (where dosage is defined as administered activity per cycle, total number of cycles and schedule), these may be investigated in selected expansion groups. In this case, up to four additional group(s) of participants may be opened (with approximately 15-20 participants each).

Dosimetry and PK data may be further acquired in a subset of participants in each expansion group. The dosimetry and PK data collected from participants treated in the dosage escalation part will be analyzed as a single population to inform on optimal schedule of SPECT/CT scans and PK assessments that are needed to accurately derive dosimetry and PK parameters in participants in dosage expansion, assuming no influence of administered activity on these parameters.

A DMC will be included for the expansion part of the study (Section 10.1.4).

4.2 Scientific rationale for study design

Table 4-1 Rationale for study design

Key Study Design Aspect	Rationale
Overall (phase I, open-label study)	The dosage escalation allows the RD(s) to be established and will be guided by BHLRM together with all available data including safety, tolerability, PK, dosimetry and preliminary efficacy. The BHLRM along with the EWOC principle (Babb et al 1998) will control the risk of DLT in future participants on study. The dosage escalation part will support the determination of a range of activities and regimens to be further explored during expansion part. The expansion part is designed to assess the preliminary anti-tumor activity, as well as to further confirm the safety and tolerability of ^{225}Ac-DWJ155 at the RD(s) established from the dosage escalation part in the defined participant populations.
Dosage expansion in selected population	The dosage expansion in breast cancer, NSCLC, GEA, and bladder cancer were selected based on the tumor molecular epidemiology demonstrating HER2 expression and preclinical efficacy data in HER2-expressing tumors. Given the strong biologic rationale in these cancer types, the safety and preliminary anti-tumor efficacy of ^{225}Ac-DWJ155 will be explored in these indications. The opening of each expansion group is optional.
Participant selection based on HER2 expression	HER2 expression (by ISH and IHC) will be used to determine participant eligibility for breast cancer and GEA as HER2 expression data is widely available for participants with these diseases. For breast and gastric cancer, the level of HER2 expression required for inclusion is based on settings where other HER2-directed therapies have shown efficacy. HER2 expression (by IHC) will also be allowed to determine participant eligibility for NSCLC given that ^{68}Ga-DWJ155 may not yet be fully characterized when participants are being enrolled during escalation. For NSCLC, a HER2 expression cutoff of IHC 1+ by ASCO/CAP scoring is used based on preclinical data showing benefit in IHC 1+ NSCLC mouse xenograft models.
Participant selection based on ^{68}Ga-DWJ155 PET imaging	Participant selection based on ^{68}Ga-DWJ155 PET uptake is in line with the theranostic principle and may increase the likelihood of ^{225}Ac-DWJ155 treatment efficacy. In this study, selection based on ^{68}Ga-DWJ155 PET uptake is implemented for NSCLC and bladder cancer as HER2 expression data (ISH/IHC) is not as widely available for these diseases.

4.2.1 Patient input into design

Patient advocates for breast cancer and NSCLC, representing different global regions, were consulted for a trial with similar indications. The results of the consultation that have been implemented are listed below:

- Collection of smoking history for all study indications in screening (no longer NSCLC-specific)
- Development of social impact strategy for dosage expansion to ensure a representative group of participants
- Participants with asymptomatic brain metastases will be included in the trial, if they are on a stable steroid dose, if steroids are indicated.

4.3 Justification for dosage

4.3.1 Dosage of ^{225}Ac-DWJ155

The starting administered activity of ^{225}Ac-DWJ155 in this study will be 2 MBq (54 µCi).

The range of mass doses anticipated to be tested in this FiH study is supported by the non-clinical safety data obtained in GLP single dose toxicity studies using a non-radioactive formulation. The mass dose of DWJ155 ligand associated with the starting administered activity is approximately 40 µg. At constant specific activity of ^{225}Ac-DWJ155, the mass dose of DWJ155 will increase with escalation of the administered activity per cycle. Based on conservative assumptions, the human exposures (Cmax and AUC) at the starting mass dose of 40 µg and at highest foreseeable mass doses up to 400 µg, are predicted to be 50-fold (AUC) and 90-fold (Cmax) lower and 5-fold (AUC) and 9-fold (Cmax) lower, respectively, than the AUC and Cmax at the no observed adverse effect level (NOAEL) of (cold) ^{175}Lu-DWJ155 in the most sensitive species (dog) (^{225}Ac-DWJ155 IB).

The selected 2 MBq starting administered activity is anticipated to have high margins in radiation dose in normal organs relative to established dose constraints for External Beam Radiation Therapy (EBRT) and/or radiopharmaceutical therapies, based on translational dosimetry from preclinical studies, as described below.

A mouse-to-human dosimetry extrapolation was derived from a biodistribution study in naive female mice conducted with ^{225}Ac-DWJ155, and two different extrapolation methods (relative mass scaling (RMS) and combined mass and time scaling (CMTS)) have been used to estimate human radiation doses for ^{225}Ac-DWJ155 in normal organs. For alpha particles emitted from Ac-225, a relative biological effectiveness (RBE) value of 5 is commonly used for an estimate of the absorbed dose-equivalent in clinical practice (Sgouros et al 2020, Kratochwil et al 2017). Based on known HER2 expression in normal organs and translational dosimetry, the organs anticipated to have the narrowest margins in regards to established organ constraints are kidney, red marrow, and ovaries. The estimated radiation absorbed dose-equivalent to major human target organs for an administered activity of 2 MBq are significantly below established dose constraints for those organs, with margins for all tissues including red marrow (25%), ovaries (9%), and kidneys (7.4%) and ≤ 2% in all remaining organs/ tissues, when using the most conservative estimate from the two extrapolation methods (^{225}Ac-DWJ155 IB). At the selected dosage in DL1, consisting of 2 MBq administered over 4 cycles and a cumulative administered activity of 8 MBq, the cumulative absorbed dose across organs is anticipated to remain within established organ tolerance constraints.

Refer to Section 6.5.1 and Section 6.5.2 for more details on the starting dosage and provisional dosage levels. Final decision regarding the escalation of the administered activity and

administration frequency will be agreed upon at the dosage escalation meetings (DEMs) between the investigators and Novartis.

Participants in DL1 will receive up to 4 cycles of ^{225}Ac-DWJ155. From DL2, a lower or higher number of cycles may be explored if deemed beneficial and safe, while maintaining comparable cumulative administered activity at that DL, as displayed (Table 6-2).

The specified maximum cumulative activity at the highest dosage level in this study will not exceed 32 MBq. At this cumulative activity, the projected cumulative radiation dose from translational dosimetry is anticipated to exceed EBRT constraints for kidney, red marrow and ovaries, while staying approximately within 30% of these constraints in all other normal organs (see ^{225}Ac-DWJ155 IB). Justification of the selected maximum cumulative activity, with a focus on risks related to organs where established constraints may be crossed in the context of the participant population treated in this study, is further provided below.

In general, EBRT dose constraints have been shown to be too restrictive for RLT, due to multiple factors including the difference in rates of radiation delivery between these modalities (Kiess et al 2024). In addition, the application of normal organ EBRT thresholds needs to be considered in the context of the limited treatment options for participants enrolled in this FiH study, who progressed after previous lines of standard of care therapy, and the short life expectancy in this participant population.

In the kidneys, the predicted absorbed dose-equivalent at the maximum planned cumulative administered activity of 32 MBq is 27 $Sv_{RBE=5}$. This is only slightly in excess of the conservative 23 Sv EBRT dose constraint for the kidneys, which has been associated with delayed renal toxicity (specifically, a ~5% risk of clinical nephritis at 5 years (Emami et al 1991)). There is a large and growing body of literature supporting a higher renal tolerance dose for radioligand therapies, which are delivered at lower dose rates than EBRT. Multiple studies of Lu-177-based radioligand therapies have demonstrated reassuring safety at renal absorbed doses significantly exceeding the 23 Sv EBRT dose constraint for the kidneys (Mader et al 2023, van der Zwan et al 2019). Similarly, Ac-225 based radioligand therapies have been reported with a favorable renal safety profile. In an analysis of 14 metastatic castration-resistant prostate cancer (mCRPC) participants receiving 4 cycles of [^{225}Ac]Ac-PSMA617 at radioactive doses up to 0.15 MBq/kg (i.e., a cumulative radioactive dose of up to 42 MBq for a participant of 70 kg), no change in serum creatinine was observed despite an absorbed dose-equivalent to the kidney of approximately 31 $Sv_{RBE=5}$ in some participants (Kratochwil et al 2017). In another report of 53 participants with advanced prostate cancer who received a median of 3 cycles (range 1-7) of [^{225}Ac]Ac-PSMA617 at administered activities of 4-8 MBq per cycle, only 3 participants experienced grade III-IV renal function impairment (2 participants grade III and 1 participant grade IV, all 3 participants presented with suboptimal renal function before start of therapy) (Sathekge et al 2022). Finally, in a study following long-term outcomes after ^{225}Ac-DOTATATE as salvage treatment for participants with gastroenteropancreatic neuroendocrine tumors, no renal toxicity was observed in participants (n=57) who received ^{177}Lu-peptide receptor radionuclide therapy (PRRT) therapy prior to ^{225}Ac-DOTATATE (median follow-up of 24 months, range 5-41 months). The mean cumulative activities of ^{177}Lu-PRRT and ^{225}Ac-DOTATATE administered in these participants was 25.7 GBq (range 5.5 - 49.5 GBq) and 39.6-48.6 MBq (range 8.9 - 100 MBq), respectively (Ballal et al 2022). No direct dosimetric analysis is presented in this study. Based on reported absorbed dose in the kidneys for ^{177}Lu-PRRT of

0.65 Sv/GBq (Novartis 2018) and for ^{225}Ac-DOTATATE of 0.45 $Sv_{RBE=5}$/MBq (Sgouros et al 2023), the estimated cumulative radioactive dose to kidneys from the two radiotherapies at the reported mean administered activities is in the range of 34 - 38 Sv.

In the case of red marrow, acute hematologic toxicity in individual participants can be used in lieu of dosimetry to guide treatment. Acute hematologic toxicity after a single cycle of radioligand therapy is expected to present as a decline in platelets and white blood cells within 4 to 6 weeks of treatment (Dorn et al 2003, Hartung-Knemeyer et al 2012, Vallabhajosula et al 2005, Meredith et al 1992, Bushnell et al 1992). Strict hold criteria and dosage modification criteria (refer to Section 6.5.5) implemented in this FiH study are designed to prevent re-dosing of participants experiencing acute hematologic toxicities. Although dose to bone marrow is correlated with hematologic toxicity at the population level prompting a suggested constraint of 2-3 Sv to minimize such toxicity (Wahl et al 2021), the red marrow tolerance dose for radioligand therapies varies significantly between individuals. This variability is influenced by factors such as prior treatments, the specific isotope used, PK, and tissue-specific target expression; when these factors are not well-characterized, as in the case of a novel radioligand therapeutic, individual tolerance doses are difficult to predict. Using acute hematologic toxicity to guide dosing decisions in individual participants (when red marrow is the dose-limiting organ) gives confidence that the administered activity will not be unnecessarily limited based on dosimetry alone in participants with a higher tolerance dose. Finally, of note, most data suggesting a red marrow dose constraint of 2-3 Sv to limit hematologic toxicity comes from single-dose treatments. When dose to red marrow is fractionated over multiple cycles, toxicity is likely lower; for example, a recent study of 20 participants treated with [^{177}Lu]Lu-DOTATATE found that when a cumulative red marrow dose of 0.6 to 2.5 Sv was delivered over 4 cycles, no grade 2 or higher toxicity for platelets, neutrophils, or hemoglobin was observed (Blakkisrud et al 2024).

In the ovaries, the predicted absorbed dose-equivalent at the maximum planned cumulative administered activity of 32 MBq is 4.3 $Sv_{RBE=5}$. An EBRT dose constraint of 3 - 6 Sv to the ovaries is applied in certain clinical situations to minimize the risk of early menopause (Wo and Viswanathan 2009, Chambers et al 1991). Early menopause is considered to be an acceptable risk given the intended participant population of this study.

Finally, the highest allowed total (cumulative) administered activity of ^{225}Ac-DWJ155 (i.e., 32 MBq or lower) will be conditional on supportive safety data.

A 6-week cycle duration has been chosen in DL1 to enable adequate safety monitoring time for onset of potential myelosuppression between cycles and to allow time to recover from any potential myelosuppression, even though hematologic toxicities are not expected at the selected starting administered activity due to anticipated safety margins. This dosing interval is also in line with that of approved RLTs such as ^{177}Lu-PSMA-617 (Pluvicto®) and of other investigational Ac-225-based RLTs such as ^{225}Ac-PSMA-R2. In addition, to minimize risk to study participants, appropriate eligibility criteria and specific dose-limiting toxicity definitions, as well as participant monitoring, specific dosage modification, and stopping rules are included in this protocol.

The safety of the selected administered activities per cycle and cumulatively across cycles and the treatment schedule will be monitored (including hematology, clinical chemistry, vital signs,

AEs, cardiac imaging, and ECGs) with the option, from DL2, of the administration cycle to be reduced to 4 weeks (Q4W) or to be extended to 8 weeks (Q8W). Administration of Q4W could provide superior benefit-risk to participants treated in this FiH study, with improved disease control by minimizing time for cancer cell repopulation between cycles, if supported by a favorable safety and tolerability profile.

Overall, based on preclinical safety studies, mouse-to-human extrapolation of biodistribution data and Novartis' experience with other RLTs (e.g., Pluvicto® and of other Ac-225-based RLTs under development such as ^{225}Ac-PSMA-R2 and ^{225}Ac-PSMA-617), the proposed starting administered activity of 2 MBq of ^{225}Ac-DWJ155 to be given over 6 weeks and for up to 4 cycles is considered a safe starting regimen. The provisional dosage levels and alternative schedules intended to be explored in this FiH trial, as supported by review of safety and tolerability data, will allow characterization of the maximum potential of ^{225}Ac-DWJ155 in this study population. Specific eligibility criteria, dosage-limiting toxicity definitions, participant monitoring, dosage modification guidelines, and stopping rules included in this protocol are intended to ensure the safety of the study participants.

4.3.2 Activity of ^{68}Ga-DWJ155

Participants will receive two ^{68}Ga-DWJ155 administrations: one after screening but before initiating treatment with ^{225}Ac-DWJ155, and a second one after the last administration of ^{225}Ac-DWJ155 to understand target expression changes after treatment with the ^{225}Ac-DWJ155. The proposed starting administered activity is 250 MBq with an acceptable range of 200 to 275 MBq and is in line with approved ^{68}Ga-based RLI agents (Giesel et al 2019, Hennrich, Eder 2021). Pending emerging data, starting administered activity may be adjusted in the range of 150-500 MBq to better capture tumor uptake. The selected administered activity range of 150 MBq to 500 MBq corresponds to a weight-based activity range of 2.5 MBq/Kg to 8.3 MBq/kg, using a reference body weight of 60 kg. A flat administered activity approach for ^{68}Ga-DWJ155 is followed in this study and emerging data will inform future studies whether body weight-based administered activity is appropriate. Considering the isotope's half-life of 68 min, the proposed range of administered activities is considered adequate for obtaining good quality imaging and is expected to be safe.

Based on extrapolation from preclinical data (refer to ^{68}Ga-DWJ155 IB for further details), the effective radiation dose is expected to be comparable to approved Ga-68 imaging agents. The effective radiation dose of ^{68}Ga-DWJ155 for the total body is estimated to be in the range of 0.031 – 0.033 mSv/MBq, which is similar to the values measured for approved Ga-68-based imaging agents such as Netspot® (0.026 mSv/MBq, Walker et al 2013) and Locametz® (0.022 mSv/MBq, Sandgren et al 2019). Translational dosimetry conducted from a biodistribution study of ^{68}Ga-DWJ155 in the mouse estimates total body effective doses of 7.6 – 8.1 mSv and 15.2 – 16.3 mSv, for clinical administered activities between 250 MBq and up to 500 MBq, respectively (for further details refer to the ^{68}Ga-DWJ155 IB). Participants will receive two administrations of ^{68}Ga-DWJ155 (prior to Cycle 1 and at EoT), with estimated total body effective doses of 15.2 – 16.2 mSv and 30.4 – 32.6 mSv, for administered activities between 250 MBq and up to 500 MBq, respectively. Per ICRP Publication 62 the level of risk corresponding to the expected maximal effective dose is moderate and justified by substantial level of societal benefit based on development of a potential non-invasive, total body HER2

expression level diagnostic which is intended to be used as theranostic pair with Ac-225 for the treatment of cancer in participants with high unmet medical need (i.e., breast cancer, lung cancer).

4.4 Rationale for choice of control drugs or combination drugs and background therapy

Not applicable.

4.5 Purpose and timing of interim analyses/design adaptations

No formal interim analysis is planned for the dosage escalation part, however, the dosage escalation design foresees that decisions on future dosage levels based on available data are taken at the end of the DLT period for each dosage level. These are described in Section 6.5.3 and Section 9.8.

At the end of dosage escalation, an integrated analysis of all available safety, tolerability, dosimetry, PK, and preliminary anti-tumor activity data by dosage level will be conducted. The RD(s) will be selected based on a review by Novartis study personnel and investigators of the results of the integrated analyses. The expansion part will then begin as specified in Section 4.1. Study stopping rules will be applied as specified in Section 7.5.

4.6 Public health emergency mitigation procedures

During a public health emergency as declared by local or regional authorities e.g., pandemic, epidemic, or natural disaster, mitigation procedures to ensure participant safety and trial integrity may be implemented. Notification of the public health emergency as declared by local or regional authorities should be discussed among investigators and Novartis. All procedures adapted to the situation must be submitted, if required per local regulations, through a protocol amendment for approval by local or regional Health Authorities and Ethics Committees prior to implementation of mitigation procedures.

4.7 Rationale for planned off-site procedures

Off-site lab assessments are possible in this study to minimize burden on participants and offer them increased flexibility to participate in the study from an off-site location (as described in Section 1.3 and defined in Section 10).

4.8 End of study definition

Study completion is defined as when the last participant finishes their Study Completion visit and any repeat assessments associated with this visit and supporting the collection of data for the primary outcome measure, have been documented and followed up appropriately by the Investigator.

The end of the study will be:

1. When at least 80% of the participants per disease group in the expansion part have completed the follow-up for disease progression or discontinued the study for any reason and all participants have completed treatment and the 36-month long-term safety follow-up period; or

2. In the event of an early study termination decision (see Section 7.6):
   - The date of that decision if no participants have been enrolled; or
   - When all enrolled participants have completed the treatment and the 36-month long-term safety follow-up period or have discontinued the study for any reason (in such cases, the study completion for disease progression follow-up evaluations might not be completed), whichever occurs first.
   - The disease follow-up evaluations might not be completed if Novartis decides to stop enrollment. In such cases, study completion will be when the treatment period and the safety follow-up have been completed for all participants.

Statistical considerations for details of timing of the primary analysis and final reporting of data are available in Section 9.

5 Study population

This study will include adult participants with histologically or cytologically confirmed advanced (defined as locally advanced (not amenable to definitive multi-modality therapy including surgery) or metastatic) cancer of one of the types defined below, for whom no standard therapy is available, tolerated, or appropriate in the judgement of the investigator. Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

5.1 Inclusion criteria

Participants eligible for inclusion in this study must meet all of the following criteria:

1. Signed informed consent must be obtained prior to participation in the study.
2. Male or female participants age ≥ 18 years.
3. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 1.
4. Participants with one of the following histologically or cytologically confirmed and documented malignancies, for whom no standard therapy is available, tolerated, or appropriate in the judgement of the investigator:
   - Dosage Escalation (first cohort only):
   a. Advanced HER2+ (last known HER2 IHC 3+ or IHC 2+/ISH+ by ASCO/CAP) breast cancer following prior therapy in the advanced setting with at least two prior systemic lines of therapy based on local standard of care and investigator assessment, with no limitations to prior lines of treatment. Tumor ER, progesterone receptor (PR), and HER2 status must be available in the participant's medical record.
   - Dosage Escalation (from second cohort onward):
   a. Advanced HER2+ (last known HER2 IHC 3+ or IHC 2+/ISH+ by ASCO/CAP) breast cancer following prior therapy in the advanced setting with at least two prior systemic lines of therapy based on local standard of care and investigator assessment, with no limitations to prior lines of treatment. Tumor ER, progesterone receptor (PR), and HER2 status must be available in the participant's medical record.
   b. Advanced HR+/HER2-low (last known HER2 IHC 1+ or IHC 2+/ISH- by ASCO/CAP) breast cancer with disease progression after at least one line of

hormone-based therapy in combination with a CDK4/6i and at least one additional line of systemic therapy administered in the advanced setting, unless participant was ineligible to receive such therapy. Participants who have received hormone-based therapy plus CDK4/6i in the adjuvant setting are not required to have received this therapy again in the advanced setting. Tumor ER, PR, and HER2 status must be available in the participant's medical record.

c. Advanced NSCLC with known HER2 IHC ≥ 1+ by ASCO/CAP, available in the participant's medical record, or confirmed tumor positivity based on local assessment of ^{68}Ga-DWJ155 uptake, without known AGAs, who have received a platinum-based chemotherapy and an immune checkpoint inhibitor therapy either in combination or in sequence, unless participant was ineligible to receive such therapy.
d. Advanced NSCLC with known HER2 IHC ≥ 1+ by ASCO/CAP, available in the participant's medical record, or confirmed tumor positivity based on local assessment of ^{68}Ga-DWJ155 uptake, with AGAs, who have received prior treatment with at least a targeted therapy or ADC (if locally approved), platinum-based chemotherapy, and immune-oncologic (IO) therapy if appropriate, unless participant was ineligible to receive such therapy.

- Dosage Expansion:

e. Advanced HER2+ (last known HER2 IHC 3+ or IHC 2+/ISH+ by ASCO/CAP) breast cancer following prior therapy in the advanced setting with at least two prior systemic lines of therapy based on local standard of care and investigator assessment, with no limitations to prior lines of treatment. Tumor ER, PR, and HER2 status must be available in the participant's medical record.
f. Advanced HR+/HER2-low (last known HER2 IHC 1+ or IHC 2+/ISH- by ASCO/CAP) breast cancer with disease progression after at least one line of hormone-based therapy in combination with a CDK4/6i and at least one additional line of systemic therapy administered in the advanced setting, unless participant was ineligible to receive such therapy. Participants who have received hormone-based therapy plus CDK4/6i in the adjuvant setting are not required to have received this therapy again in the advanced setting. Tumor ER, PR, and HER2 status must be available in the participant's medical record.
g. Advanced HR+/HER2-0 (last known HER2 IHC 0 by ASCO/CAP) breast cancer with disease progression after at least one line of hormone-based therapy in combination with a CDK4/6i and at least one additional line of systemic therapy administered in the advanced setting, unless participant was ineligible to receive such therapy. Participants who have received hormone-based therapy plus CDK4/6i in the adjuvant setting are not required to have received this therapy again in the advanced setting. Tumor ER, PR, and HER2 status must be available in the participant's medical record.
h. Advanced HR-/HER2-low breast cancer (TNBC with last known HER2 IHC 1+ or IHC 2+/ISH- by ASCO/CAP) with disease progression following, or intolerance to, at least two lines of systemic therapy (including cytotoxic chemotherapy, antibody-drug conjugate, targeted therapies and/or IO) for advanced disease, unless participant was ineligible to receive such therapy. Tumor ER, PR, and HER2 status must be available in the participant's medical record.

i. Advanced HR-/HER2-0 breast cancer (TNBC with last known HER2 IHC 0 by ASCO/CAP) with disease progression following, or intolerance to, at least two lines of systemic therapy (including cytotoxic chemotherapy, antibody-drug conjugate, targeted therapies and/or IO) for metastatic disease, unless participant was ineligible to receive such therapy. Tumor ER, PR, and HER2 status must be available in the participant's medical record.

j. Advanced NSCLC with AGAs who have received prior treatment with at least a targeted therapy or ADC and platinum-based chemotherapy, and IO therapy if indicated, unless participant was ineligible to receive such therapy. Tumor must be positive for ^{68}Ga-DWJ155 uptake based on local assessment. The number of participants with HER2 mutation will be capped at a maximum of n=5 in the Dosage Expansion part of the study.

k. Advanced NSCLC without known AGAs who have received a platinum-based chemotherapy and an immune checkpoint inhibitor therapy either in combination or in sequence, unless participant was ineligible to receive such therapy. Tumor must be positive for ^{68}Ga-DWJ155 uptake based on local assessment.

l. Advanced GEA with HER2 IHC of 3+ or 2+ (ISH + or -) following disease progression, or intolerance to, all SOC systemic therapies per local guidelines including chemotherapy, targeted agents, IO, and/or ADCs, unless participant was ineligible to receive such therapy. Tumor HER2 status must be available in the participant's medical record.

m. Advanced bladder cancer following disease progression, or intolerance to, all SOC systemic therapies per local guidelines, including chemotherapy, ADCs, and/or immunotherapy, unless participant was ineligible to receive such therapies. Tumor must be positive for ^{68}Ga-DWJ155 uptake based on local assessment.

5. Disease Assessment
   a. Dosage escalation: Measurable disease as determined by RECIST version 1.1. Tumor lesions previously irradiated or subjected to other locoregional therapy will only be considered measurable if there is documented disease progression at the treated site after completion of therapy.
   - BC only: If no measurable disease is present, then at least one predominantly lytic bone lesion must be present that can be accurately assessed at baseline and is suitable for repeated assessment (participants with no measurable disease and only one predominantly lytic bone lesion that has been previously irradiated are eligible if there is documented evidence of disease progression of the bone lesion after irradiation).

   b. Dosage expansion: Measurable disease as determined by RECIST version 1.1. Tumor lesions previously irradiated or subjected to other locoregional therapy will only be considered measurable if there is documented disease progression at the treated site after completion of therapy.
6. Biopsy collection
   a. Dosage escalation: an archival tumor sample with sufficient tissue is requested (specifications are described in Section 8.8). If unavailable, the participant must have a site of disease amenable to biopsy during screening period (post ^{68}Ga-DWJ155 PET imaging) and the participant must be willing to undergo a biopsy procedure if safe

and medically feasible. Exceptions may be considered following documented discussion with Novartis.

b. Dosage expansion: The participant must have a site of disease amenable to biopsy and be a candidate for tumor biopsy according to the treating institution's guidelines. The participant must be willing to undergo a newly obtained tumor biopsy during screening period (post ^{68}Ga-DWJ155 PET imaging) unless medically unsafe or infeasible. An archival tumor sample may be submitted in lieu of the newly obtained tumor biopsy if it was collected within 6 months prior to screening, with no intervening treatment, and sufficient tissue is available for submission (specifications for these samples are described in Section 8.8). Exceptions may be considered following documented discussion with Novartis.

5.2 Exclusion criteria

Participants meeting any of the following criteria are not eligible for inclusion in this study.

1. Out-of-range laboratory values defined as:
   a. Calculated eGFR < 60 mL/min/1.73m^2 measured or calculated using the CKD-EPI 2021 formula for adults
   b. Total bilirubin > 1.5 x ULN (except for participants with Gilbert's syndrome who are excluded if total bilirubin > 3.0 x ULN) or direct bilirubin > 1.5 x ULN
   c. Alanine aminotransferase (ALT) > 3 x ULN, except for participants with tumor involvement of the liver who are excluded if ALT > 5 x ULN
   d. Aspartate aminotransferase (AST) > 3 x ULN, except for participants with tumor involvement of the liver who are excluded if AST > 5 x ULN
   e. Lipase > 1.5 x ULN
   f. Absolute neutrophil count (ANC) < 1.5 x 10^9/L
   g. Hemoglobin < 9 g/dL
   h. Platelet count < 100 x 10^9/L
2. Ongoing treatment with thrombopoietin mimetics, erythroid stimulating agents, or hematopoietic colony stimulating factors (e.g. Granulocyte Colony-Stimulating Factor (G-CSF), Granulocyte-Macrophage Colony-Stimulating Factor (GM-CSF), Macrophage Colony-Stimulating Factor (M-CSF)).
3. Use of transfusion support ≤ 4 weeks prior to the start of treatment with ^{225}Ac-DWJ155
4. Impaired cardiac function or clinically significant cardiac disease, including any of the following:
   a. Clinically significant and/or uncontrolled heart disease such as congestive heart failure New York Heart Association (NYHA) Grade ≥ 2 and requiring treatment
   b. Uncontrolled hypertension or clinically significant arrhythmia (e.g., ventricular tachycardia), complete left bundle branch block, high-grade AV block (e.g. bifascicular block, Mobitz type II and third-degree AV block)
   c. QT interval corrected by Fridericia's formula (QTcF) > 470 msec (as a mean value of triplicates) on screening ECG or congenital long QT syndrome
   d. Acute myocardial infarction or unstable angina pectoris < 3 months prior to study entry

e. Baseline left ventricular ejection fraction (LVEF) < 50% as measured by transthoracic echocardiography (TTE). If TTE cannot be performed, multiple gated acquisition (MUGA) scan is acceptable.

5. Unmanageable urinary tract obstruction or urinary incontinence. If ureteral obstruction can be managed with the placement of ureteral stents, this exclusion criterion does not apply.
6. Participants with uptake of ^{68}Ga-DWJ155 in non-tumor tissues or organs that, in the opinion of the investigator, increase the risk associated with ^{225}Ac-DWJ155 treatment, will not be eligible for treatment with ^{225}Ac-DWJ155. This may include, but is not limited to, bone marrow or other normal tissue/organ uptake. Any decision regarding administration of ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155, and dosage(s) may be discussed with the Novartis medical team.
7. Known hypersensitivity to any of the investigational drug (^{68}Ga-DWJ155 or ^{225}Ac-DWJ155) or their excipients.
8. Prior treatment with any therapeutic radiopharmaceutical.
9. Treatment with any of the following anti-cancer therapies prior to the first injection of ^{225}Ac-DWJ155 within the stated timeframes:
   a. < 10 effective half-lives for any imaging radiopharmaceutical
   b. ≤ 4 weeks for EBRT or brachytherapy, or ≤ 2 weeks for limited field radiation for palliation (except for lung-directed radiation therapy)
   c. ≤ 6 months for lung-directed EBRT, regardless of primary site
   d. ≤ 4 weeks for major surgery or participants who have not recovered from the surgical procedure (mediastinoscopy, insertion of a central venous access device, insertion of a feeding tube, and abdominal paracentesis are not considered major surgery)
   e. ≤ 2 weeks for fluoropyrimidine therapy
   f. ≤ 6 weeks for cytotoxic agents with major delayed toxicities, such as nitrosoureas and mitomycin C
   g. ≤ 30 days for ado-trastuzumab emtansine
   h. ≤ 30 days for trastuzumab
   i. ≤ 30 days for trastuzumab-deruxtecan
   j. ≤ 90 days for pertuzumab
   k. ≤ 4 weeks or ≤ 5 half-lives (whichever is shorter) for chemotherapy or biological therapy (including monoclonal antibodies) or continuous or intermittent small molecule therapeutics or any other non-radiopharmaceutical investigational agent, unless otherwise specified. If an agent in one of the categories above has a specific longer timeframe before starting a new therapy, this longer timeframe applies.
10. Those who have received 4 or more courses of EBRT to the thorax (exceptions can be made after documented discussions with Novartis).
11. Participants with symptomatic brain metastasis, known leptomeningeal disease, or brain metastases requiring increasing doses of corticosteroids or anti-epileptics within 2 weeks prior to treatment. Participants with treated brain metastases are allowed if the brain metastases have been stable for greater than or equal to 4 weeks as documented by radiologic imaging. Participants who have had complete surgical resection or received stereotactic radiosurgery will be permitted to enroll in the study provided that they

completed treatment greater than 2 weeks prior to study treatment, have recovered from treatment, are clinically/neurologically stable and do not require escalating corticosteroid therapy as noted above. If participants require steroids for management of central nervous system (CNS) metastases, they must have been on a stable or decreasing dose of steroids for 2 weeks preceding the first administration of study treatment.

12. Participants who are receiving systemic corticosteroid treatment for indications other than brain metastases and who are not on a stable dose of corticosteroid equivalent of ≤ 10 mg daily prednisone for at least 2 weeks before ^{225}Ac-DWJ155 administration.
13. Participants with a history of or active diagnosis of interstitial lung disease or pneumonitis of ≥ Grade 2.
14. Clinically uncontrolled pleural effusion that requires pleurocentesis for drainage within 4 weeks before Cycle 1 Day 1.
15. Clinically severe pulmonary compromise, as judged by the investigator, resulting from intercurrent pulmonary illnesses including, but not limited to, any underlying pulmonary disorder, or any autoimmune, connective tissue or inflammatory disorders with pulmonary involvement or prior complete pneumonectomy.
16. Participants requiring oxygen supplementation.
17. Presence of CTCAE version 5.0 ≥ Grade 2 toxicity due to prior cancer therapy, except for neuropathy (inclusion of participants with neuropathy of ≤ Grade 2 is permitted), alopecia, and participants adequately treated for skin rash or with replacement therapy for endocrinopathies.
18. Any serious uncontrolled infection (acute or chronic), such as but not limited to those caused by bacteria, viruses, or fungi, confirmed by clinical evidence, imaging, and/or relevant positive laboratory tests (e.g., blood cultures, Polymerase Chain Reaction (PCR) for deoxyribonucleic acid (DNA)/ribonucleic acid (RNA)). Participants with hepatitis B (HBV) or hepatitis C (HCV) infection whose disease is controlled (defined as positive anti-Hepatitis B core antibody (HBc) and negative Hepatitis B virus surface antigen (HBsAg) for HBV and undetectable viral load by real-time PCR for HCV) under antiviral therapy should not be excluded. Testing for HBV or HCV status is not necessary unless clinically indicated or if the participant has a history of HBV or HCV infection. Participants with human immunodeficiency virus (HIV) infection may be enrolled if the participant is on antiviral therapy for at least 4 weeks prior to enrollment with a viral load of < 400 copies/mL, has a CD4+ T-cell count ≥ 350 cells/μL, and the participant has not had an Acquired Immunodeficiency Syndrome (AIDS)-defining infection in the prior 12 months.
19. Known additional malignancy that is progressing or requires active treatment. Exceptions include basal cell carcinoma of the skin or squamous cell carcinoma of the skin that has undergone potentially curative therapy or *in situ* cervical cancer or other tumors that will not affect life expectancy.
20. Inability to complete the needed investigational and standard imaging examinations due to any reason (e.g., severe claustrophobia, inability to lie still for the entire imaging time).
21. Any other medical condition that would, in the investigator's judgment, prevent the participant's participation in the clinical trial due to safety concerns, compliance with

clinical trial procedures (including radiation safety precautions), or interpretation of study results.

22. Pregnant or breastfeeding women.
23. WOCBP, defined as all women physiologically capable of becoming pregnant from menarche until becoming post-menopausal, unless they have had surgical bilateral oophorectomy (with or without hysterectomy), total hysterectomy or bilateral salpingectomy at least six weeks before taking study treatment. In the case of oophorectomy alone, the reproductive status of the woman needs to have been confirmed by follow-up hormone level assessment. Women are considered post-menopausal if they have had over 12 months of natural (spontaneous) amenorrhea with an appropriate clinical profile (e.g. hormonal profile confirming menopause and/or age-appropriate history of vasomotor symptoms).

    WOCBP are excluded unless they are using highly effective methods of contraception (failure rate < 1% per year) for 12 hours after the administration of ^{68}Ga-DWJ155 and, if applicable, while on ^{225}Ac-DWJ155 and for 8 months after the last administration of ^{225}Ac-DWJ155. In addition, female participants must not donate eggs for the same time periods.

    Highly effective contraception methods include:

    a. Total abstinence (when this is in line with the preferred and usual lifestyle of the participant). Note that periodic abstinence (e.g., calendar, ovulation, symptothermal, post-ovulation methods) and withdrawal are not acceptable methods of contraception.
    b. Bilateral tubal occlusion or bilateral tubal ligation (at least six weeks before taking ^{225}Ac-DWJ155 treatment).
    c. Sterilization (vasectomy) of male partner(s) of the female participant at least 6 months prior to screening provided partner(s) has(have) received medical confirmation of surgical success.
    d. Use of hormonal methods:

    - Combined (estrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation; oral, intravaginal or transdermal.
    - Progestogen-only hormonal contraception associated with inhibition of ovulation: oral, injectable or implantable.
    - Intrauterine device (IUD) or intrauterine hormone-releasing system (IUS).

    In case of use of hormonal contraception, women should have been stable on the same method for a minimum of 3 months before taking ^{225}Ac-DWJ155 treatment.

    If local regulations deviate from the contraception methods listed above to prevent pregnancy, local regulations apply and will be described in the informed consent form (ICF).
24. Sexually active males unwilling to use a condom during intercourse while taking ^{225}Ac-DWJ155 treatment and for 5 months after the last administration of ^{225}Ac-DWJ155. WOCBP partners of sexually active male participants must also use highly effective contraception methods (see above) while the study participant is taking study treatment and for 5 months after the study participant's last administration of ^{225}Ac-DWJ155. Sexually active males who receive ^{68}Ga-DWJ155 must use a condom during intercourse for 12 hours after the administration of ^{68}Ga-DWJ155. A condom is required for all

sexually active male participants to prevent them from fathering a child AND/OR to prevent delivery of ^{225}Ac-DWJ155 treatment via seminal fluid to their partner. In addition, male participants must not donate sperm for the time period specified above. If local regulations deviate from the contraception methods listed above to prevent pregnancy, local regulations apply and will be described in the ICF.

5.3 Lifestyle considerations

Participants will be given instructions in order to minimize radiation exposure to others after ^{225}Ac-DWJ155 administration. Please refer to Post Ac-225 Instructions: Reducing External Exposure to Others for details.

5.4 Screen failures

A screen failure occurs when a participant who consents to participate in the clinical trial and is subsequently found to be ineligible (including ineligible for ^{68}Ga-DWJ155 administration and ^{68}Ga-DWJ155 PET/CT or PET/MRI) and therefore not entered in the trial. The reason for screen failure should be entered on the applicable eCRF.

Participants who sign an ICF and are considered eligible for ^{68}Ga-DWJ155 imaging but fail to undergo ^{68}Ga-DWJ155 imaging, or received ^{68}Ga-DWJ155 and fail to be started on ^{225}Ac-DWJ155 treatment for any reason will be considered early terminators. The reason for early termination should be captured on the appropriate disposition eCRF.

If the participant fails to be assigned to treatment, the interactive response technology (IRT) must be notified within 2 days of the screen fail that the participant was not assigned to treatment. Participants who are assigned and fail to start treatment, e.g. participants assigned in error, will be considered an early terminator. The reason should be recorded on the appropriate Case Report Form.

In this situation of screen failure, the investigator must provide a minimal set of screen failure information. Minimal information includes reason for screen failure, demography, medical history, screen failure details, eligibility criteria, and any serious adverse event (SAE) should be recorded on the appropriate eCRF. Data and samples collected from participants prior to screen failure may still be analyzed.

Individuals who do not meet the criteria for participation in this study may be rescreened. Each case must be discussed and agreed with Novartis on a case-by-case basis. In the case where a safety laboratory assessment at screening is outside of the range specified in the eligibility criteria, the assessment may be repeated once prior to ^{68}Ga-DWJ155 administration. If the repeat value remains outside of the specified range, the participant must be excluded from the study.

5.5 Participant numbering

Upon signing the informed consent form, the participant is assigned to the next sequential Participant Number (No.) available. The Participant No. consists of the Site Number (Site No.) (as assigned by Novartis to the investigative site) with a sequential number suffixed to it, so that each participant is numbered uniquely across the entire database. This number will be

retained throughout participation in the trial. If the participant is rescreened, then a new Participant No. will be assigned.

5.6 Replacement policy

Participants will not be replaced on study. All treated participants will contribute to the final analyses. However, during dosage escalation, if a participant is considered as non-evaluable for the Dose Determining Set (DDS), enrollment of a new participant to the current cohort may be considered in order to support the benefit-risk assessment for dosage-escalation.

6 Study treatment(s) and concomitant therapy

6.1 Description of study treatment(s) and treatment arms

For this study, the terms "investigational drug" and/or "radiopharmaceutical(s)" refers to ^{68}Ga-DWJ155 and/or ^{225}Ac-DWJ155. The term "imaging drug" refers only to ^{68}Ga-DWJ155 and the term "study treatment" refers only to ^{225}Ac-DWJ155.

^{225}Ac-DWJ155 and the kits for the radiopharmaceutical preparation of ^{68}Ga-DWJ155 will be provided centrally by Novartis. All doses prescribed and administered to participants and all dosage changes during the study must be recorded on the Study Treatment eCRF.

Participants who meet all eligibility criteria (Section 5) will receive one administration of ^{225}Ac-DWJ155 at the beginning of each cycle. Further details about provisional dosage levels are outlined in Section 6.5.2.

Details on requirements for storage, management and administration of investigational drugs, and instructions for participant numbering, prescribing and dispensing are outlined in Section 6.2.

No auxiliary medicinal products (AxMP) or additional treatment beyond investigational treatment described in Table 6-1 are included in this trial.

Table 6-1 Investigational drug

Investigational radiopharmaceutical	**^{68}Ga-DWJ155** **Diagnostic investigational radiopharmaceutical**	**^{225}Ac-DWJ155** **Therapeutic investigational radiopharmaceutical**
Type	Drug	Drug
Dose Formulation	Solution for IV administration	Solution for IV administration
Activity	250 MBq (activity to order) Pending emerging data characterizing ^{68}Ga-DWJ155, starting injected activity may be adjusted in the range of 150-500 MBq to better capture tumor uptake.	Provisional activity in the range of 2 MBq to 8 MBq (activity to order)
Acceptable activity range at the time of administration	200 – 275 MBq	± 10% of the planned activity
Dosage Level(s)	N/A (prepared dose corresponds to clinical dose)	N/A (prepared dose corresponds to clinical dose)
Route of Administration	Solution for injection	Solution for injection/infusion
Use	Experimental	Experimental
Authorization status in EEA	No	No
Sourcing by	Kit for radiopharmaceutical preparation is provided by Novartis Global Clinical Supply (GCS). Solution for injection is prepared at a local or central radiopharmacy.	Provided centrally by a vendor on behalf of Novartis.
Packaging and Labeling	Kit for radiopharmaceutical preparation is a sterile kit containing DWJ155 for radiolabeling with Ga-68. Each component of the kit for radiopharmaceutical preparation, and the solution for injection will be labeled as required per country requirements. The solution for injection is stored as directed in the label with appropriate containment and shielding.	Study treatment will be provided in a Type 1 glass vial. Each lead shielded container (secondary packaging) will be labeled as required per country requirement.
Reference Safety Information (RSI)	68Ga-DWJ155 IB	225Ac-DWJ155 IB

6.1.1 Additional study treatments

Not applicable.

6.1.2 Treatment arms/group

Not applicable.

6.1.3 Treatment duration

In DL1, participants eligible for treatment with ^{225}Ac-DWJ155 will receive up to 4 cycles on a 6-week schedule. From DL2, a lower or higher number of cycles may be explored if deemed beneficial and safe, with the option to reduce the administration cycle to 4 weeks or extend to 8 weeks. Please see Section 6.5.2 for further details.

If a participant experiences drug-related toxicities, administered activity and schedule modifications will be permitted as described in Section 6.5.5.

Participants will be discontinued from treatment if they withdraw consent, experience unacceptable toxicity, have confirmed disease progression per RECIST v1.1 based on the local assessment at the site, or at the discretion of the investigator or participant (refer to Section 7).

Participants will be permitted to continue treatment beyond initial disease progression, following documented discussion with investigator (or delegate) and Novartis Medical Monitor, provided they meet each of the following criteria: (i) benefit (or reasonable potential for future benefit) in the opinion of the investigator; (ii) the absence of signs or symptoms indicating rapid progression or symptoms which may require urgent medical intervention (e.g. symptomatic pleural effusion or spinal cord compression); (iii) tolerance of study treatment including but not limited to the absence of a decline in performance status; (iv) continued treatment should not jeopardize critical interventions to treat/prevent severe complications or prevent participants from receiving adequate care; and (v) the cumulative administered activity does not exceed the allowable administered activity per respective dosage level (see Section 4.3.1 and Section 6.5.2).

Participants who are offered the opportunity to continue study treatment beyond disease progression must be informed of other treatment options with proven benefit, if those exist. The participant's consent to continue treatment beyond progression must be documented in the participant's medical record.

Participants who meet the above criteria and continue treatment beyond initial disease progression will continue all study procedures as outlined in the visit evaluation schedule. In case of clinical deterioration or suspicion of worsening disease progression, a follow-up imaging assessment should be performed promptly rather than waiting for the next scheduled assessment. Participants with evidence of further disease progression on an imaging assessment or who are no longer deriving clinical benefit will be discontinued.

6.2 Preparation, handling, storage, and accountability

Each study site will be supplied with investigational drugs in packaging as described in Table 6-1. For details related to preparation, handling and storage of investigational drugs, refer to the appropriate Pharmacy Manual. For details related to ordering of ^{225}Ac-DWJ155, refer to the ROME manual.

For ^{68}Ga-DWJ155, Investigator staff will identify the study medication kits to dispense to the participant by contacting the IRT and obtaining the medication number(s). A unique identifier is printed on the study medication label.

Investigational drugs accountability and reconciliation data is recorded in the IRT system and relevant Logs.

6.2.1 Handling of study treatment

Investigational drugs must be received by a designated person at the study site, handled and stored safely and properly and kept in a secured location to which only the Investigator and designated site personnel have access. Upon receipt, investigational drugs must be stored according to the instructions specified on the labels and/or in the current ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155 IBs. Storage of radiopharmaceuticals should be in the original package (lead shielding) to protect from ionizing radiation and in accordance with national regulation on radioactive materials. The radioactive investigational drug will also be locally discarded according to all disposal requirements and local regulations for radioactive materials.

Clinical supplies are to be dispensed only in accordance with the protocol. Technical complaints are to be reported to the respective Novartis Country Organization Quality Assurance (see Pharmacy Manual for details).

Medication labels will be in the local language and comply with the legal requirements of each country. They will include storage conditions for the study treatment and expiry date.

The investigator or designated site staff must maintain an accurate record of the shipment and dispensing of investigational drugs in a drug accountability log. The investigator must also provide accountability for locally sourced materials used for administration (e.g., i.v. bags). Monitoring of drug accountability will be performed by field monitors during site or remote monitoring visits, and at the completion of the trial.

The site may destroy and document destruction of unused investigational drugs, drug labels and packaging, as appropriate in compliance with site processes, monitoring processes, and per local regulation/guidelines. In addition, any radioactive material will be locally discarded according to all disposal requirements and local regulations and will also be documented appropriately: a copy of the completed drug accountability log and disposition certificates will be sent to the Novartis monitor or to the Novartis address provided in the Investigator folder at each site.

6.2.2 Handling of other or additional treatment

Not applicable.

6.2.3 Instruction for prescribing and taking study treatment

^{68}Ga-DWJ155

The imaging drug must be prepared for administration in accordance with the Pharmacy Manual and radiation safety regulations.

^{225}Ac-DWJ155

The therapeutic agent is a sterile radiopharmaceutical solution for injection/infusion supplied as a ready-to-use product which must be prepared for administration in accordance with the Pharmacy Manual and radiation safety regulations.

6.3 Measures to minimize bias: randomization and blinding

6.3.1 Treatment assignment, randomization

No randomization will be performed in this study. The assignment of a participant to a particular cohort and dosage level in escalation, group in expansion will be coordinated by Novartis. All participants entering the screening phase will be registered in the IRT system. The investigator or his/her delegate will then contact the IRT once all the inclusion/exclusion criteria have been assessed to confirm whether the participant will start study treatment or not.

6.4 Study treatment compliance

^{68}Ga-DWJ155 and ^{225}Ac-DWJ155 will be administered in the treating institution. This information must be captured in the source document, the appropriate eCRFs, IRT and the Drug Accountability Log.

6.5 Dosage escalation guidelines

6.5.1 Starting activity

The starting administered activity of ^{225}Ac-DWJ155 is 2 MBq, to be administered at a 6-week interval, for up to 4 cycles.

As described in Section 4.3, ^{68}Ga-DWJ155 will be given as a single administration of approximately 250 MBq with an acceptable range of 200 to 275 MBq. Pending emerging data, starting administered activity may be adjusted in the range of 150-500 MBq to better capture tumor uptake.

A range of ±10% of the ^{225}Ac-DWJ155 assigned dose should be administered to the participant (as per the pre-administration assay).

Refer to Section 4.3.1 for further details on the rational for the selected starting administered activity and schedule.

6.5.2 Provisional dosage levels

Table 6-2 describes the ^{225}Ac-DWJ155 starting administered activity and the dosages that may be evaluated during this trial. Except for the first cohort, selection of actual administered activities will be determined based on available clinical safety and toxicity data guided by the BHLRM, following a discussion with participating investigators during DEMs. Dosage escalation will continue until the RD(s) is/are determined.

The administration schedule in DL1 will be 6-week cycles. Based on the totality of data, including but not limited to the BHLRM model, safety, tolerability, and available PK and dosimetry, a different schedule, i.e., 4-week or 8-week, may be explored starting from DL2. If changing the administration schedule from 6- to 4-week, the starting administered activity per cycle on the 4-week schedule, i.e., the administered activity per cycle at the first instance when the 4-week is explored, will be no higher than the highest administered activity per cycle on the 6-week schedule that has previously been shown to be safe.

Each participant is expected to receive up to 4 cycles of ^{225}Ac-DWJ155 in DL1. A lower or higher number of cycles may be explored from DL2 if deemed beneficial and safe, while maintaining comparable cumulative administered activity at that DL (Table 6-2).

Table 6-2 Provisional administered activity levels

Dosage level	Administered activity per Cycle (MBq)	Increment from previous DL	No. Cycles*	Cumulative administered activity (MBq)
-1**	1	-50%	4	4
1	2	N/A	4	8
2	4	+100%	4	16
3	6	+50%	4	24
4	8	+33%	4	32

*At DL2 and/or higher, a lower or higher number of cycles may be explored if deemed beneficial and safe.

** Dosage level -1 represents treatment dosage for participants requiring a dosage reduction from the starting dosage level.

- 3-6 participants per DL with option to open additional cohorts at previously tested DLs. Additional and/or intermediate dosage levels may be added in order to better characterize safety and preliminary efficacy. Not all dosage levels are required to open.
- The administered activity will not exceed 100% from the previously tested safe administered activity per cycle, however, smaller increases may be recommended.
- Refer to the Dosage modification Section 6.5.5 for further details.

6.5.3 Guideline for dosage escalations and dosage selection for the Phase 1 expansion

The dosage escalation is conducted in order to establish the RD(s) of ^{225}Ac-DWJ155 to be used in the expansion part. Specifically, it is the one or more dosages that in the view of Investigators and Novartis study personnel have the most appropriate benefit-risk as assessed by the review of safety and tolerability, and any available PK, dosimetry, efficacy, and pharmacodynamics.

The maximum tolerated activity (MTA) is the highest administered activity/cycle estimated to have less than 25% risk of causing a dosage-limiting toxicity (DLT) during the DLT evaluation period in more than 33% of treated participants. The dosages selected for the expansion part can be any administered activity and schedule equal to or less than the MTA, and may be declared without identifying the MTA.

Each dosage escalation cohort will start with 3 to 6 newly treated participants. They must have adequate exposure and follow-up to be considered evaluable for dosage escalation decisions (see Section 9 for criteria in the definition of the DDS). Initiation of ^{225}Ac-DWJ155 administrations between the first two participants in the first dosage level will be staggered by at least 6 weeks to cover the first participant's DLT period prior to treating the second participant. Treatment of subsequent participants in these and subsequent dosage levels may proceed concurrently.

Dosage escalation decisions will be made when all participants in a cohort have completed the DLT evaluation period or discontinued. However, in cases where at least three participants are already evaluable for the DDS, and all the conditions below are satisfied:

- there is a maximum of one additional participant treated and ongoing in that cohort who is not evaluable for DDS yet
- the possible DLT outcome of the remaining last participant does not impact the administered activity(ies) recommendation for the next cohort as per the escalation criteria guided by BHLRM with EWOC

The DEM decision may be performed before such participant completes the DLT evaluation period or discontinues. For the 6-week schedule, the DLT evaluation period is 6 weeks starting from the first administration of ^{225}Ac-DWJ155. For the 4-week and 8-week schedule, the DLT evaluation period is 4 weeks and 8 weeks, respectively, starting from the first administration of ^{225}Ac-DWJ155.

Decisions on whether to increase the administered activity/cycle, cumulative administered activity and the treatment schedule will be made by Investigators and Novartis study personnel in a dosage-escalation teleconference, and will be based on primarily the available safety data, including toxicity. Available preliminary efficacy, PK and dosimetry data will also be reviewed. To date, dosimetry of Ac-225 remains challenging due to the low photon emission probability in the actinium decay chain at the low administered activity/cycle. Initial studies demonstrated feasibility of dosimetry of Ac-225 RLTs in experimental settings (Delker et al 2023, Liubchenko et al 2024). Detailed information on dosimetry and PK assessments is presented in Section 8.7.

Any dosage escalation decisions made by Investigators and Novartis personnel will not exceed the dosage level satisfying the EWOC principle by the BHLRM. In all cases, the administered activity per cycle for the next escalation cohort will not exceed a 100% increase from the previously tested safe administered activity. However, smaller increases in administered activity may be recommended by the Investigators and Novartis upon consideration of all the available clinical data (see Section 6.5.2 for planned increases for DL3 and DL4).

The administration schedule in DL1 will be in 6-week cycles. Based on the totality of data, including but not limited to the BHLRM model, safety and tolerability, a 4-week or 8-week schedule may be explored in DL2. The same schedule will be adopted for subsequent cohorts, or a different schedule may be adopted.

To better understand the safety, tolerability, PK, pharmacodynamics, or anti-tumor activity of ^{225}Ac-DWJ155 before or while proceeding with further escalation, additional cohorts of 1 to 6 participants (or 1 to 12 participants at administered activities that are candidates for RD declaration), (e.g., in select indications), may be enrolled at any dosage level at or below the highest administered activity previously tested and shown to be safe.

To reduce the risk of exposing participants to an overly toxic dosage, when 2 participants experience a DLT in a new cohort, the BHLRM will be updated with the most up-to-date new information from all cohorts, without waiting for all participants from the current cohort to complete the evaluation period.

- If the 2 DLTs occur in a cohort of participants treated at a new administered activity level, enrollment to that cohort will stop, and the next cohort will be opened at a lower administered activity level that satisfies the EWOC criteria. If the 2 DLTs occur in participants with the same cancer diagnosis at a given administered activity level, upon

evaluation of all relevant data, an independent dosage escalation strategy may be implemented for participants with this cancer diagnosis with 2 DLTs observed.

- If the 2 DLTs occur in a cohort of participants treated at an already tested administered activity level, then upon re-evaluation of all relevant data, additional participants may be enrolled into the open cohorts only if the administered activity still meets the EWOC criteria. Alternatively, if recruitment to the same administered activity level cannot continue, a new cohort of participants may be recruited to a lower administered activity that satisfies the EWOC criteria.

In the event that an administered activity associated with a specific dosage level is not deemed acceptable for newly enrolled but untreated participants, ongoing participants may continue treatment at that dosage level at the discretion of the Investigator and Novartis if it is in the best interests of the participant.

Besides the scenario of 2 DLTs, the current dosage level being tested may be de-escalated based on new safety findings, including but not limited to observing a DLT, before a cohort is completed. Subsequent to a decision to de-escalate, re-escalation may occur if data in subsequent cohorts satisfies the EWOC criteria.

Enrollment into dosage escalation cohorts may continue in parallel with the dosage expansion for other dosage(s) in order to support a better understanding of safety or efficacy (administered activity and schedule) and to gain more information on the exposure-response relationships. In this case, it is possible that an additional RD may be considered in the dosage expansion part.

All dosage decisions must be agreed by Investigators and Novartis study personnel.

6.5.3.1 Implementation of dosage escalation decisions

To implement dosage escalation decisions, the available toxicity information (including adverse events and laboratory abnormalities that are not DLTs), the recommendations from the BHLRM, and the available pharmacodynamics information will all be evaluated by the Investigators and Novartis study personnel (including the study physician and statistician) during a dosage escalation decision meeting by teleconference. Drug administration at the next higher dosage level may not proceed until the Investigator receives written confirmation from Novartis indicating that the results of the previous dosage level were evaluated and that it is permissible to proceed to a higher dosage level.

6.5.4 Definitions of dosage limiting toxicities (DLTs)

A dosage-limiting toxicity (DLT) is defined as any adverse event or abnormal laboratory value as defined in Table 6-3 that occurs within the DLT evaluation period (i.e., first cycle: 4 weeks, 6 weeks or 8 weeks starting from the first administration of ^{225}Ac-DWJ155, depending on the administration schedule) unless clearly and incontrovertibly assessed as due to extraneous circumstances (e.g., underlying disease, disease progression, intercurrent illness/injury, or concomitant medications).

The National Cancer Institute Common Terminology Criteria for Adverse events (NCI CTCAE) version 5.0 will be used for all grading. For the purpose of dosage-escalation decisions, DLTs will be considered and included in the BHLRM.

The investigator must notify Novartis immediately of any unexpected CTCAE Grade ≥ 3 AEs or laboratory abnormalities. Prior to enrolling participants into a higher dosage level, CTCAE Grade ≥ 2 AEs will be reviewed for all participants at the current dosage level.

Table 6-3 Criteria for defining dosage-limiting toxicities

A DLT is defined as any of the following AEs or abnormal laboratory value of CTCAE version 5.0 that occurs within the DLT evaluation period (i.e., first cycle: 4 weeks, 6 weeks or 8 weeks starting from the first administration of ^{225}Ac-DWJ155, depending on the administration schedule) unless clearly and inconvertibly assessed as due to extraneous circumstances (e.g., underlying disease, disease progression, intercurrent illness/injury, or concomitant medications):	
Hematologic	Grade 3 Neutropenia with fever or other clinical symptoms Grade 3 Neutropenia without fever or other clinical symptoms that does not resolve to ≤ Grade 1 within 7 days Grade ≥ 4 Neutropenia
	Grade 3 Thrombocytopenia with clinically significant bleeding Grade ≥ 4 Thrombocytopenia
	Grade 3 Anemia that does not resolve to Grade ≤ 2 within 7 days in the absence of transfusion Grade ≥ 4 Anemia
Nausea, vomiting, or diarrhea	Grade 3 nausea or vomiting that does not resolve to Grade ≤ 2 within 72 hours of starting optimal anti-emetic therapy Grade ≥ 4 nausea or vomiting
	Grade 3 diarrhea that does not resolve to Grade ≤ 2 within 72 hrs after starting optimal anti-diarrheal therapy Grade ≥ 4 diarrhea
Colitis	Grade 2 colitis that does not resolve to Grade ≤ 1 within 7 days of starting treatment with corticosteroids Grade ≥ 3 colitis
AST, ALT, bilirubin, or alkaline phosphatase	Isolated Grade 3 AST or ALT increase (without a combined increase in total bilirubin > 2 x ULN) that does not resolve to Grade ≤ 1 within 7 days. Grade ≥ 2 AST or ALT increase combined with total bilirubin > 2 x ULN, in the absence of cholestasis[1] Grade 4 AST or ALT increase Grade ≥ 3 total bilirubin increase in the absence of cholestasis Grade 4 serum alkaline phosphatase increase that does not resolve to Grade ≤3 within 7 days.
Amylase or lipase	Symptomatic Grade ≥ 3 amylase or lipase increase
Hypertension	Grade 3 hypertension that does not resolve to Grade ≤ 1 within 7 days after initiating or intensifying anti-hypertensive treatment Grade ≥ 4 hypertension
Electrolytes	Grade 3 electrolyte abnormalities that are associated with symptoms Grade 3 electrolyte abnormalities that are not associated with symptoms that do not resolve to Grade ≤ 1 within 72 hours after starting corrective treatment Grade ≥ 4 electrolyte abnormalities

Dermatologic	Grade 2 bullous disease that does not resolve to ≤ Grade 1 within 7 days of starting corticosteroids Grade ≥ 3 bullous disease Grade 3 non-bullous rash without epidermal detachment that does not resolve to Grade ≤ 1 within 7 days of starting treatment Grade ≥ 4 rash
Fatigue	Grade 3 Fatigue that does not resolve to Grade ≤ 2 within 7 days Grade ≥ 4 Fatigue
Creatine kinase	Grade 3 creatine kinase increase that is symptomatic Grade 3 creatine kinase increase that is associated with evidence of cardiac involvement Grade 3 asymptomatic creatine kinase increase that does not resolve to Grade ≤1 within 14 days Grade ≥ 4 creatine kinase increase
Pneumonitis	Grade ≥ 2 pneumonitis
Ophthalmologic	Grade 2 eye pain or reduction of visual acuity that does not respond to topical therapy and does not improve to ≤ Grade 1 within 14 days after initiating topical therapy Grade 2 eye pain that requires systemic treatment Grade ≥ 3 ophthalmologic AEs
Myocarditis	Grade ≥ 2 myocarditis
AEs not otherwise specified	Grade ≥ 3 AEs with the exception of lymphopenia of any grade Other clinically significant toxicities, including a single event or multiple occurrences of the same event that lead to a dosing delay may be considered to be DLTs by the investigators and Novartis based on the significance of the adverse event and the duration of the delay.
[1] "Cholestasis" defined as: Alkaline Phosphatase (ALP) elevation (> 2 × ULN and R value< 2) in participants without bone metastasis, or elevation of ALP liver fraction in participants with bone metastasis Note: The R value is calculated by dividing the ALT by the ALP, using multiples of the ULN for both values. It denotes the relative pattern of ALT and/or ALP elevation is due to cholestatic or hepatocellular liver injury.	

6.5.5 Dosage modifications

For participants who do not tolerate the protocol-specified administration schedule, adjustments are permitted in order to allow participants to continue the study treatment. The following guidelines need to be applied:

- For each participant, a maximum of two dosage reductions will be allowed, after which the participant will be discontinued from study treatment. For each participant, once a dosage reduction has occurred, the dosage may not be re-escalated during subsequent treatment cycles.
- Dosage reductions to an administered activity below 1 MBq per cycle are not permitted. If a participant does not tolerate this administered activity, treatment must be discontinued. If a participant requires a delay of > 4 weeks for a participant on a 4-week dosing cycle (or > 6 weeks for a participant on a 6-week dosing cycle or > 8 weeks for a participant on an 8-week dosing cycle) from the intended day of the next scheduled activity

administration, then the participant must be discontinued from the study unless the participant experiences clinical benefit or has reasonable potential for future clinical benefit in the opinion of the investigator. In this case, treatment can be continued at an administered activity agreed upon by Novartis and the investigator.

- The criteria for dosage modifications and dosage interruptions for toxicities considered at least possibly related to study treatment (i.e., adverse drug reactions) are summarized in Table 6-4. All interruptions or changes to study treatment administration must be recorded on the appropriate eCRF.
- Events not included in the study protocol or the reference guidance documents should be managed per institutional preference.
- Participants who discontinue the study treatment for a study-related AE or a study-related abnormal laboratory value must be followed as described in Table 6-4.
- The Sponsor may consider the totality of safety and tolerability data, including but not limited to acute safety/tolerability in later cycles as well as longitudinal safety after all administrations, when making dosage modifications.

These administered activity changes must be recorded on the appropriate CRF.

Table 6-4 Criteria for dosage reduction / interruption and re-initiation of ^{225}Ac-DWJ155 treatment for adverse drug reactions

Worst toxicity CTCAE Grade[a]	**Recommended Dosage Modification**
Investigations (Hematologic)	
Neutropenia (ANC)	
Grade 2	Consider study treatment hold and discuss with the Novartis Medical Monitor. Monitor at least weekly. Upon resolution to ≤ Grade 1, resume study treatment without dosage modification after discussion with the Novartis Medical Monitor.
Grade 3	Hold study treatment and monitor weekly. Upon resolution to ≤ Grade 1 within 7 days without growth factor support, resume treatment without dosage modification after discussion with the Novartis Medical Monitor. Upon resolution to ≤ Grade 1 in > 7 days or if growth factor support is given, resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. Upon recurrence of ≥ Grade 3 neutropenia after dosage reduction or interruption due to neutropenia, resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor.
Grade 4	Hold study treatment and manage per institutional guidelines. For grade 4 neutropenia < 7 days, upon resolution to ≤ Grade 1 resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor.

Worst toxicity CTCAE Grade[a]	**Recommended Dosage Modification**
	For grade 4 neutropenia ≥ 7 days, discontinue study treatment. Discontinue study treatment upon recurrence of ≥ Grade 3 neutropenia after one dosage reduction.
Febrile neutropenia	
Grade 3	Hold study treatment and monitor weekly. Upon resolution of fever and improvement of neutropenia to ≤ Grade 1, resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. Discontinue study treatment upon recurrence of ≥ Grade 3 neutropenia after dosage reduction due to neutropenia.
Grade 4	Discontinue study treatment. Manage per institutional practice.
Thrombocytopenia	
Grade 2	Hold study treatment and monitor weekly. Upon resolution to ≤ Grade 1, resume study treatment.
Grade 3	Hold study treatment and monitor weekly. Upon resolution to ≤ Grade 1, resume study treatment after discussion with the Novartis Medical Monitor. For Grade 3 associated with major bleeding, discontinue study treatment.
Grade 4	Hold study treatment and monitor weekly. Upon resolution to ≤ Grade 1, resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. For Grade 4 associated with major bleeding, discontinue study treatment.
Anemia	
Grade 3	Hold study treatment and monitor weekly. Upon resolution to ≤ Grade 2 within 7 days in the absence of transfusion, resume study treatment after discussion with the Novartis Medical Monitor. Upon resolution to ≤ Grade 2 after 7 days, resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. Discontinue study treatment upon recurrence of ≥ Grade 3 anemia after one dosage reduction due to anemia.
Grade 4	Hold study treatment. Upon resolution to ≤ Grade 2 within 7 days, resume study treatment at one lower dosage level. If > 7 days, discontinue study treatment.
Lymphopenia	
Any grade	Treatment-related lymphopenia does not require study treatment hold or discontinuation.

Worst toxicity CTCAE Grade[a]	**Recommended Dosage Modification**
Ocular (uveitis, eye pain, blurred vision)	
Grade 1	Continue study treatment without dosage modification. Ophthalmology consultation.
Grade 2	Hold study treatment. Urgent ophthalmology consultation. Upon resolution to ≤ Grade 1 may consider resuming study treatment without dosage reduction after discussion with the Novartis Medical Monitor and in consultation with ophthalmology. Discontinue study treatment upon recurrence of Grade 2 ocular toxicity after one dosage reduction or interruption.
Grade 3 or Grade 4	Discontinue study treatment. Urgent ophthalmology consultation.
Pulmonary (pneumonitis)	
Grade 1	Consider study treatment hold. Manage per institutional practice. Consider resuming study treatment upon radiographic evidence of improvement.
Grade 2	Hold study treatment and manage per institutional practice. Upon complete resolution, resumption of study treatment at reduced dosage may be considered following documented discussion and agreement with Novartis.
Grade 3 or Grade 4	Discontinue study treatment. Manage per institutional practice.
Cardiovascular	
ECG QTc-Interval prolonged[b]	
Grade 3	Hold study treatment. Manage per institutional practice. Upon resolution to Grade ≤ 1 or baseline or < 30 msec difference from baseline may resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. Baseline ECG refers to the ECG(s) collected at screening.
Grade 4	Discontinue study treatment.
Hypertension	
Grade 3	Hold study treatment. Manage per institutional practice. Upon resolution to Grade ≤ 1 or baseline within 7 days, may resume study treatment without dosage modification after discussion with the Novartis Medical Monitor. If > 7 days, may resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor.
Grade 4	Discontinue study treatment.

Worst toxicity **CTCAE Grade[a]**	**Recommended Dosage Modification**
Left ventricular ejection fraction (LVEF)	
Asymptomatic, absolute decrease in LVEF of 10%- < 20% from baseline and is below 50%	Hold study treatment, obtain consultation with Cardiologist, and repeat cardiac imaging after ≤ 4 weeks. If improved to 10% or less from baseline and at least at institutional lower limit of normal (LLN) within 4 weeks, then resume study treatment at one lower dosage level. Monitor LVEF on D1 of each subsequent cycle for at least 3 cycles; if no recurrence of LVEF decline and if agreed by Cardiologist, the frequency may then be decreased to every other cycle. If not improved to 10% or less from baseline and to at least 50% within 4 weeks, then discontinue participant from study treatment.
Grade 3 left ventricular systolic dysfunction	Hold study treatment and obtain consultation with Cardiologist. Following resolution of symptoms and recovery of the LVEF to the normal range within 4 weeks, study treatment may be resumed at one lower dosage level, if agreed by Cardiologist. Repeat echocardiogram 2, 4, 8, and 12 weeks after re-starting study treatment. If no recurrence of LVEF decline and if agreed by Cardiologist, the frequency of LVEF monitoring may then be decreased to the standard schedule per the Schedule of Activities (Section 1.3). If recovery of LVEF to the normal range does not occur within 4 weeks, discontinue study treatment.
Other cardiac disorders	
Grade 2 (except myocarditis)	Hold study treatment. Upon resolution to Grade ≤ 1 or baseline, may resume study treatment without dosage modification after discussion with the Novartis Medical Monitor.
Grade 2 myocarditis, or Grade ≥ 3 other cardiac disorders	Discontinue study treatment.
Gastrointestinal	
Diarrhea/colitis[c]	
Grade 1	May continue study treatment without dosage modification. Manage per institutional standard guidelines which should include anti-diarrheal treatment, consideration of corticosteroid therapy, and hydration.
Grade 2	Hold study treatment. GI consultation. Upon resolution to ≤ Grade 1 and tapering of steroid requirement to ≤ 10 mg prednisone per day, resume study treatment without dosage modification after discussion with the Novartis Medical Monitor.
Grade 3	Hold study treatment. GI consultation.

Worst toxicity CTCAE Grade[a]	**Recommended Dosage Modification**
	Upon resolution to ≤ Grade 1 within 72 hours after optimal therapy and tapering of steroid requirement to ≤ 10 mg prednisone per day, consider resuming study treatment without dosage modification after discussion with the Novartis Medical Monitor. If resolved in >72 hours, may resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. Discontinue study treatment upon recurrence of Grade 3 diarrhea/colitis after one dosage reduction or interrupt.
Grade 4	Discontinue study treatment.
AST or ALT elevation	
Grade 2 AST and/or ALT	Consider study treatment hold. Manage per institutional practice. Upon resolution to ≤ Grade 1 or baseline, if DLT criteria are not met, consider resuming study treatment without dosage modification. Upon resolution to ≤ Grade 1 or baseline, if DLT criteria are met, resume study treatment at one lower dosage level after discussion with Novartis Medical Monitor.
Grade 2 transaminitis with bilirubin > 2 x ULN (unless Gilbert's syndrome)	Discontinue study treatment.
Grade 3 AST and/or ALT	Hold study treatment hold. Manage per institutional practices. Upon resolution to ≤ Grade 1 or baseline within 7 days, if DLT criteria are not met, consider resuming study treatment without dosage modification after discussion with the Novartis Medical Monitor. If resolved in > 7 days, if DLT criteria are NOT met, consider resuming study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. If resolved in > 7 days, if DLT criteria are met, resume study treatment at one lower dosage level after discussion with the Novartis Medical Monitor.
Grade 4 AST and/or ALT	Discontinue study treatment.
Isolated total bilirubin elevation	
Grade 2	Consider study treatment hold. Upon resolution to ≤ Grade 1, may continue study treatment without dosage modification.
Grade 3[d]	Hold study treatment. Upon resolution to ≤ Grade 1, may consider resuming study treatment at one lower dosage level after discussion with the Novartis Medical Monitor. Discontinue study treatment upon recurrence of Grade 3 isolated total bilirubin elevation after one dosage reduction.
Grade 4[d]	Discontinue study treatment.

Worst toxicity **CTCAE Grade[a]**	**Recommended Dosage Modification**
Isolated Alkaline Phosphatase elevation	
Grade 3 or 4	Hold study treatment and discuss with Novartis Medical Monitor. If workup shows that radiation induced liver disease is suspected, discontinue study treatment. If workup shows that radiation induced liver disease is not suspected, upon resolution to Grade ≤ 1 or baseline, resume study treatment with dosage reduction after discussion and agreement with the Novartis Medical Monitor.
Asymptomatic amylase and/or lipase elevation[e]	
Grade 3, not associated with symptoms or clinical manifestations of pancreatitis	Continue study treatment. If levels do not resolve to ≤ Grade 2 within 14 days after the initial report, hold study treatment. Upon resolution to ≤ Grade 2, may resume study treatment without dosage modification after discussion with the Novartis Medical Monitor.
Pancreatitis	
Grade 2/radiologic evidence	Hold study treatment. Manage per institutional practice. Upon resolution, may resume study treatment without dosage modification, if no clinical evidence of pancreatitis and after discussion with the Novartis Medical Monitor.
Grade 3 or Grade 4	Discontinue study treatment.
Renal	
Serum creatinine	
Grade 2	Consider study treatment hold. Manage per institutional practice. Upon resolution to ≤ Grade 1, may resume study treatment without dosage modification after discussion with the Novartis Medical Monitor.
Grade 3 or 4	Discontinue study treatment.
Musculoskeletal	
Grade 2 or Grade 3	Consider study treatment hold. Consider resuming study treatment without dosage modification upon resolution to ≤ Grade 1 with appropriate management.
Grade 4	Discontinue study treatment. In some cases, resuming study treatment may be considered after discussion with the Novartis Medical Monitor and consultation with a rheumatologist.
Endocrine	
Hypothyroidism or hyperthyroidism	
Grade 2	May continue study treatment without dosage modification. Management per institutional practice.
Grade 3 and Grade 4	Hold study treatment.

Worst toxicity CTCAE Grade[a]	**Recommended Dosage Modification**
	May resume therapy following resolution or control with physiologic hormone replacement.
Other endocrine disorders	
Grade 2 and Grade 3	Hold study treatment. May resume study treatment following resolution or control with physiologic hormone replacement.
Grade 4	Hold study treatment. Grade 4 treatment-related endocrinopathies, such as adrenal insufficiency, adrenocorticotropic hormone (ACTH) deficiency, hyper- or hypothyroidism, or glucose intolerance, which resolve or are adequately controlled with physiologic hormone replacement (corticosteroids, thyroid hormones) or glucose-controlling agents, respectively, may not require discontinuation after discussion with and approval from the Novartis Medical Monitor.
Neurology	
Grade 1	Consider study treatment hold, particularly for clinical suspicion of Guillain-Barre syndrome, transverse myelitis, or peripheral neuropathy.
Grade 2	Hold study treatment. In some cases, resuming study treatment may be considered after discussion with the Novartis Medical Monitor. Discontinue study treatment upon recurrence of Grade 2 neurologic toxicity after one dosage reduction or interruption.
Grade 3 or Grade 4	Discontinue study treatment.
Dermatology (rash)	
Grade 1	Continue study treatment without dosage modification. Topical steroids, antihistamines, topical emollients.
Grade 2	Consider holding study treatment. Topical or oral corticosteroids, antihistamines. If study treatment is held and resolution to ≤ Grade 1, resume study treatment without dosage modification.
Grade 3 or Grade 4	Hold study treatment. Manage per institutional practice. After resolution to ≤ Grade 1, consider resuming study treatment with dosage reduction after discussion with the Novartis Medical Monitor.
Bullous dermatitis	Hold study treatment. Grade 1-2 bullous dermatitis: discussion with the Novartis Medical Monitor is required before considering resuming study treatment. Grade 3-4 bullous dermatitis: discontinue study treatment.
Stevens-Johnson syndrome (SJS), or Lyell syndrome/toxic epidermal necrolysis (TEN)	Permanently discontinue study treatment.

Worst toxicity **CTCAE Grade[a]**	**Recommended Dosage Modification**
Other laboratory adverse events, not specified elsewhere in table and not included in the consensus guidelines	
Grade 3 or Grade 4	Hold study treatment. Upon resolution to ≤ Grade 1 or baseline, resume study treatment without dosage modification. Isolated Grade 4 electrolyte abnormalities not associated with clinical sequelae and corrected after appropriate management within 72 hours of their onset do not require discontinuation. In the case of Grade 4 electrolyte imbalances associated with clinical sequelae, or not resolved to ≤ Grade 1 within 72 hours despite appropriate management, discontinue study treatment.
Other non-laboratory adverse events, not specified elsewhere in table and not included in the consensus guidelines	
Grade 2	Consider study treatment hold, at investigator discretion. Upon resolution to ≤ Grade 1, resume study treatment without dosage modification.
Grade 3	Hold study treatment. Upon resolution to ≤ Grade 1, resuming study treatment must be discussed with the Novartis Medical Monitor.
Grade 4	Discontinue study treatment.
[a] All dosage modifications should be based on the worst preceding toxicity per the Common Toxicity Criteria for Adverse Events (CTCAE) version 5.0. [b] Based on the mean of triplicate ECGs. [c] Note: anti-diarrheal medication is recommended at the first sign of abdominal cramping, loose stools, or overt diarrhea. [d] Note: If Grade 3 or 4 hyper-bilirubinemia is due to the indirect (non-conjugated) component only, and hemolysis as the etiology has been ruled out as per institutional guidelines (e.g., review of peripheral blood smear and haptoglobin determination), then delay study treatment until resolved to ≤ Grade 1, and resume study treatment at the discretion of the investigator. [e] Note: A CT scan or other imaging study to assess the pancreas, liver, and gallbladder must be performed within one week of the first occurrence of any ≥ Grade 3 of amylase and/or lipase elevation.	

6.5.6 Follow-up for toxicities

Participants whose treatment is interrupted or permanently discontinued due to an AE or clinically significant laboratory value must be followed-up at least once a week (or more frequently if required by institutional practices, or if clinically indicated) for 4 weeks, and subsequently at approximately 4-week intervals, until resolution or stabilization of the event, whichever comes first. Please refer to Section 8.9 for more detailed instructions regarding AEs, SAEs and the reporting of these events.

6.5.6.1 Follow up on potential drug-induced liver injury (DILI) cases

Participants with transaminase increase combined with total bilirubin increase may be indicative of potentially severe DILI and should be considered as clinically important events and assessed appropriately to establish the diagnosis. The required clinical information, as

detailed below, should be sought to obtain the medical diagnosis of the most likely cause of the observed laboratory abnormalities.

The threshold for potential DILI may depend on the participant's baseline AST/ALT and total bilirubin value; participants meeting any of the following criteria will require further follow-up as outlined below:

- For participants with normal ALT and AST and total bilirubin value at baseline: AST or ALT > 3.0 x ULN combined with total bilirubin > 2.0 x ULN
- For participants with elevated AST or ALT or total bilirubin value at baseline: [AST or ALT > 3.0 x baseline] OR [ALT or AST > 8.0 x ULN], whichever occurs first, combined with [total bilirubin > 2.0 x baseline AND > 2.0 x ULN]

As DILI is essentially a diagnosis of exclusion, other causes of abnormal liver tests should be considered and their role clarified before DILI is assumed to be the cause of liver injury.

A detailed history, including relevant information such as review of ethanol consumption, concomitant medications, herbal remedies, supplement consumption, history of any pre-existing liver conditions or risk factors, should be collected.

Laboratory tests should include ALT, AST, total bilirubin, direct and indirect bilirubin, Gamma-glutamyl Transferase (GGT), prothrombin time (PT)/International Normalized Ratio (INR), alkaline phosphatase, albumin, and creatine kinase. Testing of Glutamate Dehydrogenase (GLDH) is additionally recommended unless unavailable.

Evaluate status of liver metastasis (new or exacerbation) or vascular occlusion, e.g., using CT, MRI, or duplex sonography.

Perform relevant examinations (Ultrasound or MRI, Endoscopic retrograde cholangiopancreatography (ERCP)) as appropriate to rule-out an extrahepatic cause of cholestasis. Cholestasis (is defined as an ALP elevation > 2.0 x ULN with R value < 2 in participants without bone metastasis, or elevation of the liver-specific ALP isoenzyme in participants with bone metastasis).

Note: The R value is calculated by dividing the ALT by the ALP, using multiples of the ULN for both values. It denotes whether the relative pattern of ALT and/or ALP elevation is due to cholestatic ($R \leq 2$), hepatocellular ($R \geq 5$), or mixed ($R > 2$ and < 5) liver injury. For children, there are caveats to calculating the R-ratio as normal levels of ALP are higher than in adults with standard ranges varying by developmental age. In clinical situations where it is suspected that ALP elevations are from an extrahepatic source, the GGT can be used if available. GGT may be less specific than ALP as a marker of cholestatic injury, since GGT can also be elevated by enzyme induction or by ethanol consumption. It is more sensitive than ALP for detecting bile duct injury.

Table 6-5 provides guidance on specific clinical and diagnostic assessments which can be performed to rule-out possible alternative causes of observed liver function test (LFT) abnormalities.

Table 6-5 Clinical and diagnostic assessments for liver-related disorders

Disease	Assessment
Hepatitis A, B, C, E	• Immunoglobulin (Ig)M anti-hepatitis A virus (HAV); HBsAg, IgM & IgG anti-hepatitis B core antibody (HBc), HBV DNA; anti-HCV, HCV RNA, IgM & IgG anti-hepatitis E virus (HEV), HEV RNA
Cytomegalovirus (CMV), Herpes Simplex Virus (HSV), Epstein–Barr Virus (EBV) infection	• IgM & IgG anti-CMV, IgM & IgG anti-HSV; IgM & IgG anti-EBV
Autoimmune hepatitis	• Antinuclear Antibodies (ANA) & Anti-Smooth Muscle Antibody (ASMA) titers, total IgM, IgG, IgE, IgA
Alcoholic hepatitis	• Ethanol history, GGT, mean corpuscular volume (MCV), Carbohydrate deficient (CD)-transferrin
Nonalcoholic steatohepatitis	• Ultrasound or MRI
Hypoxic/ischemic hepatopathy	• Medical history: acute or chronic congestive heart failure, hypotension, hypoxia, hepatic venous occlusion. Ultrasound or MRI.
Biliary tract disease	• Ultrasound or MRI, ERCP as appropriate.
Wilson disease (if <40 yrs old)	• Caeruloplasmin
Hemochromatosis	• Ferritin, transferrin
Alpha-1-antitrypsin deficiency	• Alpha-1-antitrypsin

Other causes should also be considered based upon participants' medical history (hyperthyroidism / thyrotoxic hepatitis – triiodothyronine (T3), thyroxine (T4), thyroid-stimulating hormone (TSH); cardiovascular disease / ischemic hepatitis – ECG, prior hypotensive episodes; Type 1 diabetes mellitus / glycogenic hepatitis).

Obtain PK sample to determine exposure to study treatment and metabolites.

Following appropriate causality assessments, as outlined above, the causality of the treatment is estimated as "probable" (e.g., >50% likely), if it appears greater than all other possible causes of liver injury combined.

All cases confirmed on repeat testing that meet the laboratory criteria defined above, with no other alternative cause for LFT abnormalities identified, should be considered as "medically significant" and thus, meet the definition of SAE and should be reported as a SAE using the term "potential treatment-induced liver injury". All events should be followed up with the outcome clearly documented.

6.6 Concomitant and other therapies

In order to determine if pre-medication is necessary, participants should not receive pre-medication to prevent an infusion reaction before the first administration of ^{225}Ac-DWJ155 or ^{68}Ga-DWJ155. If a participant experiences an infusion-associated reaction, the participant may receive pre-medication on subsequent dosing days. The pre-medication should be chosen per institutional practice at the discretion of the treating physician.

Use of diuretics (e.g., furosemide) is allowed if required, to help a participant void before ^{68}Ga-DWJ155 PET/CT or PET/MRI imaging acquisition and/or after the last PET/CT or PET/MRI, at the discretion of the investigator. Use of all diuretics should be captured in the eCRF like other concomitant medications.

6.6.1 Concomitant therapies

In general, concomitant medications and therapies deemed necessary for the supportive care (e.g., anti-emetics, anti-diarrheals) and safety of the participant are allowed. The participant must be told to notify the investigational site about any new medications, herbal remedies, and dietary supplements he/she takes after the start of the study treatment. All medications, procedures, and significant non-drug therapies (including physical therapy and blood transfusions) administered after the participant was enrolled into the study must be recorded on the appropriate Case Report Forms.

Each concomitant drug must be individually assessed against all exclusion criteria and prohibited therapy. If in doubt, the Investigator should contact the Novartis medical monitor before enrolling a participant or allowing a new medication to be started. If the participant is already enrolled, contact Novartis to determine if the participant should continue participation in the study.

Vaccines against infectious diseases are allowed.

6.6.1.1 Permitted concomitant therapies requiring caution and/or action

Hematopoietic growth factors (e.g., G-CSF, GM-CSF) are not permitted during the DLT period unless the participant has already experienced a DLT. After a DLT has occurred, or after the DLT period, they may be used as per institutional practice. Treatment with erythroid stimulating agents (ESAs) must not be initiated during the DLT period unless the participant has already experienced a DLT.

Transfusions of platelets or red blood cells should be performed as per institutional practice.

Systemic corticosteroids are allowed during the study, however, should be used at the lowest dose to control symptoms if required, and discontinued as soon as possible. Topical, inhaled, nasal, and ophthalmic corticosteroids are allowed.

Anti-emetics may be used at the discretion of the attending physician. Diarrhea management is per the discretion of the treating physician. Diarrhea could be managed conservatively with medications such as loperamide. Participants with severe diarrhea should be assessed for intravenous hydration and correction of electrolyte imbalances.

Palliative radiation therapy may be used after discussion with Novartis Medical Monitor.

6.6.1.2 Use of bone modifying agents

Bisphosphonates and/or denosumab are generally allowed for the management of bone metastasis and osteoporosis provided participants have been on stable doses for at least 2 weeks prior to study entry. However, chronic concomitant bisphosphonate/denosumab therapy for the prevention of bone metastases is not permitted. If bisphosphonate/denosumab therapy is to be initiated or increased after study entry, disease progression must be excluded and clearly documented, and the rationale discussed with and approved by Novartis. Otherwise, the participant must discontinue study treatment.

6.6.2 Prohibited therapies

During the study, participants must not receive other investigational drugs, agents, devices, chemotherapy, or any other therapies that may be active against cancer.

6.6.3 Hypersensitivity

Acute allergic reactions should be treated as clinically indicated per institutional practice or the dosage modification guideline (Table 6-4). In the event of anaphylactic/anaphylactoid reactions, this includes any therapy necessary to restore normal cardiopulmonary status. If a participant experiences a Grade 4 anaphylactic/anaphylactoid reaction following administration of ^{68}Ga-DWJ155 or ^{225}Ac-DWJ155, the participant will discontinue from any further procedures. Such acute allergic reactions will be reported to the Sponsor in an expedited manner. This should be designated as reportable SAEs regardless of hospitalization, as medically important events. Please refer to SAE reporting Section 8.10.3 for more information.

6.7 Continued access to study treatment after the end of the study

Not applicable.

7 Discontinuation of study treatment and participant discontinuation/withdrawal

7.1 Discontinuation of study treatment

Discontinuation of study treatment for a participant can be initiated by either the participant or the investigator.

The investigator must discontinue study treatment for a given participant if, he/she believes that continuation would negatively impact the participant's well-being.

Study treatment must be discontinued under the following circumstances:

- Participant/guardian decision
- Any situation in which continued study participation might result in a safety risk
- Pregnancy
- Death

If discontinuation of study treatment occurs, the investigator should make a reasonable effort to understand the primary reason for the participant’s discontinuation of study treatment and record this information.

Participants who discontinue from study treatment agree to return for the end of treatment and follow-up visits indicated in Section 1.3 (Schedule of Activities). The EoT visit should be performed 28 days (for participants under 4-week schedule), 42 days (for participants under 6-week schedule), or 56 days (for participants under 8-week schedule) after the last administered dose of study treatment. However, if the decision to permanently discontinue study treatment is made later than 28 days (for participants under 4-week schedule), 42 days (for participants under 6-week schedule) or 56 days (for participants under 8-week schedule) after the last

administered dose of study treatment, the EoT visit should be as soon as possible within 14 days of the decision to permanently discontinue study treatment.

If the participant cannot or is unwilling to attend any visit(s), the site staff should maintain regular telephone contact with the participant, or with a person pre-designated by the participant. This telephone contact should preferably be done according to the study visit schedule.

The following data should be collected at minimum:

- New / concomitant treatments
- Adverse Events / Serious Adverse Events

The Investigator must contact the IRT to register the participant's discontinuation from study treatment.

7.1.1 Safety follow-up

All participants receiving ^{68}Ga-DWJ155 must have safety evaluations 3 days post ^{68}Ga-DWJ155 administration. Refer to Table 1-3 for further details.

All participants receiving ^{225}Ac-DWJ155 must be followed up for safety for 180 days after the last administration of ^{225}Ac-DWJ155. The follow-up visit should be done at least every 12 weeks (90 days) after the last ^{225}Ac-DWJ155 administration and will include the assessments outlined in Section 1.3. Following the EoT visit (i.e., 28 days (for participants under 4 week schedule), 42 days (for participants under 6 week schedule) or 56 days (for participants under 8 week schedule) after the last administration of ^{225}Ac-DWJ155), only SAEs and study treatment-related non-serious AEs are to be reported to Novartis, unless otherwise specified by local law/regulations.

Data collected should be added to the appropriate CRFs. For female participants of child bearing potential, pregnancy tests will be performed as outlined in Section 8.4.5.

7.1.2 Long-term safety follow-up

All participants will continue to be followed up (via phone call or visit) every 180 days after completion of the 180-day Safety follow-up period until completion of the follow-up period of 36 months from the last ^{225}Ac-DWJ155 administration. After the 180-day Safety follow-up period, SAEs should only be reported to Novartis if the investigator suspects a causal relationship to study treatment, unless otherwise specified by local law/regulations. SAEs of second primary malignancies, regardless of causality, will be collected until the end of the 36-month long-term follow-up period as well.

7.1.3 Disease progression follow-up

For participants who discontinue study treatment for any reasons other than death, documented disease progression while on treatment, clinical deterioration, lost to follow-up, withdrawal of consent, study termination, or initiation of subsequent anti-cancer treatment, also should return for tumor evaluation assessments and continue blood sample collection and analysis for cfDNA every 12 weeks from the last administration of study treatment until documented disease progression, death, lost to follow-up, or withdrawal of consent (as detailed in Section 1.3).

7.2 Participant discontinuation from the study

Discontinuation from study is when the participant permanently stops receiving the study treatment, and further protocol-required assessments or follow-up, for any reason.

If the participant agrees, a final evaluation at the time of the participant's study discontinuation should be made as detailed in Section 1.3 (Schedule of Activities).

7.3 Withdrawal of informed consent and exercise of participants' data privacy rights

Withdrawal of consent/opposition to use data and/or biological samples occurs when a participant:

- Explicitly requests to stop use of their data

and

- No longer wishes to receive study treatment

and

- Does not want any further visits or assessments (including further study-related contacts)

This request should be made per local regulations (e.g., in writing) and recorded in the source documentation.

Withdrawal of consent impacts ability to further contact the participant, collect follow-up data (e.g., to respond to data queries) and potentially other country-specific restrictions. It is therefore very important to ensure accurate recording of withdrawal vs. discontinuation based on the protocol definitions of these terms.

In this situation, the Investigator should make a reasonable effort (e.g., telephone, e-mail, letter) to understand the primary reason for the participant's decision to withdraw their consent/exercise data privacy rights and record this information. The Investigator shall clearly document if the participant has withdrawn his/her consent for the use of data in addition to a study discontinuation.

Study treatment must be discontinued and no further assessments conducted.

Further attempts to contact the participant are not allowed unless safety findings require communicating or follow-up.

If the participant agrees, a final evaluation at the time of the participant's withdrawal of consent/exercise data privacy rights should be made.

Further details on withdrawal of consent or the exercise of participants' data privacy rights are included in the corresponding informed consent form.

7.4 Lost to follow-up

For participants whose status is unclear because they fail to appear for study visits or fail to respond to any site attempts to contact them without stating an intention to discontinue, the Investigator must document "due diligence" steps taken to contact the participant, e.g., dates of telephone calls, registered letters, in the source documents. A participant should not be

considered as lost to follow-up until due diligence has been completed or until the end of the study.

7.5 Study stopping rules

During the dosage expansion, a DMC will be in place. Participants will be followed for safety including, but not limited to, the following;

- DLTs (as defined in Section 6.5.4)
- Clinically significant Grade 3/4 AEs,
- SAEs, or
- Long term toxicities or death other than disease progression that is possibly related to the study treatment.

This safety data will be reviewed and a recommendation from the DMC will be given to the sponsor whether to continue, modify, or terminate the trial (refer to Section 7.6). In addition, if the following occurs during the dosage expansion part, the DMC will meet immediately to discuss and make a recommendation (please refer to Section 10.1.4.1 for further details):

- More than 30% of participants experiencing a Grade 3/4 treatment-related toxicity (out of a minimum of 6 participants treated) at the same dosage level or
- Death other than disease progression that is possibly related to the study treatment.

7.6 Early study termination by the Sponsor

The study can be terminated by Novartis at any time.

Reasons for early termination (but not limited to):

- Unexpected, significant, or unacceptable safety risk to participants enrolled in the study
- Decision based on recommendations from applicable board(s) after review of safety and efficacy data
- Discontinuation of study treatment development in the indication(s) in this study

In taking the decision to terminate, Novartis will always consider participant welfare and safety. Should early termination be necessary, participant must be treated as discontinued from study treatment by completing the EoT visit. The participant must also complete the safety follow-up period as per Section 7.1. The Investigator may be informed of additional procedures to be followed in order to protect the participant's interests.

The Investigator or Novartis depending on local regulation will be responsible for informing Institutional Review Board (IRB)/Independent Ethics Committee (IEC) of the early termination of the trial.

8 Study assessments and procedures

- Study procedures and their timing are summarized in Section 1.3. Protocol waivers or exemptions are not allowed.
- Immediate safety concerns should be discussed with Novartis upon occurrence or awareness to determine if the participant should continue or discontinue the study treatment.
- Adherence to the study design requirements, including those specified in Section 1.3, is essential and required for study conduct.
- Repeat or unscheduled samples may be taken for safety reasons or for technical issues with the samples.

8.1 Screening and eligibility

8.1.1 Screening

The study IRB/IEC approved informed consent form must be signed and dated before any screening procedures are performed, except for laboratory and radiological evaluations which were performed as part of the participant's clinical standard of care within the acceptable screening window.

Participants will be evaluated against study inclusion and exclusion criteria and safety assessments (refer to Section 1.3). Screening assessments must be repeated if performed outside of the specified screening window. Participants must meet all inclusion and none of the exclusion criteria at screening to be eligible for the study.

Laboratory test results(s) or symptoms that do not satisfy the eligibility criteria may be repeated or treated during the screening visit window. If the repeated laboratory test(s) cannot be performed within the screening period, or do not meet the eligibility criteria, or other eligibility criteria have changed and are not met anymore, the participant is considered a screening failure.

Where applicable, eligibility for treatment with ^{225}Ac-DWJ155 needs to have confirmed tumor positivity based on the local assessment of ^{68}Ga-DWJ155 tumor uptake (see inclusion criterion 4 in Section 5.1).

Rescreening of a participant who has failed screening may be allowed. In such cases, a new ICF must be signed and a new participant ID must be assigned in the eCRF. All required screening assessments must be repeated if they do not meet the allowed time window for screening when the participant is rescreened for participation in the study. An individual participant can only be rescreened once for the study.

8.2 Participant demographics/other baseline characteristics

Data to be collected will include participant demographics: age, sex and race/predominant ethnicity (if permitted), relevant medical history and current medical conditions, smoking history, diagnosis and extent of cancer (including baseline tumor molecular characteristics), and details of prior anti-neoplastic treatments, prior medications, surgeries, medical procedures, and non-drug therapies, and will be recorded in the eCRF. For enrolled participants, known baseline

tumor molecular characteristics will be collected, including but not limited to (if available) HER2, ER and PR status and selected tumor genetic alterations.

Country-specific regulations should be considered for the collection of demographic and baseline characteristics in alignment with CRF.

Participant race and ethnicity (if permitted) are collected and analyzed to assess the diversity of the study population as required by certain Health Authorities and may be used to identify variations in safety or efficacy due to these factors. All prescription medications, over the counter drugs and significant non-drug therapies prior to the study start must be documented. See protocol Section 6.6 for further details.

8.3 Efficacy assessments

Time points for all imaging and efficacy assessments are provided in Section 1.3 and the imaging assessment collection plan is outlined in Table 8-1.

The study will include different types of imaging modalities. Conventional imaging (CT and/or MRI, bone scans and other imaging assessments as clinically indicated) will be used for standard tumor response assessments for all participants. ^{68}Ga-DWJ155 PET (PET/CT or PET/MRI) imaging will be performed up to two times in the study: all participants will undergo a baseline ^{68}Ga-DWJ155 (PET/CT or PET/MRI) scan at screening (before first treatment with ^{225}Ac-DWJ155), and a second ^{68}Ga-DWJ155 (PET/CT or PET/MRI) scan at EoT visit. Participants assigned to dosimetry assessments in the dosage escalation part of the study will undergo up to 5 serial SPECT/CT scans for dosimetry assessments in Cycle 1, at the imaging time points specified in Table 8-6. Additional SPECT/CT scans may be performed in subsequent cycles if required per local regulation (Section 8.7.4 and Section 8.7.5).

All images collected during the course of the study to assess the extent of disease, will be transferred to the designated imaging Contract Research Organization (CRO). The methods for imaging acquisition are specified in the imaging manual, while central reads are described in the imaging charter.

CT scans associated with PET and SPECT imaging are required for attenuation correction and to provide anatomical information for segmentation of lesions and other volumes-of-interest from PET and SPECT data. The low-dose CT used for attenuation correction of PET/CT or SPECT/CT scans should not be used for performing tumor assessments. The ALARA principle will be followed, and low-dose CT scans will be acquired using manufacturer- and institution-recommended standard parameters. The expected radiation dose from low-dose CT needed for PET/SPECT attenuation correction is expected to be up to 5 millisieverts (mSv) per scan. The cumulative expected dose of ionizing radiation from SPECT-associated low-dose CT scans is up to 25 mSv for participants in the dosimetry subgroup. The cumulative expected dose of ionizing radiation from PET-associated low-dose CT scans is less than 10 mSv for participants undergoing two ^{68}Ga-DWJ155 PET/CT scans.

The expected ionizing radiation exposure from diagnostic CT is 15-20 mSv per scan. As defined in the SoA and in line with standard-of-care, participants will receive diagnostic CT scans every 8 weeks (for 4-week or 8-week schedule) or every 6 weeks (for 6-week schedule) during the treatment period. To minimize the risk of additional radiation exposure, the CT scan frequency will be reduced to every 12 weeks during the 36-month follow-up period. This will result in up

to 9 CT scans during the treatment period (135-180 mSv) and up to 12 CT scans during the long-term follow-up (180-240 mSv).

8.3.1 Efficacy assessment

Table 8-1 Imaging assessment collection plan

Assessments	Procedure	Baseline	During Treatment/Follow-up
^{68}Ga-DWJ155 PET imaging assessment (not part of tumor response assessment)	^{68}Ga-DWJ155 PET/CT or PET/MRI	Mandated (Imaging period)	Mandated at EoT
Dosimetry assessment (not part of tumor response assessment)	^{225}Ac-DWJ155 SPECT/CT	N/A	
Conventional Imaging for tumor response assessment by RECIST v1.1	Chest, abdomen, and pelvis CT with intravenous contrast enhancement (or MRI if CT is contraindicated) and other imaging assessments as clinically indicated	Mandated	Mandated, every 6 weeks (6 week schedule) or every 8 weeks (4- or 8-week schedule) after first administration of study treatment until the EoT visit (-7 day window), and then every 12 weeks after the last administration of study treatment during the safety and disease progression follow-up period (±15 day window)
Additional imaging for tumor response assessment	CT or MRI of other metastatic sites (e.g., neck) and other imaging assessments as clinically indicated; see Section 8.3.1	If clinically indicated	If lesions were documented at baseline, follow same schedule as for CT/MRI of chest, abdomen, and pelvis

Conventional imaging (CT and/or MRI, and other imaging assessments as clinically indicated) will be performed for tumor response assessments at regular intervals as outlined in Table 8-1 and further detailed below. The local investigator's assessments will be used for deriving all efficacy endpoints on RECIST v1.1 (Eisenhauer et al 2009 and Section 10.5) and for treatment decision making. An independent central review and/or analysis of the tumor assessments may be performed if deemed necessary.

The methods for assessment and recording are specified in the imaging charter and imaging manual.

The coded medical images will be used primarily for analysis as described in this protocol; however, the images may also be used for the development and evaluation of new analysis

methods directly related to the area of research that this study covers. Incidental findings are beyond the scope of the central imaging vendor. If an investigator/radiologist recognizes any incidental finding in the images during the course of conducting the clinical trial, the investigator should follow up as part of his/her duty of care to ensure the safety and well-being of the participant, as well as compliance with local ethical regulations.

Baseline imaging assessments

CT/MRI imaging assessments for RECIST v1.1 will be performed at screening/baseline within 35 days before the start of ^{225}Ac-DWJ155 treatment (Day -35 to Day -21 prior ^{68}Ga-DWJ155 imaging and prior to C1D1).

Any imaging assessments already completed during the regular workup of the participant within 35 days prior to the start of ^{225}Ac-DWJ155 treatment, including before signing the main study ICF, can be considered as the baseline images for this study and should be submitted to the designated imaging CRO for quality control and central review. Any imaging assessments obtained after ^{68}Ga-DWJ155 imaging cannot be considered baseline images.

Any potential measurable lesion that has been previously treated with radiotherapy should be considered as a non-measurable lesion. However, if a lesion previously treated with radiotherapy has clearly progressed since the radiotherapy, it can be considered as a measurable lesion.

If a participant is known to have a contraindication to CT intravenous (i.v.) contrast media or develops a contraindication during the trial, a non-contrast CT of the chest (MRI is not recommended due to respiratory artifacts; however, if CT is not feasible per local regulations, MRI can be performed instead) plus a contrast-enhanced MRI (if possible) of the abdomen and pelvis should be performed.

If brain metastases are suspected at baseline, brain MRI or CT should be completed. Contrast enhanced brain MRI is preferred, however, if MRI contrast is contraindicated, then MRI without contrast or CT with/without contrast is acceptable.

If clinically indicated, a whole-body bone scan should be performed per institutional standard of care [e.g., Tc-99m bone scan, whole body bone MRI, FDG PET or ^{18}F-sodium fluoride (NaF) PET]. Localized CT, MRI, or X-rays should be acquired and submitted to imaging CRO for all skeletal lesions identified on the screening whole body scan, which are not visible on the chest, abdomen and pelvis CT/MRI.

If clinically indicated, CT or MRI of other areas (e.g., neck) of disease as appropriate should be performed and submitted to imaging CRO.

If skin lesions are present at screening, color photography should be acquired and submitted to imaging CRO using a digital camera in clear focus, including a scale/ruler, in such a way that the size of the lesion(s) can be determined from the photograph.

Chest X-rays and ultrasound should not be used to measure tumor lesions.

Post-baseline imaging assessments

Imaging assessments as described in Table 8-1 should be performed at the time points specified using the same imaging modality used at baseline, irrespective of study treatment interruption or actual dosing (see Section 1.3). Imaging assessments for response evaluation will be performed during the on-treatment period every 6 weeks (for 6-week schedule) or every 8 weeks (for 4- or 8-week schedule) after the first administration of study treatment, and every 12 weeks after the last administration of study treatment during the follow-up period until disease progression, death, lost to follow-up, withdrawal of consent/opposition to use data/biological samples. Imaging assessments should be scheduled using C1D1 as the reference date (not the date of the previous tumor assessment) and should be respected regardless of whether study treatment is temporarily withheld, or unscheduled assessments are performed.

Additional imaging assessments may be performed and submitted to imaging CRO at any time during the study at the investigator's discretion to support the efficacy evaluations for a participant, as clinically indicated. Clinical suspicion of disease progression at any time requires a physical examination and imaging assessments to be performed promptly rather than waiting for the next scheduled imaging assessment.

Each lesion that is measured at baseline must be measured by the same method (either the same imaging method or by photography, including a metric ruler) and when possible, the same local radiologist/physician throughout the study so that the comparison is consistent. If an off-schedule imaging assessment is performed because progression is suspected, subsequent imaging assessments should be performed in accordance with the original imaging schedule.

The CT portion of a combined PET/CT or SPECT/CT scan may be used only if the CT is of diagnostic quality, including the utilization of i.v. contrast media. The low-dose CT used for attenuation correction of PET/CT or SPECT/CT scans should not be used for performing tumor assessments.

At the discretion of the investigators, FDG-PET scans may be performed to verify progressive disease but will not be used on their own to determine radiographic progression.

All study imaging (including any off-schedule imaging studies) should be submitted to the designated imaging CRO for quality control. A central review may be performed if deemed necessary.

8.3.2 ^{68}Ga-DWJ155 PET and ^{225}Ac-DWJ155 SPECT imaging

^{68}Ga-DWJ155 will be administered twice to each participant. A safety follow-up will be performed as outlined in Section 1.3.

Baseline ^{68}Ga-DWJ155 PET/CT or PET/MRI imaging assessments will be performed in all participants at screening as described in Table 1-3, and a second ^{68}Ga-DWJ155 PET/CT or PET/MRI scan will also be performed at the EoT visit. The follow-up ^{68}Ga-DWJ155 PET imaging will be conducted in the same manner as the baseline PET imaging.

A PET scan after ^{68}Ga-DWJ155 administration will be considered positive if at least one measurable lesion (per RECIST v1.1) co-localized with anatomical imaging shows ^{68}Ga-DWJ155 PET uptake higher than the surrounding tissues based on local investigator assessment.

For ^{68}Ga-DWJ155 PET/CT or PET/MRI scan acquisition details (including imaging time points and scan duration) please refer to the Imaging Manual. All PET/CT and PET/MRI scans should be submitted to the designated imaging CRO and will be analyzed centrally based on the study specific imaging charter.

For participants with disease sites in the urinary bladder or pelvis, investigators may consider placing a urinary catheter prior to PET imaging with ^{68}Ga-DWJ155 and/or SPECT imaging with ^{225}Ac-DWJ155 for improved visualization of tumor. In such cases, investigators may elect to flush the bladder with normal saline per institutional guidelines to reduce residual urinary background signal.

Participants will be followed up for safety as described in Section 7.1.1 and as outlined in Section 1.3.

The correlation between ^{68}Ga-DWJ155 PET imaging data and ^{225}Ac-DWJ155 SPECT data (if available) will be investigated to understand whether a higher uptake in tumor based on imaging is correlated with higher absorbed radiation dose in tumor on an ongoing basis to assess potential differences in biodistribution and tumor uptake. While there is no plan to optimize PET image quality during the study, good quality images are required for both qualitative and quantitative assessments.

^{225}Ac-DWJ155 SPECT/CT images will be used for dosimetry calculations and to investigate ^{225}Ac-DWJ155 uptake in the enrolled cancer types. Imaging acquisition details for ^{225}Ac-DWJ155 SPECT/CT will be described in the Dosimetry Site Manual. All SPECT/CT scans should be submitted to the designated imaging CRO and will be analyzed centrally based on the study specific imaging charter.

For details on SPECT imaging that will be used for ^{225}Ac-DWJ155 dosimetry assessments, please refer to Section 8.7 and Table 8-4.

8.3.3 Appropriateness of efficacy assessments

The efficacy assessments selected are standard for these tumor indications. Additional information can be found in Section 10.5.

8.4 Safety assessments

Safety assessments are specified below with Section 1.3 detailing when each assessment is to be performed.

For details on AE collection and reporting, refer to Section 8.10.

As per Section 4.6, during a public health emergency as declared by local or regional authorities i.e. pandemic, epidemic or natural disaster, that limits or prevents on-site study visits, regular phone or virtual calls can occur for safety monitoring and discussion of the participant´s health status until it is safe for the participant to visit the site again.

ECOG Performance status scale will be used as described in Table 8-2.

Table 8-2 ECOG Performance status scale

Grade	ECOG
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair
5	Dead

8.4.1 Physical examinations

Physical examinations will be performed according to Section 1.3. At Screening and prior to treatment on C1D1, a complete physical examination will be performed and will include the examination of general appearance, skin, neck (including thyroid), eyes, ears, nose, throat, lungs, heart, abdomen, back, lymph nodes, and extremities.

After C1D1 onwards, a short physical examination will be performed (it will include the examination of general appearance and vital signs).

Clinically relevant findings that were present prior to signing informed consent must be recorded on the appropriate eCRF that captures medical history. Significant new findings that begin or worsen after informed consent which meet the definition of an AE must be recorded as an AE.

8.4.2 Vital signs

Vital signs include blood pressure (BP) (supine position preferred when ECG is collected), pulse measurement, and body temperature.

Height will be measured at screening.

Body weight (in indoor clothing, but without shoes) will be measured at screening and at subsequent time points as specified in Section 1.3.

For Japan only: oxygen saturation (SpO2) will be measured by pulse oximetry for participants enrolled in Japan every time vital signs are performed according to the visit assessment scheduled in Table 1-4, Table 1-5, and Table 1-6. The results of SpO2 will be recorded only in the source documentation.

8.4.3 Electrocardiograms

Triplicate 12-lead ECGs will be performed at each ECG collection time point as specified in Section 1.3. The ECG will be performed after the participant has been resting (supine) for approximately 10 minutes prior to each ECG collection time point. Individual ECGs should be recorded approximately 2 minutes apart. The preferred sequence of cardiovascular data

collection is ECG collection first, followed by vital signs, blood sampling, and any remaining assessments for that visit. The QTcF must be used for clinical decisions (e.g., to assess eligibility). The investigator must calculate QTcF if it is not auto-calculated by the ECG machine. The mean QTcF value for each time point will be calculated from the triplicate ECGs for each participant. ECGs will be collected for all participants according to Table 8-3. During dosage escalation, additional ECGs will be collected in all participants undergoing dosimetry/PK assessments (refer to Section 8.7) and as outlined in Table 8-4. In these participants, the blood sample for PK should be collected closely after the ECG collection is completed, and before any blood biomarker sample.

ECGs will be locally collected and reviewed as well as being centrally reviewed. Interpretation of the tracing must be made by a qualified physician and documented on the ECG. The original ECGs on non-heat-sensitive paper, appropriately signed, must be archived at the study site. Each ECG tracing should be labeled with the study number, participant initials (where regulations permit), participant number, date, and kept in the source documents at the study site. Instructions for collection and transmission of ECGs to the central ECG laboratory will be provided in the ECG manual.

Participant's eligibility for the study should be assessed by the investigator based on the local interpretation of results from the triplicate ECG recordings performed on the machine provided. Clinically significant abnormalities present at screening should be reported on the appropriate eCRF. Clinically significant findings must be discussed with Novartis prior to enrolling the participant in the study. New or worsened clinically significant findings occurring after first administration of ^{68}Ga-DWJ155 must be recorded as AEs.

Additional, unscheduled, safety ECGs may be repeated at the discretion of the investigator at any time during the study as clinically indicated. For any ECGs with participant safety concerns, two additional ECGs must be performed to confirm the safety finding. ECG safety monitoring, or a review process, should be in place for clinically significant ECG findings at baseline before administration of study treatment and during the study.

Table 8-3 Central ECG collection plan (all participants)

Period / cycle	Day	Time	ECG Type
Screening	-35 to ^{68}Ga-DWJ155 imaging	Anytime during the screening period	12 Lead Triplicate
1	1	Pre-dose (baseline ECG)*	12 Lead Triplicate
1	1	End of ^{225}Ac-DWJ155 administration (+10 minutes)	12 Lead Triplicate
1	1	1 hour (±10 minutes) post-EOI*	12 Lead Triplicate
1	1	4 hour (±15 minutes) post-EOI*	12 Lead Triplicate
2 and subsequent cycles	1	End of administration (+10 min)	12 Lead Triplicate
EoT		Anytime	12 Lead Triplicate

Period / cycle	Day	Time	ECG Type
* For participants also undergoing dosimetry/PK assessments EOI=end-of-infusion; If clinically indicated as per the investigator's judgment (e.g., if significant ECG abnormalities are observed at the end of the ^{225}Ac-DWJ155 administration), the collection of ECG assessments at additional post-dose time points coinciding with PK sampling time points as per Table 8-4 should be considered.			

8.4.4 Clinical safety laboratory tests

All laboratory parameters assessed for safety purposes will be evaluated locally. Refer to Table 10-1 for a summary of the parameters to be evaluated according to Section 1.3. On days of dosing, samples for these parameters will be collected prior to the administration of investigational drugs. Results from locally performed labs collected on the day of administration should be reviewed prior to administration of the investigational drug.

More frequent evaluations may be performed at the investigator's discretion if medically indicated; results should be recorded as unscheduled laboratory assessments.

During treatment and safety follow-up periods, for participants who have CTCAE grade 2 or higher neutropenia or thrombocytopenia as well as participants who have CTCAE grade 3 or higher anemia, the relevant hematology assessment must be performed as indicated in Table 6-4.

Novartis will be provided with a copy of the laboratory certification and tabulation of the normal ranges for each parameter required. In addition, if at any time a participant has laboratory parameters obtained from a different outside laboratory, Novartis must be provided with a copy of the certification and a tabulation of the normal ranges for that laboratory.

Clinically significant abnormalities must be recorded as either medical history/current medical conditions or AEs, as appropriate. All abnormal lab results must be evaluated for criteria defining an AE and reported as such if the criteria are met.

All abnormal lab results must be evaluated for criteria defining an AE and reported as such if the criteria are met. For those lab AEs, repeated evaluations are mandatory until normalization of the result(s) or until the result is no longer considered to be clinically significant.

8.4.5 Pregnancy testing

All women of childbearing potential will have serum pregnancy testing within 72 hours before any administration of ^{68}Ga-DWJ155 or ^{225}Ac-DWJ155, and at the EoT visit. Monthly urine pregnancy test will then be performed and reported to site for 8 months after the last ^{225}Ac-DWJ155 administration. Additional pregnancy testing might be performed if requested by local requirements.

Women of childbearing potential are defined as women physiologically capable of becoming pregnant from menarche until becoming post-menopausal unless they have had surgical bilateral oophorectomy (with or without hysterectomy), total hysterectomy or bilateral salpingectomy at least six weeks before taking study treatment. In the case of oophorectomy alone, the reproductive status of the woman needs to have been confirmed by follow-up hormone level assessment. A postmenopausal state is defined as no menses for 12 months without an alternative medical cause and an appropriate clinical profile.

Medical documentation of bilateral oophorectomy, hysterectomy or bilateral salpingectomy must be retained as source documents.

In absence of the medical documentation of these procedures, or if the post-menopausal status is not clear, the investigator should use medical judgment to appropriately evaluate the fertility state of the woman and document it in the source document.

In case of a positive pregnancy test, the investigator must be contacted immediately and the reporting requirements as per Section 8.10.4 must be followed. If a participant becomes pregnant, investigational drug must be stopped immediately. If a pregnancy test is positive, but the participant is thought not to be pregnant, investigational drug should be stopped until it is determined that the test was falsely positive and pregnancy is excluded.

8.4.6 Cardiac imaging - Trans-Thoracic Echocardiography (TTE)

Trans-thoracic echocardiography (TTE) will be used to assess LVEF according to the schedule described in Section 1.3. If a site is unable to perform TTE, MUGA scan is acceptable. Additional assessments may be performed if clinically indicated. The same assessment method (TTE or MUGA) is to be used for all cardiac imaging examinations throughout the study for each participant. When possible, the same cardiologist or radiologist should read and report the outcome to minimize variability in results.

8.4.7 Chest X-ray (Japan only)

For Japan only, a 2-view chest X-ray will be performed according to the visit evaluation schedules in Section 1.3.

8.4.8 Appropriateness of safety measurements

The safety assessments selected are standard for these indications/participant populations.

8.5 Additional assessments

Please refer to Table 1-7 and Section 8.7 for additional assessments for participants undergoing dosimetry assessments.

8.6 Immunogenicity assessments

No applicable.

8.7 Pharmacokinetics and dosimetry

This section describes the assessments related to blood and urine PK and SPECT/CT imaging for dosimetry.

In the dosage escalation part of the study, blood PK and dosimetry assessments will be conducted in approximately 3 participants in each cohort (preferably the first 3 participants). In the earlier dosage levels (DL1 and DL2), where injected activities per cycle $\leq$ 4 MBq are planned, imaging-based dosimetry in the first participant may be reviewed before additional participants undergo dosimetry, to confirm feasibility of the analysis. If the analysis is not feasible at DL1 and/or DL2, subsequent participants at that dosage level may be exempt from

dosimetry. From dosage levels where injected activity per cycle is ≥ 6 MBq, dosimetry will be acquired in at least 3 participants at each new dosage level.

In the event that fewer than 3 participants in a dosage level (DL3 and higher) in the dosage escalation part of the study contribute to dosimetry (or dosimetry cannot be evaluated, e.g., due to errors during the imaging), dosimetry data will be collected from further participants enrolled beyond the first 3 participants in escalation or in additional cohorts.

Urine PK will be collected in up to six participants in total, at dedicated sites in selected cohorts and relevant dosage levels (e.g., DL2 or higher) in the dosage escalation part of the study or in the expansion part of the study

The timing of PK and/or dosimetry assessments may be modified based on emerging data, in order to better define the PK profile and/or dosimetry evaluations.

The extent of PK and dosimetry collection and assessment schedule in the expansion part of the study will be defined in a future amendment, based on data generated during the dosage escalation part.

Refer to dedicated manuals for detailed instructions regarding PK sample collection, processing and shipment. Detailed instructions on imaging procedures for dosimetry and dosimetry methodology are provided in the Dosimetry Manual and Dosimetry Charter, respectively.

Residual PK samples may be used for other exploratory PK or PK/pharmacodynamics analysis at the sponsor's discretion. This could include using leftover samples for protein binding analysis, metabolite profiling or exploratory biomarker analysis. Given the exploratory nature of the work, the analytical methods used for those assessments, may not be validated.

There may be circumstances when a decision is made to stop a collection, or not perform or discontinue an analysis due to either practical or strategic decisions. Therefore, depending on the results obtained during the study, sample collection/analysis may be omitted at the discretion of Novartis.

8.7.1 Pharmacokinetic blood collection and handling

Blood samples will be collected in Cycle 1 by either direct venipuncture or an indwelling cannula inserted in a forearm vein/ foot vein (in case blood sampling is required while the participant is in scanner) from those participants in the escalation and expansion part in which dosimetry is acquired, or in participants at selected sites who undergo the urine PK collection schedule (see Section 8.7.2).

Blood will be drawn from the arm contralateral to the one in which ^{225}Ac-DWJ155 has been injected, whenever possible.

At specified time points described in Table 8-4 in Cycle 1, blood samples will be collected into tubes containing heparin as an anticoagulant and gently inverted several times to thoroughly mix the anticoagulant.

All post-dose PK blood sampling times are relative to the time of end of administration on C1D1. On days and timepoints coinciding with ECG assessments, the ECG should be done prior to drawing the PK sample, and the PK sample should be drawn before any blood biomarker sample.

After collection, two aliquots will be prepared from each PK blood sample, for PK and radiometabolite analysis, respectively.

Refer to the laboratory manual for detailed instructions on collection, handling, and shipment of blood and plasma PK samples.

All data including but not limited to actual date and time of blood PK sampling must be recorded in the appropriate blood collection eCRF page.

Table 8-4 Pharmacokinetic blood collection log

Cycle	Day	Scheduled Time Point
1	1	Pre-dose [a]
1	1	End of infusion (EOI)[b]
1	1	0.5 h (±10 min) post-EOI
1	1	1 h (±10 min) post-EOI
1	1	2 h (±15 min) post-EOI
1	1	4 h (±30 min) post-EOI
1	1	6 h (±30 min) post-EOI
1	2	24 h (±2 h) post-EOI
1	3	48 h (±4 h) post-EOI
1	4-5[c]	72-96 h (±6 h) post-EOI

[a] To be collected on the same day of the infusion, before start of infusion

[b] To be collected within 10 min of end of infusion, immediately after EOI ECG assessment

[c] Not applicable for participants in DL1

8.7.2 Pharmacokinetic urine collection and handling

Each participant should void his/her bladder immediately before the start of infusion.

Up to six participants in selected cohorts at relevant dosage levels (e.g., DL2 or higher) in the dosage escalation part of the study or in the expansion part of the study, and at dedicated sites, will undergo full urine collection over 72 hrs. In addition, should dosimetry assessments indicate ^{225}Ac-DWJ155 uptake in the gastrointestinal tract, urine PK assessments may be performed in an earlier cohort in at least 1 participant, per documented communication between Investigators and Novartis.

Participants who undergo PK urine collection and are not undergoing PK and SPECT/CT assessments for dosimetry analysis, will undergo PK blood sampling according to Table 8-4.

All urine eliminated during the first 72 hrs following the end of infusion on Cycle 1 Day 1 will be collected, dividing the collection according to the time intervals specified in Table 8-5. During each sampling interval, the participant should void his/her bladder in a dedicated collection container for that sampling interval. Participants are recommended to regularly hydrate orally.

The content of each urine fraction container should be gently mixed, and two aliquots of urine will be withdrawn from each collection container for the specified sampling intervals, for PK and radiometabolite analysis, respectively (see Section 8.7.3).

Refer to laboratory manual for detailed instructions on collection, handling and shipment of urine PK samples.

All data including but not limited to actual date and time of urine sampling must be recorded in the appropriate urine collection eCRF page.

Table 8-5 Urine Collection Log

Cycle[a]	Day	Scheduled Time[b]
1	1	0 h (start of infusion)-6 h (±30 min)
1	1-2	6 (±30 min)-24 h (±2h)
1	2-3	24 (±2h)-48 h (±4h)
1	3-4	48 (±4h)-72 h (±4h) [c]

[a] Up to 6 participants in selected cohorts at relevant dosage levels and at dedicated sites. The assignment of a participant will be coordinated by Novartis and it will be communicated to investigators at time of individual participant enrollment.

[b] Scheduled collection times to be calculated from the EOI unless otherwise noted

[c] If the first 3 participants who contribute to urine PK show negligible radioactivity (≤ 5% of IA) detected in the 48-72h collection interval, this collection interval may be omitted in the remaining participants. Novartis will notify investigators accordingly, at time of participant enrollment.

8.7.3 Analytical method

Radioactivity in blood and urine will be measured with an appropriately certified gamma counting system, calibrated with a certified source of Ac-225 of known activity counted in the same geometry as that of the biological samples (i.e., appropriate volume, in an radioimmunoassay (RIA) tube suitable for the gamma counter) according to methodology described in a separate document.

Exploratory radiometabolite assessments (detection) on radioactive plasma and urine samples will be performed using HPLC with radioactivity detection, by a qualified vendor. If at any time point the amount of metabolite data is deemed sufficient, sample collection, shipment and analysis of these data may be discontinued upon the sponsor's request.

8.7.4 Dosimetry

Only calibrated and qualified SPECT/CT scanner(s) can be used to perform ^{225}Ac-DWJ155 SPECT/CT imaging. SPECT/CT acquisition will be standardized across sites.

SPECT/CT images will be acquired to assess dosimetry to critical organs and to selected tumor lesions.

Participants undergoing dosimetry assessments should void their bladder just prior to each SPECT/CT assessment.

Participants assigned to PK/dosimetry assessments in the dosage escalation part of the study will undergo assessments after the first ^{225}Ac-DWJ155 administration in Cycle 1, i.e., up to 5 serial SPECT/CT scans in Cycle 1, at the time points specified in Table 8-6, to construct the time activity curves. Participants in DL1 will undergo scans up to 48 hours post-administration, participants in DL2 will undergo scans up to 72 hours post-administration, and all remaining participants at higher DLs will undergo scans for the full schedule. Pending emerging data, the timing of each scan relative to ^{225}Ac-DWJ155 administration may be adjusted to better capture

the uptake, retention and washout kinetics in the key critical organs and/or the number of imaging time-points may also be reduced. The timing (timepoint with smaller time window) of each acquisition will be communicated to investigators at the time of enrollment allocation.

As Ac-225 dosimetry requires longer imaging times with current SPECT/CT technology (typically ≥ 30 min per field of view (FOV)) than beta-emitters, a limited number of FOVs can be acquired to reduce overall scan time. The required anatomic coverage of the SPECT scan for ^{225}Ac-DW155 dosimetry will be optimized based on the results from the initial participants undergoing dosimetry and will be such that key dose-limiting organ(s) are within the field-of-view(s). Additional fields-of-view may be acquired to capture tumor lesions. For SPECT/CT scan acquisition details refer to the Dosimetry Site Manual. All SPECT/CT scans and raw projection data should be submitted to the designated imaging CRO and will be analyzed centrally based on the study specific Dosimetry Charter.

The dosimetry assessments will be omitted in case ^{225}Ac-DWJ155 administration has not occurred.

Table 8-6 SPECT/CT Imaging Collection Table

Cycle	Day after administration[a]	Provisional time window
1	1	2 h - 4 h ±1h post-EOI
1	2	24 h ±4h post-EOI
1	3	48 h ±4h post-EOI
1	4-5[b]	72 h - 96 h ±6h post-EOI
1	8-15[b]	168 h - 312 h ±6h post-EOI

EOI=end-of-infusion

[a] Participants in DL1 will undergo scans up to 48 hours, participants in DL2 will undergo scans up to 72-96 hours. The remaining participants will undergo scans for the full schedule.

[b] A precise time window of each acquisition will be informed by emerging data and will be communicated to investigators at the time of individual participant enrollment.

The time windows for SPECT/CT imaging as shown are recommended. However, if an assessment cannot be done within specified windows, it can be done at the closest feasible time, and the actual time must be properly recorded in the eCRF.

8.7.5 Dosimetry with respect to local regulations

Should local radioprotection or other regulations require additional information, necessary dosimetry assessments can be performed in addition to the schedule in Section 8.7.1 and Section 8.7.4. These may include additional single time point (e.g., 24-48 h post-infusion) SPECT/CT with the required fields of view (verification scans) for qualitative distribution assessment of radioactivity, at subsequent cycles.

For further scan acquisition details refer to the Dosimetry Site Manual.

8.8 Biomarkers

Biomarker analyses will be used to investigate the effect of ^{225}Ac-DWJ155 in participants with solid tumors at the molecular and cellular level as well as to determine how changes in the markers may relate to exposure and clinical outcomes. In addition, potential predictive markers of efficacy, as well as mechanisms of resistance to ^{225}Ac-DWJ155 may be explored.

While the goal of the biomarker assessments is to provide supportive data for the clinical trial, there may be circumstances when a decision is made to stop a collection, or not perform or discontinue an analysis due to either practical or strategic reasons (e.g., inadequate sample number, issues related to the quality of the sample or issues related to the assay that preclude analysis, impossibility to perform correlative analyses, enrollment halt etc.). Therefore, depending on the results obtained during the study, sample collection analysis may be omitted at the discretion of Novartis. The samples collected as described in Table 8-7, may be used to meet any of the biomarker related trial objectives. Additionally, other related biomarkers may also be analyzed from the samples listed in Table 8-7, depending on sample availability, resources, participant's outcomes and as new scientific evidence or alternative methods become available.

The sample collection information must be entered on the appropriate sample collection eCRF page(s) and requisition form(s). Detailed instructions for the collection, processing, and shipment of biomarker samples are outlined in the laboratory manual for the study. Sample(s) should be collected at the visit/time point(s) defined in the biomarker table; Table 8-7.

Table 8-7 Biomarker sample collection plan

Sample Type	Visit/ Time point	Approx. volume	Marker	Purpose
Dosage escalation: Archival tumor sample (preferred) or newly obtained tumor biopsy Dosage expansion: Newly obtained tumor biopsy*	Screening period (post ^{68}Ga-DWJ155 imaging)	**Archival tumor**: Block (preferred) or a minimum of 25 newly cut sections (4 µm thickness) **Newly obtained tumor:** 3-6 passes of a core needle biopsy	HER2 expression and expression of other markers such as ER and PR Whole transcriptome analysis; expression of immune and cancer related genes Cancer-related genes by next generation DNA sequencing	Correlation of target distribution with imaging, dosimetry and efficacy Retrospective confirmation of locally assessed HER2 status Assess potential predictive markers of response & mechanisms of resistance
Blood sample for soluble factors	C1D1 Pre-dose C1D15 C1D29*** C2D1 Pre-dose C3D1 Pre-dose End of treatment	3.5 mL	Soluble HER2 Soluble protein profiling for research**	Assess potential predictive markers of response, recurrence & resistance

Sample Type	Visit/ Time point	Approx. volume	Marker	Purpose
Blood sample for cfDNA	C1D1 Pre-dose C1D15 C1D29*** C2D1 Pre-dose C3D1 Pre-dose End of treatment Every 12 weeks after the last dose of study treatment during disease progression FU	10 mL	DNA sequencing in cell free DNA	Assessing cfDNA for potential prognostic markers and/or predictive markers of efficacy and/or resistance and correlating cfDNA abundance and features with measures of clinical efficacy
Urine sample for renal biomarkers	Pre-dose collection at every cycle (including cycle 1)	10 mL	Novel renal biomarkers (e.g. CLU, Cystatin-C, Kidney Injury Molecule-1 (KIM-1), Neutrophil Gelatinase-Associated Lipocalin (NGAL), osteopontin, uCr, nephrin, podocin, podocalyxin, retinol-binding protein 4, vascular-cell adhesion molecule 1, matrix metalloproteinase 3, alpha-1-macroglobulin)	Identify potential renal damage and understand the underlying mechanism of potential nephrotoxicity. Provide correlation with administered activity and renal dosimetry, potential AEs and standard measures of renal function.
*Newly obtained biopsy can be substituted with an archival tumor sample collected within 6 months prior to screening with no intervening treatment and a copy of the corresponding de-identified pathology report. **Soluble protein profiling will use remnant sample from blood for soluble factors sample collection. ***C1D29 visit only applicable to 6-week and 8-week schedules.				

Biomarker assessments in tissue samples

During dosage escalation, the most recent archival tumor sample and a copy of the corresponding de-identified pathology report is requested at the post ^{68}Ga-DWJ155 imaging visit; if unavailable a newly obtained tumor biopsy may be collected during screening period (post ^{68}Ga-DWJ155 PET imaging). During dosage expansion, collection of a newly obtained biopsy is mandatory for participants who will receive ^{225}Ac-DWJ155 treatment. This sample can be substituted with a recent archival tumor sample and a copy of the corresponding de-identified pathology report, provided the archival tumor sample was collected within 6 months prior to the screening visit with no intervening treatment. The archival tumor samples

must meet specifications as described in Table 8-7, otherwise a newly obtained tumor biopsy must be collected. If a core needle biopsy is performed, 3-6 tumor biopsy passes are requested. In the event that an inadequate tumor sample is received following a new biopsy procedure (e.g., found to have low tumor content), an archival tumor sample may be requested if available to allow for the analysis described in Table 8-7. Exceptions may be granted following documented discussion with Novartis.

Archival and newly obtained tumor samples may be used to examine HER2 expression for potential correlation with ^{68}Ga-DWJ155 imaging and uptake, and to compare centrally assessed HER2 expression with locally assessed HER2 status. HER2 expression levels as potential predictors of response may also be explored. Next generation DNA sequencing, whole transcriptome analysis and additional protein-based analyses may also be performed on these samples. Additionally, other predictive markers may be analyzed on these samples.

Biomarker assessments in blood samples

Peripheral blood samples will be collected pre-treatment, during treatment, at end of treatment and during disease progression follow-up as indicated in Table 8-7.

Collection of blood for soluble factors will allow for the assessment of target and pathway modulation by measuring HER2, as well as assessment of additional soluble markers to assess the effect of the study treatment before, during, and at end of treatment. Furthermore, soluble factors like HER2 may be correlated with response to ^{225}Ac-DWJ155. Analysis of additional soluble markers is for research purposes and data from this analysis will not be entered into the clinical database or the clinical study report (CSR).

Blood will be collected at C1D1 pre-dose, C1D15, C1D29 (6-week and 8-week schedules), C2D1 pre-dose, C3D1 pre-dose, end of treatment and every 12 weeks during post-end of treatment disease progression follow-up until participant progresses to allow for sequence analysis of cfDNA. This analysis will interrogate potential prognostic markers and/or predictive markers of efficacy and/or resistance and correlate cfDNA abundance and features with measures of clinical efficacy.

Biomarker assessments in urine

Urine samples will be collected pre- and during treatment as indicated in Table 8-7. Novel renal biomarkers may be analyzed to investigate the biological effects of ^{225}Ac-DWJ155 on renal functions for exploratory purposes only. This analysis may include CLU, Cystatin-C, KIM-1, NGAL, osteopontin, nephrin, podocin, podocalyxin, retinol-binding protein 4, vascular-cell adhesion molecule 1, matrix metalloproteinase 3 and alpha-1-macroglobulin.

The list of biomarkers to be tested may be adapted in case additional promising renal toxicity biomarkers will receive support from health authorities for their exploratory use in clinical trials. The data from this analysis is for research purposes and will not be used for participant management decisions or any regulatory decision-making. The results will be used to explore correlation between expression change of the biomarkers and potential clinical manifestation of renal injury indicated by traditional laboratory assessments of renal function.

8.9 Health economics OR Medical resource utilization and health economics

Not applicable.

8.10 Adverse events (AEs), serious adverse events (SAEs), and other safety reporting

The Investigator and any qualified designees are responsible for managing the safety of individual participants and for appropriately treating and/or referring the participant to appropriate care and follow up for all medical events occurring in participants. They are also responsible for detecting, recording and reporting medical events that meet the definition of an AE or SAE.

Information about adverse drug reactions and how to manage them may be found in this protocol, the IB, or equivalent documentation. Information about adverse drug reactions can also be found in the product information for marketed products.

Novartis qualified medical personnel will be readily available to advise on trial-related medical questions or problems.

8.10.1 Definition and recording of adverse events

Definition of AEs

In accordance with the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines, an AE is "any untoward medical occurrence in a participant or clinical investigation participant administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment".

An AE can therefore be any of the following untoward medical occurrences temporally associated with the use of study treatment (i.e., during and/or after administration of study treatment), whether or not considered related to study treatment;

- Any unfavorable and unintended sign(s), symptom(s) or medical condition,
- Any abnormal laboratory values or other test results (e.g., ECG, X-ray) that fulfill at least one of the following criteria:
- They are associated with clinical signs or symptoms,
- They are considered clinically significant,
- They require therapy,
- They require a change in study treatment, including discontinuation from study treatment.
- Any new disease or exacerbation of an existing disease (a worsening in the character, frequency, or severity of a known medical condition or the underlying medical condition that is under study),
- Any recurrence of an intermittent medical condition (e.g., headache) not present at screening.

In addition, any medical occurrence related to a protocol-mandated requirement after informed consent (e.g., intervention or procedure or assessment), are considered AEs. This includes, but is not limited to:

- A pharmaceutical/medicinal product that is used in this study as described in this protocol (e.g., AxMP used as background therapy, rescue medication, diagnostic or challenge agent, in a run-in phase or to assess a study endpoint),
- Wash-out or change of dose of a previous treatment,
- A procedure performed such as biopsies or medical device use

Collecting and assessing AEs

The occurrence of AEs must be sought by non-directive questioning of the participant at each visit during the study. To prevent reporting bias, participants should not be questioned regarding the specific occurrence of one or more AEs. AEs also may be detected when they are volunteered by the participant during or between visits or through physical examination findings, laboratory test findings, or other assessments.

Clinically significant abnormal laboratory values or test results must be identified through a review of values outside of normal ranges/clinically notable ranges, significant changes from baseline or the previous visit, or values considered to be non-typical in participant with the underlying disease. Alert ranges for laboratory and other test abnormalities are included in Section 8.10.3.

Adverse events may be reported by the participant or, when appropriate, by a caregiver, surrogate, or the participant's legally authorized representative (if allowed according to local law).

The investigator will attempt to establish a diagnosis of the medical occurrence (including lab abnormalities that constitute AEs) based on signs, symptoms, and/or other clinical information. Whenever possible, the diagnosis (not the individual signs/symptoms) will be documented as the AE. However, if the signs and/or symptoms cannot be medically characterized as a single diagnosis or syndrome at the time of reporting, each individual event should be recorded.

Medical occurrences that begin before the start of study treatment but after obtaining informed consent and are not related to a protocol-mandated intervention/procedure/assessment will be recorded as current medical conditions on the Medical History CRF.

For each AE, the investigator must assess:

1. The Common Toxicity Criteria (CTC)

AEs will be assessed and graded according to the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0.

2. The causality

The investigator is obligated to assess the relationship between any treatment used in the study (^{68}Ga-DWJ155 or ^{225}Ac-DWJ155) and each occurrence of each AE. The investigator will use clinical judgment at the time to determine the relationship. A reasonable possibility of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out. Alternative causes, such as

underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study intervention administration, will be considered and investigated.

If the event is due to lack of efficacy or progression of underlying illness (i.e., progression of the study indication) the assessment of causality will usually be 'Not suspected.' The rationale for this guidance is that the symptoms of a lack of efficacy or progression of underlying illness are not caused by the trial drug, they happen in spite of its administration and/or both lack of efficacy and progression of underlying disease can only be evaluated meaningfully by an analysis of cohorts, not on a single participant.

For causality assessment, the investigator will also consult the IB and/or product information, for marketed products.

The causality assessment is one of the criteria used when determining regulatory reporting requirements.

Handling of AEs

All adverse events must be treated appropriately. More information about how to manage AEs may be found in this protocol and in the IB. Information about adverse drug reactions can also be found in product information for marketed products.

Once an AE is detected, the Investigator must pro-actively follow up the participant, until resolution of the AE, or until it is judged to be not recovered/not resolved (e.g., continuing at the end of the study), or until stabilization, or until the participant is lost to follow-up or withdraws their consent. Any change in severity or suspected relationship to study treatment must be assessed at each visit (or more frequently, if necessary).

Progression of malignancy (including fatal outcomes), if documented by use of appropriate method (for example, per RECIST criteria for solid tumors or per Cheson's guidelines for hematological malignancies), should not be reported as an adverse/serious adverse event, except if the Investigator considers that progression of malignancy is related to study treatment.

Adverse events separate from the progression of malignancy (for example deep vein thrombosis at the time of progression or hemoptysis concurrent with finding of disease progression) will be reported per usual guidelines used for such events with proper attribution regarding causality to the study treatment.

Timeframe of recording of AEs

For participants who receive ^{68}Ga-DWJ155 but do not proceed with ^{225}Ac-DWJ155 treatment, AE recording should be continued until 3 days post ^{68}Ga-DWJ155 administration. For participants who receive ^{225}Ac-DWJ155 treatment, AE recording should be continued for at least 180 days following the last administration of ^{225}Ac-DWJ155. Following the last ^{225}Ac-DWJ155 administration, all AEs, irrespective of causality will be collected for 28 days (for participants under 4-week schedule), 42 days (for participants under 6-week schedule) or 56 days (for participants under 8-week schedule), after which only AEs suspected to be related to the study treatment will be collected in the Adverse Events CRF. For SAEs occurring during follow-up, refer to Section 8.10.3.

If a suspicion that medical occurrence could be related to study treatment (and/or interaction with study treatment) cannot be ruled out, the reporting rules for AEs related to study treatment apply.

In addition, any medical occurrence related to a trial-mandated requirement (e.g., intervention or procedure or assessment) will be collected as AEs from signing of informed consent.

8.10.2 Definition and recording of serious adverse events

An SAE is defined as any untoward medical occurrence that, at any dosage, meets any one of the following criteria:

- Results in death
- Is life-threatening

Life-threatening in the context of a SAE refers to a medical occurrence in which the participant was at risk of death at the time of the reaction; it does not refer to a medical occurrence that hypothetically might have caused death if it were more severe (please refer to the ICH-E2D Guidelines).

- Results in persistent or significant disability/incapacity
- Constitutes a congenital anomaly or birth defect
- Requires inpatient hospitalization or prolongation of existing hospitalization, unless hospitalization is for:
- Routine treatment or monitoring of the studied indication, not associated with any deterioration in condition
- Elective or pre-planned treatment for a pre-existing condition that is unrelated to the indication under study and has not worsened since signing the informed consent
- Social reasons and respite care in the absence of any deterioration in the participant's general condition
- Treatment on an emergency outpatient basis for an event not fulfilling any of the definitions of a SAE given above and not resulting in hospital admission
- Is medically significant, e.g., defined as an event that jeopardizes the participant or may require medical or surgical intervention to prevent one of the outcomes listed above (e.g., Hy's Law events)

Medical and scientific judgment should be exercised in deciding whether other situations should be considered a serious adverse event, such as important medical events that might not be immediately life-threatening or result in death or hospitalization but might jeopardize the participant or might require intervention to prevent one of the other outcomes listed above. Such events should be considered as "medically significant." Examples of such events are intensive treatment in an emergency room or at home for allergic bronchospasm, blood dyscrasias, or convulsions that do not result in hospitalization or development of dependency. Any suspected transmission via a medicinal product of an infectious agent is also considered a serious adverse event.

All new malignant neoplasms will be assessed as serious under "medically significant" if other seriousness criteria are not met and the malignant neoplasm is not a disease progression of the study indication.

For participants with normal AST and ALT values at baseline, these thresholds used to define Hy's Law are defined as

- AST or ALT > 3 x ULN, combined with
- total bilirubin > 2 x ULN or jaundice in the absence of cholestasis (defined as ALP < 2 ULN), or other causes of hyperbilirubinemia.

For participants with abnormal AST and ALT values at baseline, these thresholds are modified to

- ALT or AST > 3 x baseline AND/OR ALT or AST >8.0 x ULN, whichever occurs first, combined with
- total bilirubin > 2 x baseline AND > 2.0 x ULN or jaundice in the absence of cholestasis (defined as ALP < 2 ULN), or other causes of hyperbilirubinemia.

For this reason, a potential Hy's Law case requires expedited reporting, and will be handled as a serious unexpected adverse event (assessing it as medically significant in the absence of any other seriousness criteria). It must be reported as an SAE to the sponsor promptly (i.e., even before all other possible causes of liver injury have been excluded). Reporting should include all available information, especially that needed for evaluating the diagnosis, severity and likelihood that the study treatment caused the reaction. For participant monitoring and to better understand potential etiologies, the investigator must initiate a close follow-up until complete resolution of the problem and completion of all attempts to obtain supplementary data.

8.10.3 SAE reporting

To ensure participant safety, every SAE, regardless of causality, occurring after the start of study treatment or associated with any other study procedure until 180 days after the last administration of ^{225}Ac-DWJ155 must be reported to Novartis/Sponsor safety immediately, without undue delay, but under no circumstances later than within 24 hours of obtaining knowledge of the events (Note: If more stringent, local regulations regarding reporting timelines prevail).

In addition, any SAEs related to a protocol-mandated requirement from signing of informed consent will be collected and reported to Novartis as described above.

Any SAEs experienced after the completion of the 180 days safety follow-up period should be reported to Novartis if the investigator suspects a causal relationship to study treatment until end of 36-month long-term follow-up, unless otherwise specified by local law/regulations. SAEs of second primary malignancies regardless of causality will be collected until the end of the 36-month long-term follow-up period.

Detailed instructions regarding the submission process and requirements are to be found in the Investigator folder provided to each site. Information about all SAEs is collected and recorded on the electronic Serious Adverse Event Report Form. All applicable sections of the form must be completed in order to provide a clinically thorough report.

The investigator must review and provide an assessment of causality for each SAE. There may be situations in which an SAE has occurred, and the investigator has minimal information to include in the initial report to Novartis. However, it is very important that the investigator always make an assessment of causality for every event before the initial transmission of the

SAE data to Novartis. The investigator may change their opinion of causality in light of follow-up information and send an SAE follow-up report with the updated causality assessment.

All follow-up information for the SAE including information on complications, progression of the initial SAE and recurrent episodes must be reported as follow-up to the original episode immediately, without undue delay, but under no circumstances later than within 24 hours of the Investigator receiving the follow-up information (Note: If more stringent, local regulations regarding reporting timelines prevail).

Any potential Hy's Law case will be handled as a serious adverse event associated with the use of the study treatment (assessing it as medically significant in the absence of any other seriousness criteria) and must be reported as SAE to the sponsor promptly (i.e., even before all other possible causes of liver injury have been excluded and even if abnormal transaminases values at baseline).

Reporting will include all available information, especially that needed for evaluating the severity and likelihood that the study treatment caused the reaction. The investigator must initiate a close follow-up until complete resolution of the problem and completion of all attempts to obtain supplementary data.

Any SAEs experienced after the safety follow-up period (as described above) should only be reported to Novartis Safety if the Investigator suspects a causal relationship to study treatment, unless otherwise specified by local law/regulations.

Reporting of Suspected Unexpected Serious Adverse Reactions (SUSARs)

If an SAE is not previously documented in the IB or product information for marketed products and is thought to be related to any study treatment, it constitutes a Suspected Unexpected Serious Adverse Reaction (SUSAR). Novartis may urgently require further information from the investigator for HA reporting. Novartis may need to issue an Investigator Notification (IN) to inform all investigators involved in any study with the same study treatment. Novartis will comply in reporting of SUSARs to the competent authorities and relevant ethics committees in accordance with national regulatory requirements in participating countries, including EU Clinical Trial Regulation 536/2014.

8.10.4 Pregnancy

For all women of childbearing potential, a pregnancy test will be taken before ^{68}Ga-DWJ155 administration, before each ^{225}Ac-DWJ155 administration, and during post-treatment follow-up as indicated in Section 1.3.

Details of all pregnancies in female participants and, if indicated, female partners of male participants will be collected after the start of study treatment and until period for reporting pregnancies should align with the time period for post treatment contraception determined in Exclusion criteria.

The investigator must report to Novartis each pregnancy within 24 hours of learning of its occurrence.

If a female study participant becomes pregnant, study treatment must be stopped.

The pregnancy consent form must be presented to the study participant. The participant must be given adequate time to read, review and sign the pregnancy consent form. This consent form is necessary to allow the investigator to collect and report information regarding the pregnancy.

The investigator must counsel the participant, discuss the risks of the pregnancy and the possible effects on the fetus and monitor the participant.

As per regulatory requirement, the pregnancy must be followed, up to one year after the estimated date of delivery to determine outcome, including spontaneous or voluntary termination, details of the birth, and the presence or absence of any birth defects, congenital abnormalities, or maternal and/or newborn complications.

Pregnancy follow-up is to be recorded on the same form and must include an assessment of the relationship to study treatment for any pregnancy outcome.

Any pregnancy complication must be reported as an AE or SAE. Elective termination of a pregnancy for medical reasons, abnormal pregnancy outcomes (e.g., spontaneous abortion, fetal death, stillbirth, congenital anomalies, birth defect, ectopic pregnancy) will be reported as SAEs. Any post study pregnancy-related SAE considered reasonably related to the study treatment by the investigator must be reported to Novartis.

While the investigator is not obligated to actively seek this information in former study participants/pregnant female partner, they may learn of an SAE through spontaneous reporting.

If a female partner of a male trial participant who took study treatment in this study becomes pregnant, consent to report information regarding these pregnancy outcomes must be requested from the mother. After consent is provided, the pregnancy reporting will occur up to one year after the estimated date of delivery.

8.10.5 Disease-related events and/or disease-related outcomes not qualifying as AEs or SAEs

Progression of underlying malignancy is not reported as an adverse event and/or serious adverse event if it is clearly consistent with the suspected progression of the underlying cancer as defined by RECIST v1.1 criteria, or other criteria as determined by protocol. Hospitalization due solely to the progression of underlying malignancy should NOT be reported as a serious adverse event. Clinical symptoms of progression may be reported as adverse events if the symptom cannot be determined as exclusively due to the progression of the underlying malignancy or does not fit the expected pattern of progression for the disease under study. If it is unclear whether an adverse event is due only to the disease under study, it should be reported as an AE and/or SAE.

8.10.6 Adverse events of special interest

Any AE that constitutes a DLT should be reported like a grade 3 and 4 adverse event.

8.10.7 Reporting of study treatment errors, study treatment misuse/abuse and overdosage

Study treatment errors including study treatment overdosage are unintentional errors in the prescribing, dispensing, and administration of study treatment. Study treatment misuse refers to situations where the study treatment is intentionally and inappropriately used not in accordance with the protocol.

Study treatment abuse corresponds to the persistent or sporadic, intentional excessive use of study treatment, which is accompanied by harmful physical or psychological effects.

Study treatment errors, including study treatment overdosage and uses outside of what is foreseen in the protocol, including misuse or abuse, must be reported on the AE (or SAE, if the event meets the definition of an SAE) CRF, irrespective of it being associated with AE/SAE.

8.11 Optional Additional Research

If the participant agrees, by signing the optional consent for Additional Research, data and/or biological samples that remain after analysis is completed may be used for additional research to help develop ways to detect, monitor or treat human diseases and/or learn more about related human diseases. A decision to perform such exploratory research studies would be based on outcome data from this study or from new scientific findings related to the drug class or disease, as well as assay availability.

9 Statistical considerations

Data from participating centers in this protocol will be combined, so that an adequate number of participants will be available for analysis. Data will be summarized using descriptive statistics (continuous data) and/or contingency tables (categorical data) for demographic and baseline characteristics, and efficacy, safety, PK, and pharmacodynamics measurements.

Dosage escalation decisions will be based on a synthesis of all relevant data available from all dosage levels evaluated in the ongoing study including safety information, available pharmacodynamics and preliminary anti-tumor activity data. In particular, the following core data must be reviewed for each participant in each cohort to be discussed in the DEM, prior to deciding the next steps:

- Safety: AEs, SAEs, DLTs.
- Laboratory parameters: hematological parameters (hemoglobin, platelets, white blood cells, neutrophils), renal function test (blood urea nitrogen (BUN)/urea, creatinine, sodium, potassium), liver function tests (AST, ALT, total bilirubin, ALP).

Dosing information collected during the DLT observation period, prior and concomitant medications, demographics and diagnosis and extent of cancer (which are relevant to inclusion/exclusion criteria), date of visit and end of treatment phase disposition (if end of treatment has already occurred for a participant who is in a cohort to be discussed in the DEM).

Study data may be analyzed and reported in a primary CSR. The cut-off date for the primary analysis is defined as the date when all participants have completed at least two post-baseline tumor assessments or completed the EoT visit (see Section 7.1 for EoT definition), whichever

comes later. Any additional data for participants continuing to receive study treatment past the data cutoff date for the primary CSR, as allowed by the protocol, will be reported at completion of the study in a final CSR.

All summaries, listings, figures and analyses will be performed by study arms and treatment groups (unless otherwise specified). Additional summaries by indication and by dosage may be produced as relevant for selected endpoints (e.g. clinical efficacy activity).

Screen failure participants, as described in Section 5.4, and the reasons for not starting the study treatment will be reported in a listing but will not be included in any analyses.

Details of the statistical analysis and data reporting will be provided in the Statistical Analysis Plan (SAP).

9.1 Analysis sets

9.1.1 Full Analysis Set and Safety Set

The Full Analysis Set (FAS) and Safety Set (SS) comprise all participants who received at least one administration of study treatment ^{225}Ac-DWJ155. Participants will be analyzed according to the study treatment received, defined as the assigned treatment if the participant took at least one administration of that treatment or the first treatment received if the assigned treatment was never received.

9.1.2 Dose-determining Analysis Set

The DDS for ^{225}Ac-DWJ155 includes all FAS participants who met the minimum exposure criterion and have sufficient safety evaluations (as determined by Novartis/Sponsor and Investigator) or experienced a DLT during the DLT-evaluation period. The DLT evaluation period for the 6-week schedule is the first 6 weeks of dosing, while for the 4-week and 8-week schedules it is first 4 and 8 weeks of dosing, respectively. A participant has met the minimum exposure criterion if she/he received one administration as per the received schedule (i.e. at least 80% of the planned administered activity of ^{225}Ac-DWJ155) during the DLT evaluation period. Participants in the DDS will be analyzed according to the actual study treatment received during the first cycle.

9.1.3 Pharmacokinetic Analysis Set

The pharmacokinetic analysis set (PAS) will include all participants from the SS with at least one available valid (not flagged for exclusion) PK sample and with no protocol deviations that impact the interpretability of the data. This analysis set will be used for all PK data analysis.

9.1.4 ^{68}Ga-DWJ155 Safety Set

^{68}Ga-DWJ155 Safety Set (SS-^{68}Ga-DWJ155) includes all participants who received the administration of ^{68}Ga-DWJ155. This analysis set will be the basis of the safety and imaging summaries for ^{68}Ga-DWJ155. This includes screened participants who receive ^{68}Ga-DWJ155 but not ^{225}Ac-DWJ155.

9.1.5 Dosimetry Analysis Set

The dosimetry analysis set (DAS) will include all participants from the SS who provided at least one set of dosimetry of data. A set of dosimetry data is defined as having sufficient imaging assessment and/or related blood PK assessment to enable a dosimetry estimate. The DAS will be used for summaries (tables and figures) and listings of dosimetry.

9.2 Statistical analyses

9.2.1 General considerations

Not applicable.

9.2.2 Participant demographics and other baseline characteristics

Demographic and other baseline data including disease characteristics will be summarized descriptively by treatment group for the FAS and ^{68}Ga-DWJ155 SS.

Categorical data will be presented as frequencies and percentages. For continuous data, mean, standard deviation, median, minimum, and maximum will be presented. For selected parameters, 25th and 75th percentiles will also be presented.

Relevant medical histories and current medical conditions at baseline will be summarized by system organ class and preferred term, by treatment group.

9.2.3 Treatments

The SS will be used for the analyses below. Categorical data will be summarized as frequencies and percentages. For continuous data, mean, standard deviation, median, 25th and 75th percentiles, minimum, and maximum will be presented.

The duration of exposure in days to ^{225}Ac-DWJ155 will be summarized by means of descriptive statistics using the SS.

The number of participants with dosing adjustments (reductions, interruption, or permanent discontinuation) and the reasons will be summarized by treatment group and all dosing data will be listed.

Concomitant medications and significant non-drug therapies prior to and after the start of the study treatment will be listed by treatment group.

9.3 Primary endpoint(s)/estimand(s) analysis

9.3.1 Safety endpoints

9.3.1.1 Definition of primary endpoint(s)

The primary endpoints are described in Table 3-1.

9.3.1.2 Statistical hypothesis, model, and method of analysis

For summaries of DLTs, the DDS will be used. For all safety analyses, except the DLT analyses, the safety set (SS) will be used. All listings and tables will be presented by treatment group.

The overall observation period will be divided into four mutually exclusive segments:

Pre-treatment period for ^{225}Ac-DWJ155

- Starting 4 days after first ^{68}Ga-DWJ155 administration (including ^{68}Ga-DWJ155 administration day) to the day before the first ^{225}Ac-DWJ155 administration if participant is dosed with ^{225}Ac-DWJ155 > 4 days after ^{68}Ga-DWJ155 administration (including ^{68}Ga-DWJ155 administration day).

On-treatment period for ^{225}Ac-DWJ155

- From date of first administration of study treatment to 42 days for 6-week schedule (28 days for 4-week schedule or 56 days for 8-week schedule) after date of last actual administration of ^{225}Ac-DWJ155 (including last administration date).

Short-term Post-treatment period for ^{225}Ac-DWJ155

- Starting at day 43 to day 180 for a 6-week schedule, and day 29 to day 180 for a 4-week schedule after last administration of ^{225}Ac-DWJ155 (or day 57 to day 180 for a 8-week schedule).

Long-term Post-treatment period for ^{225}Ac-DWJ155

- Starting at day 181 after last administration of ^{225}Ac-DWJ155.

Dosage-limiting toxicities

Dosage escalation will be based upon the estimation of the probability of DLT in DLT evaluation period (4 weeks for 4-week schedule, 6 weeks for 6-week schedule and 8 weeks for 8-week schedule) for participants in the DDS. This probability will be estimated by a Bayesian adaptive approach described below. In addition, all other safety, clinical, PK, and pharmacodynamics data will be considered for dosage escalation.

Bayesian adaptive approach

The dosage escalation part of this study will be guided by a Bayesian analysis of DLT data for ^{225}Ac-DWJ155 in DLT evaluation period. The relationship between administered activity and the probability of DLT is modelled using logistic regression. Details of the model are given in Section 10.4.

Additional dosing regimens

In case any additional dosing schedule is to be tested or indication specific dosage escalation strategy is needed during the study, separate Bayesian Logistic Regression Model (BLRM) or BHLRM may be implemented to guide the dosage escalation by the EWOC principle for the new dosing regimen or by indication. A full description including parameter specification, operating characteristics and hypothetical dosing scenarios of the BLRM/BHLRM will be given in a stand-alone document(s), prior to the first participants is treated on the cohort with new dosing regimen or for indication specific strategy. The additional model specification will also be presented in the primary study report.

Assessment of participant risk

After each cohort of participants, the posterior distribution for the risk of DLT for new participants at administered activities of interest will be evaluated. The posterior distributions

will be summarized to provide the posterior probability that the risk of DLT lies within the following intervals:

Table 9-1 Under-dose, target, and excessive toxicity intervals of posterior probabilites of the DLT rates

Under-dosing:	[0 , 0.16)
Targeted toxicity:	[0.16 , 0.33)
Excessive toxicity:	[0.33 , 1]

The Escalation with Overdose Control (EWOC) Principle

Dosing decisions are guided by the EWOC principle (Rogatko et al 2007). An administered activity may only be used for newly enrolled participants if the risk of excessive toxicity at that dosage level is less than 25%.

Prior distributions

There are no available pre-clinical data informing the radioactivity-DLT relationship for ^{225}AC-DWJ155. Therefore, weakly informative priors will be used for the associated BLRM model parameters.

Reporting

Bayesian analyses will be based on the DDS. DLTs will be listed, and their incidence summarized by primary system organ class and worst grade (CTCAE version 5.0), using the DDS.

Adverse events

All information obtained on adverse events will be displayed by treatment group and participant.

The number (and percentage) of participants with treatment emergent AEs (events started after the first administration of study medication or events present prior to start of treatment but increased in severity based on preferred term) will be summarized in the following ways:

- by treatment, primary system organ class and preferred term.
- by treatment, primary system organ class, preferred term and maximum severity.
- by treatment, Standardized MedDRA Query (SMQ) and preferred term

Separate summaries will be provided for study treatment related AEs, death, SAEs, other significant AEs leading to discontinuation from study and adverse events leading to adjustment in administered activity or schedule.

A participant with multiple adverse events within a primary system organ class is only counted once towards the total of the primary system organ class.

AEs which will be counted for a specific treatment period are those which are treatment emergent. These events are those with an onset after the start of the treatment period, or which were present prior to the start of the treatment period but increased in severity, changed from being not suspected to being suspected of study treatment relationship, or developed into SAEs after the start of the treatment period.

Summary tables for AEs will include only AEs that started or worsened during the on-treatment period, the **treatment-emergent** AEs.

The incidence of treatment-emergent adverse events (new or worsening from baseline) will be summarized by system organ class and or preferred term, severity (based on CTCAE grades), type of adverse event, and relationship to study treatment.

Serious adverse events, non-serious adverse events during the on-treatment period will be tabulated.

All deaths (on-treatment and post-treatment) will be summarized.

All AEs, deaths and serious adverse events (including those from the pre and post-treatment periods) will be listed and those collected during the pre-treatment and post-treatment period will be flagged.

Vital signs

Data on vital signs will be listed, with notable values flagged, and the incidence of abnormal vital signs will be summarized by treatment group.

- summary of notable vital sign values
- table with descriptive statistics at baseline, one or several post-baseline time points and change from baseline to this/these post-baseline time points

12-lead ECG

12-lead ECGs including PR, QRS, QT, QTcF, and HR intervals will be obtained for each participant during the study. ECG data will be read and interpreted locally

- summary of notable ECG values.
- listing of ECG evaluations for all participants with at least one abnormality.

Clinical Laboratory Evaluations

Summary statistics will be provided by treatment and visit/time. Shift tables using the low/normal/high/ (low and high) classification will be used to compare baseline to the worst on-treatment value.

Grading of laboratory values will be assigned programmatically as per NCI CTCAE version 5.0. The calculation of CTCAE grades will be based on the observed laboratory values only, clinical assessments will not be taken into account.

CTCAE Grade 0 will be assigned for all non-missing values not graded as 1 or higher. Grade 5 will not be used.

For laboratory tests where grades are not defined by CTCAE version 5.0, results will be categorized as low/normal/high based on laboratory normal ranges.

The following summaries will be generated separately for hematology, and biochemistry tests:

For laboratory tests where grades are defined by CTCAE version 5.0

- Worst post-baseline CTCAE grade (regardless of the baseline status). Each participant will be counted only once for the worst grade observed post-baseline.

- Shift tables using CTCAE version 5.0 grades to compare baseline to the worst on-treatment value

For laboratory tests where grades are not defined by CTCAE version 5.0,

- Shift tables using the low/normal/high/ (low and high) classification to compare baseline to the worst on-treatment value.

Tolerability

Tolerability of study treatment will be assessed by summarizing the number of and reasons for dosage delays and dosage reductions. Dosage intensity will also be tabulated by treatment group. Refer to Section 9.2.3.

9.3.1.3 Handling of missing values/censoring/discontinuations

Participant safety data will be reported up to the cut-off date, end of post-treatment (if discontinued), death, loss to follow up, or withdrawal of consent/opposition to use data/biological samples, whichever happened first. Missing values will not be imputed and will be reported as such.

9.4 Secondary endpoint(s)/estimand(s) analysis

9.4.1 Efficacy and/or pharmacodynamic endpoint(s)

Best overall response (BOR) rate

BOR is defined as the best response recorded from the start of the treatment until disease progression/recurrence as per local review and according to RECIST v1.1. BOR rate will be summarized along with the corresponding 90% exact CI using FAS.

Overall response rate (ORR)

ORR is defined as the proportion of participants with a BOR of complete response (CR) or partial response (PR) as per local review and according to RECIST v1.1. It will be summarized along with the corresponding 90% exact CI using FAS.

Duration of Response (DOR)

DOR is the time between the first documented response (CR or PR) and the date of progression as per local review and according to RECIST v1.1, or death due to any cause. Here, death due to any cause is considered as an event to be conservative and align with PFS event definition. DOR may be presented graphically if enough events are available for analysis, using Kaplan-Meier plots for all participants who achieved a CR/PR in the study. The median DOR and corresponding 90% CI will be presented. Analysis will include responders with confirmed responses.

Disease control rate (DCR)

DCR is defined as the proportion of participants with a BOR of CR, PR, or stable disease as per local review and according to RECIST v1.1. It will be summarized along with the corresponding 90% exact CI, using FAS.

Progression-free survival (PFS) rate at pre-specified time points and median PFS

PFS is defined as the time from the date of start of treatment to the date of the first documented progression as per local review and according to RECIST v1.1 or death due to any cause.

If a participant has not experienced an event at the time of the analysis, PFS will be censored at the date of the last adequate tumor assessment.

PFS may be summarized at the RD(s) and for all other dosages with at least 5 participants treated, using the Kaplan-Meier method. Median PFS, with corresponding 95% CI, and 25th and 75th percentiles (Brookmeyer and Crowley 1982, Klein and Moeschberger 1997) may be presented. KM estimates for PFS rate at pre-specified time points (defined in the SAP), along with 90% CI (Greenwood's formula, (Kalbfleisch and Prentice 2002)) may also be provided. The number (%) of progressions, deaths and censored participants will also be summarized.

9.4.2 Pharmacokinetics

9.4.2.1 Blood pharmacokinetics

For PK analysis, the PAS will be used.

While Ac-225 decays physically without gamma radiation, PK samples can still be measured by a standard gamma counter due to the small gamma radiation coming from the decay of daughter isotopes of Ac-225 (i.e., Fr-221 and Bi-213). As daughter isotopes are known to get released from the chelator moiety of a radioligand peptide due to nuclear recoil, radiation measurements at the time of PK blood sampling cannot differentiate between free daughter isotopes in systemic circulation (unbound to ligands) and Ac-225-bound-peptide. At the secular equilibrium (at a time > 6 h) initial free daughter isotopes will have decayed, and the Ac-225 originally bound to peptide which is reflective of DWJ155 concentration in blood, will be measurable due the gamma radiation emitting by the daughters.

The ^{225}Ac-DWJ155 PK analysis will be performed based on decay-corrected blood radioactivity concentration data measured at secular equilibrium (> 6 h post-collection of the sample) using a calibrated gamma-counting device and converted to mass units, pending confirmation of metabolic stability. Radioactivity measurements are decay-corrected to the date and time of start of infusion and conversion to mass units is based on the actual volumetric activity and peptide mass content as indicated in the certificate of release of the administered activity and collected in a dedicated eCRF.

Blood radioactivity concentration and mass concentration data will be listed by administered activity/cycle or dosage, participant and visit/sampling time point, separately for the dosage escalation and expansion part of the study, and overall (i.e., pooled across escalation and expansion). Descriptive summary statistics will be provided by administered activity/cycle or dosage and study part (dosage escalation and expansion study part) and overall, visit/sampling time point, including the frequency (n, %) of concentrations below the lower limit of quantification (LLOQ) and reported as zero, mean (arithmetic and geometric), standard deviation, coefficient of variation (CV%) (arithmetic and geometric), median, minimum, and maximum.

The following PK parameters may be determined, as appropriate, by non-compartmental method(s) using the actual recorded sampling times, with Phoenix WinNonlin (Version 8.3.4 or higher): the maximum blood concentrations (Cmax) along with the times at which they occurred (Tmax), as well as the area under curve (AUC), systemic clearance (CL), volume of distribution during the terminal phase (Vz), effective and biologic terminal half-life (Table 9-2).

Early analysis of preliminary PK may be conducted based on preliminary data prior to database lock. Nominal time instead of actual elapsed time may be used.

Drug concentrations below LLOQ will be treated as missing for the calculation of the geometric means and geometric CV%, and as zero for all other calculations including calculation of PK parameters. Missing values for any PK parameters or concentrations will not be imputed and will be treated as missing.

Blood PK parameters will be listed by administered activity/cycle or dosage and participant, separately for the dosage escalation and expansion part of the study, and overall and will include mean (arithmetic and geometric), standard deviation, coefficient of variation (CV%) (arithmetic and geometric), median, minimum, and maximum. An exception to this is Tmax where median, minimum, and maximum will be presented.

Table 9-2 Non-compartmental pharmacokinetic parameters

AUClast	The AUC from time zero to the last measurable concentration sampling time (tlast) (mass x time x volume-1)
AUCinf	The AUC from time zero to infinity (mass x time x volume-1)
Cmax	The maximum (peak) observed plasma, blood, serum, or other body fluid drug concentration after single administration (mass x volume-1)
Tmax	The time to reach maximum (peak) plasma, blood, serum, or other body fluid drug concentration after single administration (time)
T1/2	The biological elimination half-life associated with the terminal slope (λz) of a semi logarithmic concentration-time curve (time).
T1/2eff	Effective half-life derived from biological half-life and physical half-life of Ac-225
CL	The total body clearance of drug from the plasma (volume x time-1)
Vz	The apparent volume of distribution during terminal phase (associated with λz) (volume)

9.4.3 Safety properties of ^{68}Ga-DWJ155

For all safety analyses, the ^{68}Ga-DWJ155 safety set (SS-^{68}Ga-DWJ155) will be used.

The overall observation period will be divided into two mutually exclusive segments:

Pre-evaluation period for ^{68}Ga-DWJ155**:**

- from day of participant's first informed consent to the day before first administration of ^{68}Ga-DWJ155

Within-evaluation period for ^{68}Ga-DWJ155:

- from the date of the first administration of ^{68}Ga-DWJ155 to 3 days after first ^{68}Ga-DWJ155 administration (including ^{68}Ga-DWJ155 administration day)

Adverse events for assessing the safety of ^{68}Ga-DWJ155 will be summarized in a similar way as in Section 9.3.1.

9.5 Exploratory endpoint(s)/estimand(s) analysis

Analysis of dosimetry, PK/pharmacodynamics relationships and imaging properties of ^{68}Ga-DWJ155 are presented below, all remaining exploratory endpoints will be detailed in the Statistical Analysis Plan (SAP).

9.5.1 Dosimetry

For dosimetry analysis, the DAS will be used.

Dosimetry analysis and results will be done by a central specialized vendor and reported in a separate document by the vendor.

Details on the analysis are described in the Dosimetry Charter.

Briefly, dosimetry assessments will be performed from radioactivity quantification in organs receiving the highest activity, identified visually. Regions or Volumes of interest (ROIs/VOIs) will be drawn on those selected organs and tumor lesions on the SPECT images serially acquired post ^{225}Ac-DWJ155 infusion. ROI will be placed over these organs to determine relative radiotracer uptake calculated as a percentage of the injected activity (%IA), separately for Fr-221 and Bi-213 and/or combined across energy windows. Dosimetry in red marrow may be estimated using imaging (SPECT/CT) and/or blood-based methodologies, as appropriate, from the serial venous blood sampling and gamma counter measurements, as described in Section 8.3. For graphical representation of %IA over time, the radioactivity values at each timepoint/organ are decay corrected to the time of start of infusion. Time activity curves (TACs) will be generated from the amount of radioactivity in one given tissue at a given moment. TACs will be fitted to mono- or poly-exponential curves, as appropriate. Other methods may be considered and specified in the Dosimetry Charter.

Tumor lesions selected for dosimetry assessment should match the lesions used for RECIST v1.1 assessment, whenever feasible (e.g., adequate size, uptake observed in the SPECT), accounting for the technical limitations in dosimetric analysis of Ac-225 radio-therapies.

Selected clinical participant data as required for dosimetry analysis will be transferred from the study eCRF to the imaging CRO, for review by the independent dosimetrist. More details on the analysis are included in the Dosimetry Charter.

Individual listings and descriptive summaries (arithmetic mean, standard deviation (SD), median, Min, Max, CV%) will be presented by administered activity/cycle or dosage for biodistribution and dosimetry derived parameters, such as TACs obtained from the radiotracer

uptake (as percentage of injected activity), absorbed dose-equivalent coefficients, theoretical cumulative absorbed dose-equivalent, and effective half-life in selected organs and lesions, as measured by image quantification.

9.5.2 Urine pharmacokinetics

Urine elimination data for ^{225}Ac-DWJ155 will be assessed based on decay-corrected urine radioactivity concentration and converted to mass units. The percentage of injected activity (%IA) in each specified time interval and cumulative over all time intervals and renal clearance will be calculated, will be listed by participant and summarized descriptively. Descriptive summary statistics will include mean (arithmetic and geometric), SD, CV (arithmetic and geometric), median, minimum, and maximum.

9.5.3 PK/pharmacodynamics relationships

Descriptive analysis and exploratory modeling analysis may be performed to assess the relationship between administered activity of ^{225}Ac-DWJ155 PK parameters, dosimetry and pharmacodynamics markers of safety and efficacy endpoints.

9.5.4 Imaging Properties of ^{68}Ga-DWJ155

The statistical analyses of imaging properties of ^{68}Ga-DWJ155 will be descriptive in nature and will include summaries and graphical presentations of data. No formal testing will be performed. Categorical data, if available, will be summarized as frequencies and percentages. For continuous data (e.g. SUV, SUVr), mean, standard deviation, median, minimum, and maximum will be presented.

SUVs (SUV_{mean} and SUV_{max}) of lesions and different potential reference organs will be provided by the imaging vendor. SUVr will be calculated by dividing the SUV of the lesion by the SUV of the different organs in order to identify the optimal reference organ and the optimal SUVr (i.e., using SUV_{mean} or SUV_{max}) that will be used. TACs will be generated by plotting the mean SUV and SUVr values (and standard deviation) of the lesions over time. Graphical presentation of the individual lesions' SUVs over time may be provided.

9.6 Other Safety analyses

Not applicable.

9.7 Other analyses

Not applicable.

9.8 Interim analysis

No interim analyses are planned. However, the dosage-escalation design allows decisions based on the current data to be taken before the end of the study. More precisely, after each cohort in the dosage escalation, the next administered activity of study treatment will be selected based on review by Novartis study personnel and Investigators of all available safety and tolerability information (including the DLT risk assessment from the BHLRM using EWOC) along with , PK, PD and efficacy data.

During the dosage expansion stopping rules are applied (please refer to Section 7.5) and a DMC will be in place (please refer to Section 10.1.4.1)

9.9 Sample size determination

9.9.1 Primary endpoint(s)

Dosage-escalation part

Cohorts of 3 to 6 evaluable participants will be enrolled in the dosage-escalation part including at least six participants at the MTA or a suitable lower activity at or above the RD(s) (where dosage is defined as administered activity per cycle, total number of cycles and schedule). Multiple cohorts may be sequentially enrolled to the same dosage level. Additional cohorts of 1 to 6 participants (or 1 to 12 participants at doses that are candidates for RD declaration), may be enrolled at any dosage level at or below a previously tested dosage level, and satisfying EWOC, for further elaboration of safety, PK, and dosimetry parameters as required. At least 21 participants should be treated in the dosage escalation part, for the model to have good operating characteristics relating to the identification of the MTA amongst those administered activities tested (details in Section 10.4).

It is not mandatory to continue escalating administered activity until the MTA is reached and it is possible to determine the RD(s) without reaching the MTA. However, the following rules will be used to stop the dosage escalation in case a MTA is identified by the design:

1. At least 6 treated participants at the administered activity to be determined as the MTA

2. This administered activity satisfies one of the following conditions:
   - the posterior probability of targeted toxicity at this administered activity exceeds 50% and is the highest among potential administered activities, or
   - minimum of 21 treated participants on the trial.

Dosage expansion part

In the expansion part, each group (HER2+ breast cancer, HR+/HER2-low breast cancer, HR+/HER2-0 breast cancer, HR-/HER2-low breast cancer, HR-/HER2-0 breast cancer, AGA NSCLC, non-AGA NSCLC, HER2 3+ or 2+ (ISH + or -) GEA and bladder cancer) will recruit approximately the number of participants shown in Figure 1-1. If cumulative study data (safety, tolerability, PK, dosimetry, pharmacodynamics and/or preliminary anti-tumor activity) from dosage escalation support more than one administration schedule, these may be investigated in selected expansion groups. In this case, up to four additional group(s) of participants may be opened (with approximately 15-20 participants each).

Section 9.2 provides probability of detecting an AE with a specified incidence given the true incidence rate under different sample size scenarios. For an adverse event with a true incidence of 2.5%, 5%, 10%, and 15%, the probability of observing at least one participant with the AE in an expansion group of 10 participants treated at the RD is 22%, 40%, 65%, and 80%, respectively; the probability of observing at least one participant with the AE in an expansion group of 15 participants treated at the RD is 32%, 54%, 79%, and 91%, respectively; the probability of observing at least one participant with the AE in an expansion group of 40

participants (in case of pooling NSCLC participants) treated at the RD is 64%, 87%, 99%, and 100%, respectively.

Table 9-3 Probability of detecting an AE with a specified incidence at the RD as a function of the sample size

Number of participants	AE incidence rate			
	0.025	**0.05**	**0.10**	**0.15**
10	0.22	0.40	0.65	0.80
15	0.32	0.54	0.79	0.91
20	0.40	0.64	0.88	0.96
40	0.64	0.87	0.99	1.00

10 Supporting documentation and operational considerations

10.1 Appendix 1: Regulatory, ethical, and study oversight considerations

10.1.1 Regulatory and ethical considerations

This study was designed and will be implemented, executed and reported in accordance with the protocol and the following:

a. Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences (CIOMS) international ethical guidelines
b. Applicable ICH Tripartite Guidelines for Good Clinical Practice (GCP)
c. Applicable local regulations (including European Clinical Trial Regulation 536/2014, US CFR 21)

The protocol, protocol amendments, ICF, Investigator's Brochure, Pharmacy Manual and other relevant documents (e.g., advertisements) must be submitted to an IRB/IEC by the Investigator and reviewed and approved by the IRB/IEC before the study is initiated at that specific site.

Any amendments to the protocol will require IRB/IEC and Health Authority approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants.

The Investigator will be responsible for the following:

- Signing a protocol signature page confirming his/her agreement to conduct the study in accordance with these documents and all of the instructions and procedures found in this protocol and to give access to all relevant data and records to Novartis monitors, auditors, Novartis Quality Assurance representatives, designated agents of Novartis, IRBs/IECs, and regulatory authorities as required
- Providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC
- Notifying the IRB/IEC of SAEs or other significant safety findings as required by IRB/IEC procedures
- Taking any immediate action required for the safety of any participant included in this study, even if this action represents a deviation from the protocol. In such cases, Novartis must be notified of this action and the IRB/IEC at the study site must be informed according to local regulations.
- Providing oversight of the conduct of the study at the site and adherence to requirements of 21 CFR, ICH guidelines, the IRB/IEC, European regulation 536/2014 for clinical studies (if applicable), European Medical Device Regulation 2017/745 for clinical device research (if applicable), and all other applicable local regulations
- Informing Novartis immediately if an inspection of the clinical site is requested by a regulatory authority

10.1.2 Informed consent process

The Investigator or their representative will explain the nature of the study, including the risks and benefits, to the participant or their legally authorized representative, if allowed according to local regulations and answer all questions regarding the study.

Participants must be informed that their participation is voluntary. Participants or their legally authorized representatives will be required to sign a statement of informed consent that meets the requirements of 21 CFR 50, local regulations, ICH guidelines, privacy and data protection, where applicable, and the IRB/IEC or study center.

Informed consent must be obtained before conducting any study-specific procedures (e.g., all of the procedures described in the protocol). The process of obtaining informed consent must be documented in the participant source documents.

The medical record must include a statement that written informed consent was obtained before the participant was enrolled in the study and the date the written consent was obtained. The authorized person obtaining the informed consent must also sign the ICF.

A copy of the ICF(s) must be provided to the participant or their legally authorized representative.

Participants who are rescreened are required to sign a new ICF.

The ICF will contain a separate section that addresses the use of remaining mandatory samples for optional research. Participants will be told that they are free to refuse to participate and may withdraw their consent at any time and for any reason during the storage period. A separate signature will be required to document a participant’s decision.

If applicable, in cases where the participant's representative(s) gives consent (if allowed according to local requirements), the participant must be informed about the study to the extent possible given his/her level of understanding. If the participant is capable of doing so, the participant must indicate agreement by personally signing and dating the written informed consent document.

Common side effects already known about ^{68}Ga-DWJ155 and ^{225}Ac-DWJ155 will be included in the ICF and must be discussed with the participant upon obtaining consent and during the study as needed. Any new information related to the safety profile that is identified between IB updates will be communicated as appropriate, for example, via an Investigator notification or an aggregate safety finding. New information might require an update to the ICF and discussion with the participant.

The following informed consents are included in this study:

- Main study consent,
- Optional consent including:
- For 'additional research' to allow future research on data/samples collected during this study
- For visits and activities that may be done outside of the study site
- Pregnancy follow up Consent for a participant or individual who got pregnant by a Trial participant

- An information sheet for individual who could get pregnant by a Trial participant
- Post Ac-225 Instructions: Reducing External Exposure to Others

A copy of the approved version of all consent forms must be provided to Novartis after IRB/IEC approval.

The consent process including re-consent may take place either in-person or remotely (e.g., via videoconferencing) per discretion of the investigator and with the agreement of the participant or their legally authorized representative. Remote consenting will be used as permitted by local regulations, EC/IRB approval, and with a GCP compliant process that considers relevant requirements (e.g., privacy, verification of identity, Informed Consent process is appropriately documented etc.). At the next in-person visit, the investigator should reaffirm the consent obtained remotely and document the reaffirmation process.

10.1.3 Data protection

Participants will be assigned a unique identifier. Any participant records or datasets that are transferred to Novartis will contain the identifier only; participant names or any information which would make the participant identifiable will not be transferred.

The participant must be informed that his/her personal study-related data will be used by Novartis in accordance with local data protection law. The level of disclosure must also be explained to the participant who will be required to give consent for their data to be used as described in the informed consent.

The participant must be informed that his/her medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by Novartis, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.

Novartis has appropriate processes and policies in place to handle personal data breaches according to applicable privacy laws.

Any data security breach that may impact the confidentiality and protection of study participants' data is recorded and investigated by Novartis as required by applicable laws or regulations and as adequate to the nature of the breach.

Upon identification of any potential data security breach, a Novartis team will perform an initial assessment. Depending on the nature and type of breach, including the categories of data impacted, the data security breach management process may include investigation of the root cause of the incident, and, on a basis of its outcome, determination of the adequate mitigation actions to address any immediate impact, and corrective and preventive measures to prevent recurrence and mitigate any adverse effects, as applicable. The data security breach will be reported to the regulatory authorities and Ethic Committees or IRBs as required by the applicable laws or regulations.

For countries with serious breach reporting regulations, and deviation of the approved protocol that is considered a serious breach, which would likely affect the safety, rights of trial participants and/or data reliability and robustness to a significant degree, will be promptly reported to Novartis.

10.1.4 Committees structure

10.1.4.1 Data monitoring committee

This study will include a data monitoring committee (DMC) in the expansion part of the study, which will function independently of all other individuals associated with the conduct of this clinical trial, including the site Investigators participating in the study. The DMC will assess and provide feedback at defined intervals the progress of a clinical trial, safety data, and critical efficacy variables if applicable and recommend to Novartis whether to continue, modify, or terminate a trial. In case the dosage expansion stopping rules criteria are met, the DMC will meet immediately to discuss and make a recommendation. Please refer to Section 7.5 for additional details.

Specific details regarding composition, responsibilities, data monitoring, and meeting frequency, and documentation of DMC reports, minutes, and recommendations will be described in a separate charter that is established between Novartis and the DMC.

10.1.5 Data quality assurance

Monitoring strategy, methods, responsibilities, and requirements are provided in the monitoring plan. Details may include definition of study critical data items and processes (e.g., risk-based initiatives in operations and quality such as risk management and mitigation strategies and analytical risk-based monitoring), including handling of noncompliance issues and monitoring techniques (central, remote, or on-site monitoring)

Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the Investigator for 15 years (25 years for countries under EU CTR) after study completion unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of Novartis. No records may be transferred to another location or party without written notification to Novartis.

10.1.5.1 Data collection

Designated Investigator staff will enter the data required by the protocol into the Electronic Case Report Forms (CRF). The CRFs have been built using fully validated secure web-enabled software that conforms to 21 CFR Part 11 requirements, Investigator site staff will not be given access to the electronic data capture (EDC) system until they have been certified as trained. Automatic validation programs check for data discrepancies in the CRFs, allow modification and/or verification of the entered data by the Investigator staff.

The Investigator/designee is responsible for assuring that the data entered into CRF is complete, accurate, and that entry and updates are performed in a timely manner e.g., within 5 days of visit. The Investigator must certify that the data entered are complete and accurate.

After final database lock, the Investigator will receive copies of the participant data for archiving at the investigational site.

All data should be recorded, handled, and stored in a way that allows its accurate reporting, interpretation, and verification.

10.1.5.2 Database management and quality control

Novartis personnel (or designated CRO) will review the data entered by investigational staff for completeness, accuracy and scientific/medical plausibility. Electronic data queries stating the nature of the problem and requesting clarification will be created for discrepancies and missing values and sent to the investigational site via the EDC system. Designated Investigator site staff are required to respond promptly to queries and to make any necessary changes to the data.

Concomitant treatments and prior medications entered into the database will be medically coded using the World Health Organization (WHO) Drug Reference List, which employs the Anatomical Therapeutic Chemical classification system. Medical history/current medical conditions and adverse events will be coded using the Medical Dictionary for Regulatory Activities (MedDRA) terminology.

Dates of screenings, visits, screen failures, early terminations, dosage escalation/dosage expansion and cohort assignment, as well as data about all ^{225}Ac-DWJ155 dispensed to the participant and all dosage changes will be tracked using an Interactive Response Technology (IRT). The system will be supplied by a vendor, who will also manage the database. The data will be sent electronically to Novartis (or a designated CRO) at specific timelines.

Once all the necessary actions have been completed and the database has been declared to be complete and accurate, it will be locked and made available for data analysis.

10.1.6 Source documents

Data reported on the CRF or entered in the CRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The Investigator may need to request previous medical records or transfer records.

The Investigator must give the monitor access to all relevant source documents to confirm their consistency with the data capture and/or data entry. The Investigator must also keep the original informed consent form signed by the participant (a signed copy is given to the participant).

Study monitors will perform source data verification to confirm that data entered into the CRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

Key study personnel must be available to assist the field monitor during these visits.

10.1.7 Publication policy

The protocol will be registered in a publicly accessible database such as clinicaltrials.gov and as required in CTIS public website. In addition, after study completion (defined as last patient last visit) and finalization of the study report the results of this trial will be submitted for publication and posted in a publicly accessible database of clinical trial results, such as the Novartis clinical trial results website and all required health authority websites (e.g., Clinicaltrials.gov, EU CTIS public website etc.).

Summary results of primary and secondary endpoints will be disclosed based upon the global Last Patient Last Visit (LPLV) date, since multinational studies are locked and reported based upon the global LPLV.

For details on the Novartis publication policy including authorship criteria, please refer to the Novartis publication policy training materials that were provided.

Any data analysis carried out independently by the Investigator must be submitted to Novartis before publication or presentation.

10.1.8 Protocol adherence and protocol amendments

This protocol defines the study objectives, the study procedures and the data to be collected on study participants. Additional assessments required to ensure safety of participants should be administered as deemed necessary on a case by case basis. Under no circumstances including incidental collection is an Investigator allowed to collect additional data or conduct any additional procedures for any purpose involving any investigational drugs under the protocol, other than the purpose of the study. If despite this prohibition, data, information, observation would be incidentally collected, the Investigator shall immediately disclose it to Novartis and not use it for any purpose other than the study, except for the appropriate monitoring on study participants.

Investigators ascertain they will apply due diligence to avoid protocol deviations. If an Investigator feels a protocol deviation would improve the conduct of the study this must be considered a protocol amendment, and unless such an amendment is agreed upon by Novartis and approved by the IRB/IEC and health authorities, where required, it cannot be implemented.

10.1.9 Off-site procedures

At the Investigator's discretion and based on benefit-risk considerations of the participant's clinical condition, qualifying participants may be offered the option to have certain clinical trial procedures according to Section 1.3 Schedule of Activities performed at an off-site location.

The following element may be offered to support off-site visits where allowed by national and local regulations: use of biosample collection centers separate to the Investigative Site.

Procedures for off-site visits utilizing biosample collection centers separate to the Investigative Site is only possible for chemistry, hematology and urinalysis assessments at the visits indicated in Section 1.3.

The off-site procedures may be utilized in certain countries, Sites and visits as determined by trial needs, if Sponsor allows, and based on national and local/site regulations.

Participants, that the investigator identifies as suitable for off-site visits, must provide informed consent.

The following conditions must be considered for off-site visits to occur:

- If the participant suffers from either (1) a severe AE or an SAE, and/or (2) any concurrent medical conditions which, in the opinion of the Investigator, could cause unacceptable safety risks, then the participant must resume the on-site visits. The participant may resume the off-site visits when, based on the Investigator’s judgment, there are no further safety risks for the participant.

- If other assessments are required, then off-site visits are not possible.

10.1.9.1 Responsibility of investigator oversight of off-site activities

Procedures that are performed off-site remain under the oversight of the Investigator, who retains accountability for the conduct of all safety assessments and will ensure the rights, safety, and wellbeing of participants. This includes the following (including, but not limited to):

- the identification, management and reporting of AEs and SAEs are performed in accordance with the protocol and applicable regulations
- source data collected off-site are reviewed and evaluated in a timely manner

10.2 Appendix 2: Abbreviations and definitions

10.2.1 List of abbreviations

%IA	Percent of Injected Activity
ACTH	Adrenocorticotropic Hormone
ADC	Antibody-Drug Conjugate
AE	Adverse Event
AGA	Actionable Gene Alteration
AIDS	Acquired Immunodeficiency Syndrome
ALARA	As Low as Reasonably Achievable
ALP	Alkaline Phosphatase
ALT	Alanine Aminotransferase
ANA	Antinuclear Antibodies
ANC	Absolute neutrophil count
APTT	Activated partial thromboplastin time
ASCO	American Society of Clinical Oncology
ASMA	Anti-smooth muscle antibody
AST	Aspartate Aminotransferase
AUC	Area Under the Curve
AxMP	Auxiliary Medicinal Product
BHLRM	Bayesian Hierarchical Logistic Regression Model
BLRM	Bayesian Logistic Regression Model
BOR	Best Overall Response
BP	Blood Pressure
BUN	Blood Urea Nitrogen
CAP	College of American Pathologists
CBR	Clinical Benefit Rate
CD	Carbohydrate deficient
CDK4/6	Cyclin-Dependent Kinases 4 and 6
cfDNA	Circulating cell-free DNA
CIOMS	Council for International Organizations of Medical Sciences
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CL	Systemic Clearance
Cmax	Maximum Concentration
CMTS	Combined mass and time scaling
CMV	Cytomegalovirus
CNS	Central Nervous System
CR	Complete Response
CRF	Case Report/Record Form (paper or electronic)
CRO	Contract Research Organization
CSR	Clinical study report
CT	Computed tomography
CTC	Common Terminology Criteria
CTCAE	Common Terminology Criteria for Adverse Events
CV	Coefficient of variation

DAS	Dosimetry analysis set
DCR	Disease Control Rate
DDS	Dose determining set
DEMs	Dosage Escalation Meetings
DILI	Drug-induced liver injury
DL	Dosage Level
DL1	Dosage Level 1
DL2	Dosage Level 2
DLT	Dosage Limiting Toxicity
DMC	Data Monitoring Committee
DNA	Deoxyribonucleic acid
DOR	Duration of Response
EBRT	External beam radiation therapy
EBV	Epstein–Barr Virus
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group Performance Status
EDC	Electronic Data Capture
EEA	European Economic Area
eGFR	Estimated Glomerular Filtration Rate
EOI	End of infusion
EPR	Early progression rate
ER	Estrogen Receptor
ERCP	Endoscopic retrograde cholangiopancreatography
ESAs	Erythroid stimulating agents
eSource	Electronic Source
EWOC	Escalation with overdose control
FAS	Full Analysis Set
FDA	Food and Drug Administration
FDG	Fluorodeoxyglucose
FDG-PET	Fluorodeoxyglucose Positron Emission Tomography
FIH	First in Human
FOV	Field of View
G-CSF	Granulocyte Colony-Stimulating Factor
GCP	Good Clinical Practice
GEA	gastroesophageal adenocarcinoma
GGT	Gamma-glutamyl transferase
GLDH	Glutamate Dehydrogenase
GM-CSF	Granulocyte-Macrophage Colony-Stimulating Factor
h	Hour
HAV	Hepatitis A virus
HBc	Hepatitis B core antibody
HBsAg	Hepatitis B virus surface antigen
HBV	Hepatitis B Virus
hCG	Human Chorionic Gonadotropin
HCV	Hepatitis C Virus

HER2	Human Epidermal Growth Factor Receptor 2
HEV	Hepatitis E virus
HIV	Human immunodeficiency virus
HPLC	High-performance liquid chromatography
HSV	Herpes simplex virus
I.V.	intravenous
IB	Investigator's Brochure
ICF	Informed Consent Form
ICH	International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use
ICRP	International Commission on Radiological Protection
IEC	Independent Ethics Committee
Ig	Immunoglobulin
IHC	Immunohistochemistry
IMP	Investigational Medical Product
INR	International Normalized Ratio
IO	Immuno-Oncologic
IRB	Institutional Review Board
IRT	Interactive Response Technology
ISH	in situ hybridization
ITT	Intention-to-Treat
IUD	Intrauterine device
IUS	Intrauterine hormone-releasing system
KIM-1	Kidney Injury Molecule-1
LDH	lactate dehydrogenase
LET	linear energy transfer
LFT	Liver function test
LLN	lower limit of normal
LLOQ	lower limit of quantification
LVEF	Left ventricular ejection fraction
M-CSF	Macrophage Colony-Stimulating Factor
mCRPC	Metastatic Castration-Resistant Prostate Cancer
MCV	Mean Corpuscular Volume
MedDRA	Medical dictionary for regulatory activities
mg	milligram(s)
min	minute(s)
mL	milliliter(s)
mm	millimeter(s)
MRI	Magnetic Resonance Imaging
MTA	Maximum Tolerated Activity
MUGA	Multiple gated acquisition
N	number
NA	Not Applicable
NCI	National Cancer Institute
NDL	Next dosage level
NGAL	Neutrophil Gelatinase-Associated Lipocalin

NOAEL	No observed adverse effect level
NSCLC	Non-Small Cell Lung Cancer
NYHA	New York Heart Association
ORR	Overall Response Rate
OS	Overall Survival
PAS	Pharmacokinetic Analysis Set
PCR	Polymerase Chain Reaction
PD	Pogressive disease
PET	Positron emission tomography
PFS	progression-free survival
PK	Pharmacokinetic(s)
PR	Progesterone Receptor
PRRT	Peptide receptor radionuclide therapy
PT	Prothrombin time
Q4W	every 4 weeks
Q8W	Every 8 weeks
QTcF	QT interval corrected by Fridericia's formula
RAP	The Report and Analysis Plan
RBE	Relative biological effectiveness
RD	Recommended dosage(s) (where dosage is defined as administered activity per cycle, total number of cycles and schedule)
RECIST	Response Evaluation Criteria In Solid Tumors
RIA	Radioimmunoassay
RLI	Radioligand Imaging
RLT	Radioligand therapy
RMS	Relative mass scaling
RNA	Ribonucleic acid
ROIs	Regions of interest
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SD	standard deviation
SMQ	Standardized MedDRA Query
SoA	Schedule of Activities
SOC	Standard of Care
SOD	Sum of Diameters
SPECT	Single-photon emission computed tomography
SpO2	Oxygen saturation
SS	Safety Set
SUV	Standardized Uptake Value
T3	Triiodothyronine
T4	Thyroxine
TACs	Time Activity Curves
Tmax	Time to Maximum Concentration
TNBC	Triple negative breast cancer
TSH	Thyroid-Stimulating Hormone
TTE	Transthoracic echocardiography

TTF	Time to Treatment Failure
TTP	Time to progression
ULN	Upper limit of normal
VOIs	Volumes of interest
Vz	Volume of Distribution (in the terminal phase)
WOCBP	Women of child-bearing potential

10.2.2 Definitions

Additional Study treatment	Medicinal products that may be used during the clinical trial as described in the protocol, but not as an investigational drug/treatment (e.g., any background therapy)
Assessment	A procedure used to generate data required by the study
Auxiliary Medicinal Product (AxMP)	Medicinal product used for the needs of a clinical trial as described in the protocol, but not as an investigational medicinal product (e.g., rescue medication, challenge agents, background treatment or medicinal products used to assess endpoints in the clinical trial). Concomitant therapy is not considered as AxMP.
Background Therapy	Generally considered to be the current standard of care for a particular condition / disease, in addition to which study treatment is given
Biologic Samples	A biological specimen including, for example, blood (plasma, serum), saliva, tissue, urine, stool, etc. taken from a study participant
Coded Data	Personal Data which has been de-identified by the investigative center team by replacing personal identifiers with a code.
Cohort	A group of individuals who share a common exposure, experience or characteristic, or a group of individuals followed-up or traced over time
Control drug	A study treatment (active or placebo) used as a comparator to reduce assessment bias, preserve blinding of investigational drug, assess internal study validity, and/or evaluate comparative effects of the investigational drug
Cycles	Number and timing or recommended repetitions of therapy are usually expressed as number of days (e.g., q28 days)
Discontinuation from study	Point/time when the participant permanently stops receiving the study treatment and further protocol required assessments or follow-up, for any reason. No specific request is made to stop the use of their samples or data.
Discontinuation from study treatment	Point/time when the participant permanently stops receiving the study treatment for any reason (prior to the planned completion of study intervention administration, if any). Participant agrees to the other protocol required assessments including follow-up. No specific request is made to stop the use of their samples or data.
Dosage	Adminstered activity per cycle, cycle length and number of cycles of the investigational RLI/RLT
Electronic Data Capture (EDC)	Electronic data capture (EDC) is the electronic acquisition of clinical trial data using data collection systems, such as Web-based applications, interactive voice response systems and clinical laboratory interfaces. EDC includes the use of Electronic Case Report Forms (CRFs) which are used to capture data transcribed from source data/documents used at the point of care
End of the clinical trial	The end of the clinical trial is defined as the last visit of the last participant.
Enrollment	Point/time of participant entry into the study at which informed consent must be obtained. The action of enrolling one or more participants.
eSource Direct Data Capture (DDC)	eSource Direct Data Capture (DDC) refers to the capture of clinical trial data electronically, at the point of care. eSource Platform/Applications combine source documents and case report forms (CRFs) into one application, allowing for the real time collection of clinical trial information to Novartis and other oversight authorities, as appropriate
Estimand	As defined in the ICH E9(R1) addendum, estimand is a precise description of the treatment effect reflecting the clinical question posed by the trial objective. It summarizes

	at a population-level what the outcomes would be in the same participants under different treatment conditions being compared. Attributes of an estimand include the population, variable (or endpoint) and treatment of interest, as well as the specification of how the remaining intercurrent events are addressed and a population-level summary for the variable.
Healthy volunteer	A person with no known significant health problems who volunteers to be a study participant
Intercurrent events	Events occurring after treatment initiation that affect either the interpretation or the existence of the measurements associated with the clinical question of interest.
Medication number	A unique identifier on the label of medication kits
Off-site	Describes trial activities that are performed at a remote location by an off-site healthcare professional, such as procedures performed at the participant's home.
Off-site healthcare Professional (OHP)	A qualified healthcare professional, who performs certain protocol procedures for the participant in an off-site location such as a participant's home.
Other treatment	Treatment that may be needed/allowed during the conduct of the study (e.g., concomitant or rescue therapy)
Part	A sub-division of a study used to evaluate specific objectives or contain different populations. For example, one study could contain a single dose part and a multiple dose part, or a part in participants with established disease and in those with newly-diagnosed disease
Participant number	A unique number assigned to each participant upon signing the informed consent. This number is the definitive, unique identifier for the participant and should be used to identify the participant throughout the study for all data collected, sample labels, etc., sample labels, etc.
Patient-Reported Outcome (PRO)	A measurement based on a report that comes directly from the participant about the status of a participant's health condition without amendment or interpretation of the participant's report by a clinician or anyone else
Period	The subdivisions of the trial design (e.g., Screening, Treatment, Follow-up) which are described in the Protocol. Periods define the study phases and will be used in clinical trial database setup and eventually in analysis
Personal data	Participant information collected by the Investigator that is coded and transferred to Novartis for the purpose of the clinical trial. This data includes participant identifier information, study information and biological samples.
Remote	Describes any trial activities performed at a location that is not the investigative site.
Rescreening	If a participant fails the initial screening and is considered as a Screen Failure, the participant can be invited once for a new Screening visit after medical judgment and as specified by the protocol
Rescue Medication	A medication used to control symptoms that are not adequately controlled on investigational and other study treatment.
Screen Failure	A participant who did not meet one or more criteria that were required for participation in the study
Source Data/Document	Source data refers to the initial record, document, or primary location from where data comes. The data source can be a database, a dataset, a spreadsheet or even hard-coded data, such as paper or eSource
Stage in cancer	The extent of a cancer in the body. Staging is usually based on the size of the tumor, whether lymph nodes contain cancer, and whether the cancer has spread from the original site to other parts of the body
Start of the clinical trial	The start of the clinical trial is defined as the signature of the informed consent by the first participant
Study device	Study device is a medical device (marketed or investigational) that is used in a circumstance that makes it part of the investigation.
Study treatment	Any drug or combination of drugs, device or intervention administered to the study participants as part of the required study procedures; includes, for example, investigational drug(s).

Tele-visit	Procedures or communications conducted using technology such as telephone or video-conference, whereby the participant is not at the investigative site where the Investigator will conduct the trial.
Treatment arm/group	A treatment arm/group defines the dosage and regimen or the combination, and may consist of 1 or more cohorts.
Variable (or endpoint)	The variable (or endpoint) to be obtained for each participant that is required to address the clinical question. The specification of the variable might include whether the participant experiences an intercurrent event.
Withdrawal of consent	Withdrawal of consent from the study occurs when the participant explicitly requests to stop use of their data and/or biological samples AND no longer wishes to receive study treatment, AND does not agree to further protocol required assessments. This request should be in writing (depending on local regulations) and recorded in the source documentation. This request should be distinguished from a request to discontinue the study. Other study participant's privacy rights are described in the corresponding informed consent form.

10.3 Appendix 3: Clinical laboratory tests

Table 10-1 Clinical laboratory tests and parameters

Laboratory Tests	Parameters
Hematology	Hematocrit, Hemoglobin, Platelets, Leukocytes, Differential (Basophils, Eosinophils, Lymphocytes, Monocytes, Neutrophils, Bands (if required), Other (absolute value preferred, percentages are acceptable)), Reticulocytes
Chemistry	Albumin, Alkaline phosphatase (ALP), ALT, Amylase, AST, Bicarbonate, Calcium, Chloride, Creatinine, Direct Bilirubin, Glucose (non-fasting or fasting), Lipase, Magnesium, Phosphate, Potassium, Sodium, Total Bilirubin, Urea Nitrogen or Urea
Urinalysis	Macroscopic panel (Dipstick) (Bilirubin, Blood, Glucose, Ketones, Leukocyte esterase, Nitrite, pH, Protein, Specific Gravity, Urobilinogen), Urine protein-creatinine ratio If macroscopic panel is abnormal, then perform microscopic panel (Erythrocytes, Leukocytes, Casts, Crystals, Bacteria, Epithelial cells)
Coagulation	Prothrombin time (PT), International normalized ratio (INR), Activated partial thromboplastin time (APTT)
Thyroid	Free T4 and/or T3, TSH
Pregnancy Test	Serum pregnancy testing, Urine pregnancy test

10.4 Appendix 4: Statistical details of the Bayesian hierarchical logistic regression model (BHLRM): priors, design, operating characteristics and hypothetical data scenarios

10.4.1 Design specification

This trial explores dosage escalation of ^{225}Ac-DWJ155 single agent Q6W, Q8W and Q4W administration schedules. The DLT observation period is first cycle: 6 weeks, 8 weeks or 4 weeks starting from the first administration of ^{225}Ac-DWJ155, depending on the administration schedule.

For each administration schedule of Q6W, Q8W and Q4W, the administered radioactivity levels in the first cycle are 2, 4, 6 and 8 MBq, (see Section 6.5.2 for provisional administered activity levels). The adaptive Bayesian design nature of the BHLRM allows additional and/or intermediate dosage levels to be added. In that case, this same BHLRM will be able to support dosage escalation for all dosage levels for each schedule and to provide inferences of posterior probability of DLT rates across dosage levels for identifying MTA/RD.

Proposed statistical model uses a meta-analytic framework (Spiegelhalter et al 2004, Neuenschwander et al 2016) to combine all data from the Q6W, Q8W and Q4W schedules, which allows robust exchangeability among strata. The administered activity in the DLT observation period (first 4 weeks for Q4W, first 6 weeks for Q6W and first 8 weeks for Q8W) is used in the model to incorporate data from both Q4W, Q6W and Q8W schedules.

Let π_j (d) be the risk of DLT for ^{225}Ac-DWJ155 at administered activity d during the first cycle on the jth (schedule). The single agent administered activity-DLT model is logistic:

^{225}Ac-DWJ155: $\text{logit}(\pi_j(d)) = \log(\alpha_j) + \beta_j \log(d/d^*)$, j=1,2

where d*=6 MBq is used to scale the doses of Q6W, Q8W and Q4W, respectively for each strata. α_j (>0) is the single-agent odds of a DLT at d* mg; and β_j is the increase in the log-odds of a DLT by a unit increase in log-administered activity.

In order to borrow information, we assume that $\theta_j=(\log(\alpha_j), \log(\beta_j))$ follows a bivariate normal distribution:

$$\theta_j \sim BVN(\mu, \Gamma),$$

and the parameters are set as:

$$\mu = (\mu_\alpha, \mu_\beta) \text{ and } \Gamma = \begin{bmatrix} \tau_\alpha^2 & \rho\tau_\alpha\tau_\beta \\ \rho\tau_\alpha\tau_\beta & \tau_\beta^2 \end{bmatrix}.$$

where μ and Γ are the mean vector and covariance matrix.

Prior distributions for the parameters of the distribution are as follows:

- normal distributions for the means μ_α and μ_β

- log-normal distributions for the standard deviations τ_α and τ_β
- a uniform distribution for the correlation ρ

A full description including parameter specification, operating characteristics and hypothetical administered activity scenarios of the model will be given in a stand-alone document(s), prior to the first participant is treated on cohort with priming administered activity or alternative administration schedule(s).

10.4.2 Prior specifications

The Bayesian approach requires the specification of prior distributions for all model parameters. Weakly informative normal priors are assumed for μ_α and μ_β, with means corresponding to a risk of DLT at the reference administered activity of 6 MBq, and a doubling in administered activity leading to a doubling in the odds of the risk of a DLT, respectively. Priors for t_α and t_β are assigned such that (1) their medians correspond to substantial between administration schedule heterogeneity, and (2) their uncertainty (95% prior interval) cover plausible between-administration schedule standard deviations (Schmidli et al 2014). The prior distributions for the model are specified in Table 10-2.

Table 10-2 Prior distributions for the parameters of the BHLRM model used to derive the prior for the ^{225}Ac-DWJ155 single-agent model parameters

Parameter	Prior distribution
μ_α	N(mean = logit(0.17), sd =1)
μ_β	N(mean = 0, sd= log(4)/1.96)
τ_α	log-normal(mean = log(0.25), sd = log(2)/1.96)
τ_β	log-normal(mean = log(0.125), sd = log(2)/1.96)
ρ	uniform(-1,1)

10.4.3 Summary of prior distributions

Summaries of the prior distribution of the probability of DLT given provisional administered activities of ^{225}Ac-DWJ155 are given in Table 10-3.

Table 10-3 Summary of the prior distribution of DLT probability for the provisional administered activities of ^{225}Ac-DWJ155

^{225}Ac-DWJ155 dosage level (total cycle, MBq)	Prior probabilities that P(DLT) is in the interval:			Quantiles:				
	[0, 0.16]	[0.16, 0.33]	[0.33, 1]	Mean	SD	2.5%	50%	97.5%
Q6W								
1	0.906	0.075	0.019	0.058	0.082	0.0	0.027	0.302
2	0.829	0.131	0.041	0.089	0.102	0.001	0.054	0.388
4	0.659	0.240	0.101	0.149	0.130	0.012	0.109	0.502
6	0.474	0.333	0.194	0.210	0.155	0.026	0.170	0.602
8	0.345	0.343	**0.312**	0.269	0.184	0.035	0.228	0.706
Q4W								
1	0.905	0.077	0.018	0.058	0.083	0.0	0.027	0.299
2	0.829	0.132	0.039	0.090	0.103	0.001	0.053	0.377

^{225}Ac-DWJ155 dosage level (total cycle, MBq)	**Prior probabilities that P(DLT) is in the interval:**					**Quantiles:**		
4	0.659	0.238	0.103	0.149	0.130	0.012	0.109	0.497
6	0.474	0.327	0.199	0.210	0.155	0.025	0.168	0.598
8	0.347	0.338	**0.315**	0.270	0.184	0.035	0.227	0.711
Q8W								
1	0.902	0.078	0.020	0.058	0.083	0.0	0.027	0.298
2	0.826	0.134	0.041	0.090	0.103	0.001	0.053	0.386
4	0.659	0.238	0.103	0.149	0.131	0.012	0.108	0.507
6	0.480	0.319	0.201	0.210	0.155	0.026	0.168	0.597
8	0.345	0.340	**0.315**	0.270	0.184	0.035	0.226	0.712

DLT probabilities in bold indicate that the administered activity does not satisfy the EWOC criterion.

10.4.4 Hypothetical on-study scenarios

To illustrate the performance of the Bayesian hierarchical model used to guide dosage escalation, a selection of hypothetical dosage escalation scenarios is considered. The results of inference in these scenarios are displayed in Table 10-4, which uses the starting administered activity of 2 MBq ^{225}Ac-DWJ155 Q6W. For each scenario, the next dosage level indicated in the table is the maximum administered activity that is allowable for the next cohort based on the assessment of DLT risk at the administered activity, the EWOC criteria, and the dosage escalation rules described in Section 6.5.3. In practice, a administered activity below the maximum might be chosen based on additional safety, PK or pharmacodynamics information.

Summaries of the prior distribution of the probability of DLT given provisional dosage of ^{225}Ac-DWJ155 are given in Table 10-3.

Table 10-4 Hypothetical dosage escalation scenarios

Scenario	**Schedule**	Dosage level (MBq)	**Ntox / Npat**	**Next administration Schedule**	**Next dosage level (NDL) (MBq)**	**P(target) at NDL**	**P(over) at NDL**	**Mean at NDL**
1.1	Q6W	2	0/6	Q4W	4	0.174	0.026	0.104
				Q6W	4	0.166	0.022	0.101
				Q8W	4	0.178	0.027	0.105
1.2	Q6W	2	1/4	Q4W	4	0.386	0.179	0.211
				Q6W	4	0.402	0.185	0.215
				Q8W	4	0.390	0.181	0.211
2.1	Q6W	2	0/4	Q4W	6	0.412	0.208	0.227
	Q6W	4	1/3	Q6W	6	0.425	0.209	0.228
				Q8W	6	0.418	0.208	0.227
2.2	Q6W	2	0/4	Q4W	6	0.386	0.138	0.194
	Q6W	4	1/6	Q6W	6	0.405	0.125	0.192
				Q8W	6	0.394	0.134	0.194
2.3	Q6W	2	0/4	Q4W*	4	0.444	0.172	0.217
	Q6W	4	2/4	Q6W	2	0.255	0.033	0.126
				Q8W*	4	0.448	0.170	0.216
3.1	Q6W	2	0/4	Q4W	4	0.325	0.051	0.149

Scenario	Schedule	Dosage level (MBq)	Ntox / Npat	Next administration Schedule	Next dosage level (NDL) (MBq)	P(target) at NDL	P(over) at NDL	Mean at NDL
	Q6W	4	0/4	Q6W*	6	0.481	0.173	0.222
	Q4W	6	2/4	Q8W*	6	0.470	0.206	0.233
4.1	Q6W	2	0/4	Q4W	6	0.441	0.136	0.202
	Q6W	4	0/4	Q6W	6	0.471	0.120	0.201
	Q6W	6	2/4	Q8W	6	0.438	0.136	0.203
	Q6W	4	0/5	-	-	-	-	-

*The BHLRM allows increasing the administered activity to a higher administered activity, however, clinical considerations such as timing of DLT, type of DLT from the current regimen may not recommend exploring the higher administered activity for a different regimen.

10.4.5 Operating characteristics

A simulation study is used to illustrate the performance of the Bayesian dosage escalation model for identifying a safe recommended administered activity in simulated clinical trials. The data of each cohort for ^{225}Ac-DWJ155 Q4W, Q6W and Q8W schedules are simulated based on scenarios representing the true DLT risk at each dosage level.

10.4.6 Scenarios of the true DLT probability

In order to test the performance of the design, six hypothetical scenarios are investigated:

- Scenario 1: True DLT rates of Q6W, Q8W and Q4W schedules based on prior median
- Scenario 2: True DLT rates of Q6W, Q8W and Q4W assumes a higher toxicity at higher administered activities
- Scenario 3: All dosage levels in both schedules are excessively toxic (more than 33% DLT-rate)
- Scenario 4: The Q4W stratum assumes a lower toxicity than Q6W and Q8W

Table 10-5 Hypothetical true DLT rates for ^{225}Ac-DWJ155 single agent

^{225}Ac-DWJ155 dosage level (total cycle administered activity, GBq)	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Q6W				
1	0.027	0.0525	**0.45**	0.08
2	0.054	0.123	**0.52**	0.14
4	0.109	0.236	**0.60**	0.22
6	0.170	0.360	**0.70**	0.27
8	0.228	0.454	**0.80**	0.36
Q8W				
1	0.027	0.0525	**0.45**	0.08
2	0.053	0.122	**0.52**	0.14
4	0.108	0.231	**0.60**	0.22
6	0.168	0.360	**0.70**	0.27
8	0.226	0.452	**0.80**	0.36

^{225}Ac-DWJ155 dosage level (total cycle administered activity, GBq)	**Scenario 1**	**Scenario 2**	**Scenario 3**	**Scenario 4**
Q4W				
1	0.027	0.0525	**0.45**	0.04
2	0.053	0.122	**0.52**	0.07
4	0.109	0.230	**0.60**	0.14
6	0.168	0.360	**0.70**	0.22
8	0.227	0.453	**0.80**	0.27
DLT rates in bold indicates the rates are in excessive toxicity interval				

10.4.7 Design to simulate trials

In the simulation study, dosage escalation decisions are simulated based on a rule that selects the administered activity with the highest possible administered activity, participant to the EWOC criterion, and the additional rules described in Section 6.5.3.

The starting administered activity in each simulated trial is ^{225}Ac-DWJ155 2 MBq. The Q6W schedule will start first.

In the simulated trials, escalation continues until an MTA is identified, or until the trial is stopped because no dosage levels satisfy the EWOC criteria. The simulated trials are stopped to declare an MTA if the following criteria are met at the selected administered activity:

- At least 6 participants are treated at the administered activity
- The administered activity satisfies one of the following conditions:
- The posterior probability of targeted toxicity at this administered activity exceeds 50% and is the highest among the potential administered activities, or
- A minimum of 21 participants have already been treated in the dosage escalation part.

Participant enrollment is simulated as follows:

- Cohort size between 3 and 6 evaluable participants (with equal probability)
- Minimum number of participants: 21
- Maximum number of participants: 48

If the maximum number of participants is reached, then the highest administered activity satisfying the EWOC is declared as the MTA.

10.4.8 Metrics to assess the operating characteristics

Operating characteristics are estimated from the simulated trials under the four scenarios. The metrics estimated are:

I. Average number of participants receiving a administered activity associated with a true DLT probability in the underdosing interval, [0, 0.16)

II. Average number of participants receiving a administered activity associated with a true DLT probability in the targeted toxicity interval, [0.16, 0.33)

III. Average number of participants receiving a administered activity associated with a true DLT probability in the overdosing interval, [0.33, 1]

IV. Probability that identified MTA at the end of the trial is in the underdose interval, with true DLT probability in the underdosing interval, [0, 0.16)

V. Probability that identified MTA at the end of the trial is in the target toxicity interval, with true DLT probability in the targeted toxicity interval, [0.16, 0.33)

VI. Probability that identified MTA at the end of the trial is in the overdose interval, with true DLT probability in the overdosing interval, [0.33, 1]

VII. Probability of stopping the trial before declaring MTA because all dose levels are toxic.

VIII. Probability of stopping the trial before declaring MTA because the maximum number of treated participants have been reached.

IX. Average number of DLTs observed.

10.4.9 Results

Table 10-6 presents the operating characteristics estimated from the simulated trials. The following conclusions can be reached:

Scenario 1 corresponds to the prior median for all model parameters. Results are favorable in this setting, with 5.4% Q6W trials, 3.9% Q8W and 4.9% of simulated Q4W trials declaring an MTA in the underdosing range. 94.5% Q6W trials, 96.0% Q8W and 95.1% Q4W declared an MTA in the targeted toxicity range. The remaining simulated trials were stopped for toxicity (Q6W 0.1%, Q8W 0.1% and Q4W 0%).

Scenario 2 corresponds to slightly higher toxicity, with the 2 higher administered activities in the overdosing interval. A little more than half simulated trials declared an MTA in the target toxicity range (Q6W 66.0%, Q8W 67.4% and Q4W 68.4%). 0.7% (Q6W) 0.1% (Q8W) and 0.1% (Q4W) of the simulated trials were stopped for excessive toxicity.

Scenario 3 corresponds to very high toxicity, with all administered activities falling in the overdosing interval. Under this scenario, most of the simulated trials are appropriately stopped for excessive toxicity (Q6W 94.2%, Q8W 96.1%, Q4W 96.5% of trials), with an average sample size of 9.1 (Q6W), 3.18 (Q8W) and 3.15 (Q4W) participants exposed.

Scenario 4 corresponds to lower toxicity in Q4W schedule than in Q6W and Q8W schedules. Results are favorable in this setting, with 71.1% (Q6W) 71.6% (Q8W) and 57.6% (Q4W) of simulated trials declaring an MTA in the targeted toxicity range. The remaining simulated trials either declared an MTA in the underdosing interval (Q6W 9.7%, Q8W 9.6% and 42.4% Q4W) and were stopped for toxicity (Q6W 1.9%, Q8W 0.3% and Q4W 0%).

The simulation results suggest that the model-based dosage-escalation design is effective at identifying an MTA with toxicity in the target range, when such a administered activity exists. In Scenarios 1, moderate sample sizes were required to select an MTA. Furthermore, the design appropriately responds to high toxicity, with the majority of trials in Scenario 3 being stopped early, and with relatively small sample sizes of participants being exposed to highly toxic administered activities.

Table 10-6 Operating characteristics of the adaptative Bayesian approach

Scenario	Metric									
		I Average number of participants at under dose	**II Average number of participants at target dose**	**III Average number of participants at excessive dose**	**IV % MTA declared in under dose**	**V % MTA declared in target dose**	**VI % MTA declared in excessive dose**	**VII % Stopped for toxicity**	**VIII % Stopped because maximum number of participants have been reached**	**IX Average number of DLTs**
1	Q6W	10.58	12.7	0	5.4	94.5	0	0.1	0	3.55
	Q8W	10.52	11.70	0	3.9	96.0	0	0.1	0	3.20
	Q4W	10.57	11.56	0	4.9	95.1	0	0.0	0	3.25
2	Q6W	7.23	8.14	5.64	17.8	66.0	15.5	0.7	0	4.91
	Q8W	7.39	8.19	3.86	16.8	67.4	15.7	0.1	0	4.22
	Q4W	7.44	7.99	3.82	16.1	68.4	15.3	0.1	0.1	4.16
3	Q6W	0	0	9.10	0	0	5.8	94.2	0	4.55
	Q8W	0	0	3.18	0	0	3.9	96.1	0	1.62
	Q4W	0	0	3.15	0	0	3.5	96.5	0	1.56
4	Q6W	7.05	11.80	3.46	9.7	71.1	17.3	1.9	0	4.95
	Q8W	7.18	11.81	2.86	9.6	71.6	18.3	0.3	0.2	4.77
	Q4W	12.81	7.79	0	42.4	57.6	0	0.0	0	3.29

10.5 Appendix 5: RECIST v1.1

10.5.1 Introduction

The purpose of this document is to provide the working definitions and rules necessary for a consistent and efficient analysis of efficacy for oncology studies in solid tumors. This document is based on the RECIST criteria for tumor responses (Therasse et al 2000) and the revised RECIST v1.1 guidelines (Eisenhauer et al 2009).

The efficacy assessments described in Section 10.5.2 and the definition of best response in Section 10.5.3.1 are based on the RECIST v1.1 criteria but also give more detailed instructions and rules for determination of best response. Section 10.5.3.2 is summarizing the "time to event" variables and rules which are mainly derived from internal discussions and regulatory consultations, as the RECIST criteria do not define these variables in detail. Section 10.5.4 of this guideline describes data handling and programming rules. This section is to be referred to in the SAP (Statistical Analysis Plan) to provide further details needed for programming.

10.5.2 Efficacy assessments

Tumor evaluations are made based on RECIST criteria (Therasse et al 2000), and revised RECIST guidelines (version 1.1) (Eisenhauer et al 2009).

10.5.2.1 Definitions

10.5.2.1.1 Disease measurability

In order to evaluate tumors throughout a study, definitions of measurability are required in order to classify lesions appropriately at baseline. In defining measurability, a distinction also needs to be made between nodal lesions (pathological lymph nodes) and non-nodal lesions.

- **Measurable disease** - the presence of at least one measurable nodal or non-nodal lesion. If the measurable disease is restricted to a solitary lesion, its neoplastic nature should be confirmed by cytology/histology.

For participants without measurable disease see Section 10.5.3.2.8.

Measurable lesions (both nodal and non-nodal)

- Measurable non-nodal - As a rule of thumb, the minimum size of a measurable non-nodal target lesion at baseline should be no less than double the slice thickness or 10mm whichever is greater - e.g. the minimum non-nodal lesion size for CT/MRI with 5 mm cuts will be 10 mm, for 8 mm contiguous cuts the minimum size will be 16 mm.
- Lytic bone lesions or mixed lytic-blastic lesions with identifiable soft tissue components, that can be evaluated by CT/MRI, can be considered as measurable lesions, if the soft tissue component meets the definition of measurability.
- Measurable nodal lesions (i.e. lymph nodes) - Lymph nodes ≥15 mm in short axis can be considered for selection as target lesions. Lymph nodes measuring ≥10 mm and <15 mm are considered non-measurable. Lymph nodes smaller than 10 mm in short axis at baseline, regardless of the slice thickness, are normal and not considered indicative of disease.

- Cystic lesions:
- Lesions that meet the criteria for radiographically defined simple cysts (i.e., spherical structure with a thin, non-irregular, non-nodular and non-enhancing wall, no septations, and low CT density [water-like] content) should not be considered as malignant lesions (neither measurable nor non-measurable) since they are, by definition, simple cysts.
- 'Cystic lesions' thought to represent cystic metastases can be considered as measurable lesions, if they meet the definition of measurability described above. However, if noncystic lesions are present in the same participant, these are preferred for selection as target lesions.
- Non-measurable lesions - all other lesions are considered non-measurable, including small lesions (e.g. longest diameter <10 mm with CT/MRI or pathological lymph nodes with ≥10 to <15 mm short axis), as well as truly non-measurable lesions e.g., blastic bone lesions, leptomeningeal disease, ascites, pleural/pericardial effusion, inflammatory breast disease, lymphangitis cutis/pulmonis, abdominal masses/abdominal organomegaly identified by physical exam that is not measurable by reproducible imaging techniques.

10.5.2.1.2 Eligibility based on measurable disease

If no measurable lesions are identified at baseline, the participant may be allowed to enter the study in some situations (e.g. in Phase III studies where PFS is the primary endpoint). However, it is recommended that participants be excluded from trials where the main focus is on the Overall Response Rate (ORR). Guidance on how participants with just non-measurable disease at baseline will be evaluated for response and also handled in the statistical analyses is given in Section 10.5.3.2.8.

10.5.2.2 Methods of tumor measurement - general guidelines

In this document, the term "contrast" refers to intravenous (i.v.) contrast.

The following considerations are to be made when evaluating the tumor:

- All measurements should be taken and recorded in metric notation (mm), using a ruler or calipers. All baseline evaluations should be performed as closely as possible to the beginning of treatment and never more than **4 weeks** before the beginning of the treatment.
- Imaging-based evaluation is preferred to evaluation by clinical examination when both methods have been used to assess the antitumor effect of a treatment.
- For optimal evaluation of participants, the same methods of assessment and technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Contrast-enhanced CT of chest, abdomen and pelvis should preferably be performed using a 5 mm slice thickness with a contiguous reconstruction algorithm. CT/MRI scan slice thickness should not exceed 8 mm cuts using a contiguous reconstruction algorithm. If, at baseline, a participant is known to have a medical contraindication to CT contrast or develops a contraindication during the trial, the following change in imaging modality will be accepted for follow up: a non-contrast CT of chest (MRI not recommended due to respiratory artifacts) plus contrast-enhanced MRI of abdomen and pelvis.

- A change in methodology can be defined as either a change in contrast use (e.g. keeping the same technique, like CT, but switching from with to without contrast use or vice-versa, regardless of the justification for the change) or a major change in technique (e.g. from CT to MRI, or vice-versa), or a change in any other imaging modality. A change from conventional to spiral CT or vice versa will not constitute a major "change in method" for the purposes of response assessment. A change in methodology will result by default in a NE overall lesion response assessment as per Novartis calculated response. However, another response assessment than the Novartis calculated NE response may be accepted from the investigator or the central blinded reviewer if a definitive response assessment can be justified, based on the available information.
- FDG-PET: can complement CT scans in assessing progression (particularly possible for 'new' disease). New lesions on the basis of FDG-PET imaging can be identified according to the following algorithm:

- Negative FDG-PET at baseline, with a positive FDG-PET at follow-up is a sign of PD based on a new lesion.
- No FDG-PET at baseline with a positive FDG-PET at follow-up:
  - If new disease is indicated by a positive PET scan but is not confirmed by CT (or some other conventional technique such as MRI) at the same assessment, then follow-up assessments by CT will be needed to determine if there is truly progression occurring at that site. In all cases PD will be the date of confirmation of new disease by CT (or some other conventional technique such as MRI) rather than the date of the positive PET scan. If there is a positive PET scan without any confirmed progression at that site by CT, then a PD cannot be assigned.
  - If the positive FDG-PET at follow-up corresponds to a pre-existing site of disease on CT that is not progressing on the basis of the anatomic images, this is not PD.
  - Chest x-ray: Lesions on chest x-ray are acceptable as measurable lesions when they are clearly defined and surrounded by aerated lung. However, CT is preferable.
  - Physical exams: Evaluation of lesions by physical examination is accepted when lesions are superficial, with at least 10mm size, and can be assessed using calipers.
  - Ultrasound: When the primary endpoint of the study is objective response evaluation, ultrasound (US) should not be used to measure tumor lesions, unless pre-specified by the protocol. It is, however, a possible alternative to clinical measurements of superficial palpable lymph nodes, subcutaneous lesions and thyroid nodules. US might also be useful to confirm the complete disappearance of superficial lesions usually assessed by clinical examination.
  - Endoscopy and laparoscopy: The utilization of endoscopy and laparoscopy for objective tumor evaluation has not yet been fully and widely validated. Their uses in this specific context require sophisticated equipment and a high level of expertise that may only be available in some centers. Therefore, the utilization of such techniques for objective tumor response should be restricted to validation purposes in specialized centers. However, such techniques can be useful in confirming complete pathological response when biopsies are obtained.

- Tumor markers: Tumor markers alone cannot be used to assess response. However, some disease specific and more validated tumor markers (e.g. CA-125 for ovarian cancer, PSA for prostate cancer, alpha-FP, lactate dehydrogenase (LDH) and Beta-human chorionic gonadotropin (hCG) for testicular cancer) can be integrated as non-target disease. If markers are initially above the upper normal limit they must normalize for a participant to be considered in complete clinical response when all lesions have disappeared.
- **Cytology and histology**: Cytology and histology can be used to differentiate between PR and CR in rare cases (i.e., after treatment to differentiate between residual benign lesions and residual malignant lesions in tumor types such as germ cell tumors). Cytologic confirmation of neoplastic nature of any effusion that appears or worsens during treatment is required when the measurable tumor has met the criteria for response or stable disease. Under such circumstances, the cytologic examination of the fluid collected will permit differentiation between response and stable disease (an effusion may be a side effect of the treatment) or progressive disease (if the neoplastic origin of the fluid is confirmed).
- **Clinical examination**: Clinical lesions will only be considered measurable when they are superficial (i.e., skin nodules and palpable lymph nodes). For the case of skin lesions, documentation by color photography, including a ruler to estimate the size of the lesion, is recommended.

10.5.2.3 Baseline documentation of target and non-target lesions

For the evaluation of lesions at baseline and throughout the study, the lesions are classified at baseline as either target or non-target lesions:

- Target lesions: All measurable lesions (nodal and non-nodal) up to a maximum of five lesions in total (and a maximum of two lesions per organ), representative of all involved organs should be identified as target lesions and recorded and measured at baseline. Target lesions should be selected on the basis of their size (lesions with the longest diameter) and their suitability for accurate repeated measurements (either by imaging techniques or clinically). Each target lesion must be uniquely and sequentially numbered on the CRF (even if it resides in the same organ).

Minimum target lesion size at baseline

- **Non-nodal target**: Non-nodal target lesions identified by methods for which slice thickness is not applicable (e.g. clinical examination, photography) should be at least 10 mm in longest diameter. See Section 10.5.2.1.1.
- **Nodal target**: See Section 10.5.2.1.1.

A sum of diameters (long axis for non-nodal lesions, short axis for nodal) for all target lesions will be calculated and reported as the baseline sum of diameters (SOD). The baseline sum of diameters will be used as reference by which to characterize the objective tumor response. Each target lesion identified at baseline must be followed at each subsequent evaluation and documented on eCRF.

- **Non-target lesions**: All other lesions are considered non-target lesions, i.e. lesions not fulfilling the criteria for target lesions at baseline. Presence or absence or worsening of non-target lesions should be assessed throughout the study; measurements of these lesions are not required. Multiple non-target lesions involved in the same organ can be assessed as a group and recorded as a single item (i.e. multiple liver metastases). Each non-target lesion identified at baseline must be followed at each subsequent evaluation and documented on eCRF.

10.5.2.4 Follow-up evaluation of target and non-target lesions

To assess tumor response, the sum of diameters for all target lesions will be calculated (at baseline and throughout the study). At each assessment response is evaluated first separately for the target (Table 10-2) and non-target lesions (Table 10-3) identified at baseline. These evaluations are then used to calculate the overall lesion response considering both the target and non-target lesions together (Table 10-4) as well as the presence or absence of new lesions.

10.5.2.4.1 Follow-up and recording of lesions

At each visit and for each lesion the actual date of the scan or procedure which was used for the evaluation of each specific lesion should be recorded. This applies to target and non-target lesions as well as new lesions that are detected. At the assessment visit all of the separate lesion evaluation data are examined by the investigator in order to derive the overall visit response. Therefore all such data applicable to a particular visit should be associated with the same assessment number.

Non-nodal lesions

Following treatment, lesions may have longest diameter measurements smaller than the image reconstruction interval. Lesions smaller than twice the reconstruction interval are subject to substantial “partial volume” effects (i.e., size may be underestimated because of the distance of the cut from the longest diameter; such lesions may appear to have responded or progressed on subsequent examinations, when, in fact, they remain the same size).

If the lesion has completely disappeared, the lesion size should be reported as 0 mm.

Measurements of non-nodal target lesions that become 5 mm or less in longest diameter are likely to be non-reproducible. Therefore, it is recommended to report a default value of 5 mm, instead of the actual measurement. This default value is derived from the 5 mm CT slice thickness (but should not be changed with varying CT slice thickness). Actual measurement should be given for all lesions larger than 5 mm in longest diameter irrespective of slice thickness/reconstruction interval.

In other cases where the lesion cannot be reliably measured for reasons other than its size (e.g., borders of the lesion are confounded by neighboring anatomical structures), no measurement should be entered and the lesion cannot be evaluated.

Nodal lesions

A nodal lesion less than 10 mm in size by short axis is considered normal. Lymph nodes are not expected to disappear completely, so a "non-zero size" will always persist.

Measurements of nodal target lesions that become 5 mm or less in short axis are likely to be non-reproducible. Therefore, it is recommended to report a default value of 5 mm, instead of the actual measurement. This default value is derived from the 5 mm CT slice thickness (but should not be changed with varying CT slice thickness). Actual measurement should be given for all lesions larger than 5 mm in short axis irrespective of slice thickness/reconstruction interval.

However, once a target nodal lesion shrinks to less than 10 mm in its short axis, it will be considered normal for response purpose determination. The lymph node measurements will continue to be recorded to allow the values to be included in the sum of diameters for target lesions, which may be required subsequently for response determination.

10.5.2.4.2 Determination of target lesion response

Table 10-7 Response criteria for target lesions

Response Criteria	Evaluation of target lesions
Complete Response (CR):	Disappearance of all non-nodal target lesions. In addition, any pathological lymph nodes assigned as target lesions must have a reduction in short axis to < 10 mm [1]
Partial Response (PR):	At least a 30% decrease in the sum of diameter of all target lesions, taking as reference the baseline sum of diameters.
Progressive Disease (PD):	At least a 20% increase in the sum of diameter of all measured target lesions, taking as reference the smallest sum of diameter of all target lesions recorded at or after baseline. In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm [2].
Stable Disease (SD):	Neither sufficient shrinkage to qualify for PR or CR nor an increase in lesions which would qualify for PD.
NE (Not Evaluable)	Progression has not been documented and one or more target lesions have not been assessed or have been assessed using a different method than baseline.[3]

[1]SOD for CR may not be zero when nodal lesions are part of target lesions

[2]Following an initial CR, a PD cannot be assigned if all non-nodal target lesions are still not present and all nodal lesions are <10 mm in size. In this case, the target lesion response is CR

[3]In exceptional circumstances an NE response due to change in method could be over-ruled by the investigator or central reviewer using expert judgment based on the available information (see Notes on target lesion response and methodology change in Section 10.5.2.2).

Notes on target lesion response

Reappearance of lesions: If the lesion appears at the same anatomical location where a target lesion had previously disappeared, it is advised that the time point of lesion disappearance (i.e., the "0 mm" recording) be re-evaluated to make sure that the lesion was not actually present

and/or not visualized for technical reasons in this previous assessment. If it is not possible to change the 0 value, then the investigator/radiologist has to decide between the following possibilities:

- The lesion is a new lesion, in which case the overall tumor assessment will be considered as progressive disease
- The lesion is clearly a reappearance of a previously disappeared lesion, in which case the size of the lesion has to be entered in the CRF and the tumor assessment will remain based on the sum of tumor measurements as presented in Table 10-2 above (i.e., a PD will be determined if there is at least 20% increase in the sum of diameters of **all** measured target lesions, taking as reference the smallest sum of diameters of all target lesions recorded at or after baseline with at least 5 mm increase in the absolute sum of the diameters). Proper documentation should be available to support this decision. This applies to participants who have not achieved target response of CR. For participants who have achieved CR, please refer to last bullet in this section.
- For those participants who have only one target lesion at baseline, the reappearance of the target lesion which disappeared previously, even if still small, is considered a PD.
- Missing measurements: In cases where measurements are missing for one or more target lesions it is sometimes still possible to assign PD based on the measurements of the remaining lesions. For example, if the sum of diameters for 5 target lesions at baseline is 100 mm at baseline and the sum of diameters for 3 of those lesions at a post-baseline visit is 140 mm (with data for 2 other lesions missing) then a PD should be assigned. However, in other cases where a PD cannot definitely be attributed, the target lesion response would be NE.
- Nodal lesion decrease to normal size: When nodal disease is included in the sum of target lesions and the nodes decrease to “normal” size they should still have a measurement recorded on scans. This measurement should be reported even when the nodes are normal in order not to overstate progression should it be based on increase in the size of nodes.
- Lesions split: In some circumstances, disease that is measurable as a target lesion at baseline and appears to be one mass can split to become two or more smaller sub-lesions. When this occurs, the diameters (long axis - non-nodal lesion, short axis - nodal lesions) of the two split lesions should be added together and the sum recorded in the diameter field on the case report form under the original lesion number. This value will be included in the sum of diameters when deriving target lesion response. The individual split lesions will not be considered as new lesions, and will not automatically trigger a PD designation.
- Lesions coalesced: Conversely, it is also possible that two or more lesions which were distinctly separate at baseline become confluent at subsequent visits. When this occurs a plane between the original lesions may be maintained that would aid in obtaining diameter measurements of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the maximal diameters (long axis - non-nodal lesion, short axis - nodal lesions) of the “merged lesion” should be used when calculating the sum of diameters for target lesions. On the case report form, the diameter of the “merged lesion” should be recorded for the size of one of the original lesions while a size of “0”mm should be entered for the remaining lesion numbers which have coalesced.

- The **measurements for nodal lesions**, even if less than 10 mm in size, will contribute to the calculation of target lesion response in the usual way with slight modifications.
- Since lesions less than 10 mm are considered normal, a CR for target lesion response should be assigned when all nodal target lesions shrink to less than 10 mm and all non-nodal target lesions have disappeared.
- Once a CR target lesion response has been assigned a CR will continue to be appropriate (in the absence of missing data) until progression of target lesions.
- Following a CR, a PD can subsequently only be assigned for target lesion response if either a non-nodal target lesion "reappears" or if any single nodal lesion is at least 10 mm and there is at least 20% increase in sum of the diameters of all nodal target lesions relative to nadir with at least 5 mm increase in the absolute sum of the diameters.
- A change in method for the evaluation of one or more lesions will usually lead to an NE target lesion response unless there is progression indicated by the remaining lesions which have been evaluated by the same method. In exceptional circumstances an investigator or central reviewer might over-rule this assignment to put a non-NE response using expert judgment based on the available information. E.g. a change to a more sensitive method might indicate some tumor shrinkage of target lesions and definitely rule out progression in which case the investigator might assign an SD target lesion response; however, this should be done with caution and conservatively as the response categories have well defined criteria.

10.5.2.4.3 Determination of non-target lesion response

Table 10-8 Response criteria for non-target lesions

Response Criteria	Evaluation of non-target lesions
Complete Response (CR):	Disappearance of all non-target lesions. In addition, all lymph nodes assigned a non-target lesions must be non-pathological in size (< 10 mm short axis)
Progressive Disease (PD):	Unequivocal progression of existing non-target lesions.[1]
Non-CR/Non-PD:	Neither CR nor PD
NE (Not Evaluable)	Progression has not been documented and one or more non-target lesions have not been assessed or have been assessed using a different method than baseline[2].

[1]The assignment of PD solely based on change in non-target lesions in light of target lesion response of CR, PR or SD should be exceptional. In such circumstances, the opinion of the investigator or central reviewer does prevail.

[2]It is recommended that the investigator and/or central reviewer should use expert judgment to assign a Non-NE response wherever possible (see notes section for more details)

Notes on non-target lesion response

- The investigator and/or central reviewer can use expert judgment to assign a non-NE response wherever possible, even where lesions have not been fully assessed or a different method has been used. In many of these situations it may still be possible to identify equivocal progression (PD) or definitively rule this out (non-CR/Non-PD) based on the

available information. In the specific case where a more sensitive method has been used indicating the absence of any non-target lesions, a CR response can also be assigned.

- The response for non-target lesions is **CR** only if all non-target non-nodal lesions which were evaluated at baseline are now all absent and with all non-target nodal lesions returned to normal size (i.e. < 10 mm). If any of the non-target lesions are still present, or there are any abnormal nodal lesions (i.e. ≥ 10 mm) the response can only be '**Non-CR/Non-PD**' unless there is unequivocal progression of the non-target lesions (in which case response is **PD**) or it is not possible to determine whether there is unequivocal progression (in which case response is NE).
- Unequivocal progression: To achieve "unequivocal progression" on the basis of non-target disease there must be an overall level of substantial worsening in non-target disease such that, even in presence of CR, PR or SD in target disease, the overall tumor burden has increased sufficiently to merit discontinuation of therapy. A modest "increase" in the size of one or more non-target lesions is usually not sufficient to qualify for unequivocal progression status. The designation of overall progression solely on the basis of change in non-target disease in the face of CR, PR or SD of target disease is therefore expected to be rare. In order for a PD to be assigned on the basis of non-target lesions, the increase in the extent of the disease must be substantial even in cases where there is no measurable disease at baseline. If there is unequivocal progression of non-target lesion(s), then at least one of the non-target lesions must be assigned a status of "Worsened". Where possible, similar rules to those described in Section 10.5.2.4.2 for assigning PD following a CR for the non-target lesion response in the presence of non-target lesions nodal lesions should be applied.

10.5.2.4.4 New lesions

The appearance of a new lesion is always associated with Progressive Disease (PD) and has to be recorded as a new lesion in the New Lesion CRF page.

- If a new lesion is **equivocal**, for example because of its small size, continued therapy and follow-up evaluation will clarify if it represents truly new disease. If repeat scans confirm there is definitely a new lesion, then progression should be declared using the date of the first observation of the lesion
- If new disease is observed in a region which was **not scanned at baseline** or where the particular baseline scan is not available for some reason, then this should be considered as a PD. The one exception to this is when there are no baseline scans at all available for a participant in which case the response should be NE, as for any of this participant's assessment (see Section 10.5.2.5).
- A **lymph node is considered as a "new lesion"** and, therefore, indicative of progressive disease if the short axis increases in size to ³ 10 mm for the first time in the study plus 5 mm absolute increase.
  FDG-PET: can complement CT scans in assessing progression (particularly possible for 'new' disease). See Section 10.5.2.2.

10.5.2.5 Evaluation of overall lesion response

The evaluation of overall lesion response at each assessment is a composite of the target lesion response, non-target lesion response and presence of new lesions as shown below in Table 10-9.

Table 10-9 Overall lesion response at each assessment

Target lesions	Non-target lesions	New Lesions	Overall lesion response
CR	CR	No	CR[1]
CR	Non-CR/Non-PD[3]	No	PR
CR, PR, SD	NE	No	NE
PR	Non-PD and not NE	No	PR[1]
SD	Non-PD and not NE	No	SD[1, 2]
NE	Non-PD or NE	No	NE[1]
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

[1]This overall lesion response also applies when there are no non-target lesions identified at baseline.

[2]Once confirmed PR was achieved, all these assessments are considered PR.

[3]As defined in Section 10.5.2.4.

10.5.3 Efficacy definitions

The following definitions primarily relate to participants who have measurable disease at baseline. Section 10.5.3.2.8 outlines the special considerations that need to be given to participants with no measurable disease at baseline in order to apply the same concepts.

10.5.3.1 Best overall response

The best overall response is the best response recorded from the start of the treatment until disease progression/recurrence (taking as reference for PD the smallest measurements recorded since the treatment started). In general, the participant's best response assignment will depend on the achievement of both measurement and confirmation criteria.

The best overall response will usually be determined from response assessments undertaken while on treatment. However, if any assessments occur after treatment withdrawal the protocol should specifically describe if these will be included in the determination of best overall response and/or whether these additional assessments will be required for sensitivity or supportive analyses. As a default, any assessments taken more than 30 days after the last dose of study treatment will not be included in the best overall response derivation. If any alternative cancer therapy is taken while on study any subsequent assessments would ordinarily be excluded from the best overall response determination. If response assessments taken after withdrawal from study treatment and/or alternative therapy are to be included in the main endpoint determination, then this should be described and justified in the protocol.

Where a study requires confirmation of response (PR or CR), changes in tumor measurements must be confirmed by repeat assessments that should be performed not less than 4 weeks after the criteria for response are first met.

Longer intervals may also be appropriate. However, this must be clearly stated in the protocol. The main goal of confirmation of objective response is to avoid overestimating the response rate observed. In cases where confirmation of response is not feasible, it should be made clear when reporting the outcome of such studies that the responses are not confirmed.

- For non-randomized trials where response is the primary endpoint, confirmation is needed.
- For trials intended to support accelerated approval, confirmation is needed
- For all other trials, confirmation of response may be considered optional.

The best overall response for each participant is determined from the sequence of overall (lesion) responses according to the following rules:

- CR = at least two determinations of CR at least 4 weeks apart before progression where confirmation required or one determination of CR prior to progression where confirmation not required
- PR = at least two determinations of PR or better at least 4 weeks apart before progression (and not qualifying for a CR) where confirmation required or one determination of PR prior to progression where confirmation not required
- SD = at least one SD assessment (or better) > 6 weeks after randomization/start of treatment (and not qualifying for CR or PR).
- PD = progression £ 12 weeks after randomization/ start of treatment (and not qualifying for CR, PR or SD).
- NE = all other cases (i.e. not qualifying for confirmed CR or PR and without SD after more than 6 weeks or early progression within the first 12 weeks)

The time durations specified in the SD/PD/NE definitions above are defaults based on a 6 week tumor assessment frequency. However these may be modified for specific indications which are more or less aggressive. In addition, it is envisaged that the time duration may also take into account assessment windows. E.g. if the assessment occurs every 6 weeks with a time window of +/- 7 days, a BOR of SD would require a SD or better response longer than 5 weeks after randomization/start of treatment.

Overall lesion responses of CR must stay the same until progression sets in, with the exception of a NE status. A participant who had a CR cannot subsequently have a lower status other than a PD, e.g. PR or SD, as this would imply a progression based on one or more lesions reappearing, in which case the status would become a PD.

Once an overall lesion response of PR is observed (which may have to be a confirmed PR depending on the study) this assignment must stay the same or improve over time until progression sets in, with the exception of an NE status. However, in studies where confirmation of response is required, if a participant has a single PR (³30% reduction of tumor burden compared to baseline) at one assessment, followed by a <30% reduction from baseline at the next assessment (but not ³20% increase from previous smallest sum), the objective status at that assessment should be SD. Once a confirmed PR was seen, the overall lesion response should be considered PR (or NE) until progression is documented or the lesions totally disappear in which case a CR assignment is applicable. In studies where confirmation of response is not required after a single PR the overall lesion response should still be considered PR (or NE) until

progression is documented or the lesion totally disappears in which case a CR assignment is applicable.

Example: In a case where confirmation of response is required the sum of lesion diameters is 200 mm at baseline and then 140 mm - 150 mm - 140 mm - 160 mm - 160 mm at the subsequent visits. Assuming that non-target lesions did not progress, the overall lesion response would be PR - SD - PR - PR - PR. The second assessment with 140 mm confirms the PR for this participant. All subsequent assessments are considered PR even if tumor measurements decrease only by 20% compared to baseline (200 mm to 160 mm) at the following assessments.

If the participant progressed but continues study treatment, further assessments are not considered for the determination of best overall response.

Note: these cases may be described as a separate finding in the CSR but not included in the overall response or disease control rates.

The best overall response for a participant is always calculated, based on the sequence of overall lesion responses. However, the overall lesion response at a given assessment may be provided from different sources:

- Investigator overall lesion response
- Central Blinded Review overall lesion response
- Novartis calculated overall lesion response (based on measurements from either Investigator or Central Review)

The primary analysis of the best overall response will be based on the sequence of investigator/central blinded review/calculated (investigator)/calculated (central) overall lesion responses.

Based on the participants' best overall response during the study, the following rates are then calculated:

Overall response rate (ORR) is the proportion of participants with a best overall response of CR or PR. This is also referred to as 'Objective response rate' in some protocols or publications.

Disease control rate (DCR) is the proportion of participants with a best overall response of CR or PR or SD. The objective of this endpoint is to summarize participants with signs of "activity" defined as either shrinkage of tumor (regardless of duration) or slowing down of tumor growth.

Clinical benefit rate (CBR) is the proportion of participants with a best overall response of CR or PR , or an overall lesion response of SD or Non-CR/Non-PD which lasts for a minimum time duration (with a default of at least 24 weeks in breast cancer studies). This endpoint measures signs of activity taking into account duration of disease stabilization.

Another approach is to summarize the progression rate at a certain time point after baseline. In this case, the following definition is used:

Early progression rate (EPR) is the proportion of participants with progressive disease within 8 weeks of the start of treatment.

The protocol should define populations for which these will be calculated. The timepoint for EPR is study specific. EPR is used for the multinomial designs of Dent et al 2001 and counts

all participants who at the specified assessment (in this example the assessment would be at 8 weeks ± window) do not have an overall lesion response of SD, PR or CR. Participants with an not evaluable (NE) assessment at that time point and no PD before, will not be counted as early progressors in the analysis but may be included in the denominator of the EPR rate, depending on the analysis population used. Similarly when examining overall response and disease control, participants with a best overall response assessment of not evaluable (NE) will not be regarded as "responders" but may be included in the denominator for ORR and DCR calculation depending on the analysis population (e.g. populations based on an intention to treat (ITT) approach).

10.5.3.2 Time to event variables

10.5.3.2.1 Progression-free survival

Usually in all Oncology studies, participants are followed for tumor progression after discontinuation of study medication for reasons other than progression or death. If this is not used, e.g. in Phase I or II studies, this should be clearly stated in the protocol. Note that randomized trials (preferably blinded) are recommended where PFS is to be the primary endpoint.

Progression-free survival (PFS) is the time from date of randomization/start of treatment to the date of event defined as the first documented progression or death due to any cause. If a participant has not had an event, progression-free survival is censored at the date of last adequate tumor assessment.

PFS rate at x weeks is an additional measure used to quantify PFS endpoint. It is recommended that a Kaplan Meier estimate is used to assess this endpoint.

10.5.3.2.2 Overall survival

All participants should be followed until death or until participant has had adequate follow-up time as specified in the protocol whichever comes first. The follow-up data should contain the date the participant was last seen alive / last known date participant alive, the date of death and the reason of death ("Study indication" or "Other").

Overall survival (OS) is defined as the time from date of randomization/start of treatment to date of death due to any cause. If a participant is not known to have died, survival will be censored at the date of last known date participant alive.

10.5.3.2.3 Time to progression

Some studies might consider only death related to underlying cancer as an event which indicates progression. In this case the variable "Time to progression" might be used. TTP is defined as PFS except for death unrelated to underlying cancer.

Time to progression (TTP) is the time from date of randomization/start of treatment to the date of event defined as the first documented progression or death due to underlying cancer. If a participant has not had an event, time to progression is censored at the date of last adequate tumor assessment.

10.5.3.2.4 PFS2

A recent EMA guidance (EMA Guidance 2012) recommends a substitute end point intermediate to PFS and OS called PFS2, a surrogate for OS when OS cannot be measured reliably, which assesses the impact of the experimental therapy on next-line treatment. The main purpose of this endpoint is to assess long term maintenance strategies, particularly of resensitizing agents and where it is necessary to examine the overall "field of influence".

PFS2, which could be termed PFS deferred, PFS delayed, tandem PFS, or PFS version 2.0, is the time from date of randomization/start of treatment to the date of event defined as the first documented progression on next-line treatment or death from any cause. The censoring rules for this endpoint will incorporate the same principles as those considered for PFS in this document, and in addition may involve other considerations which will need to be detailed in the protocol.

Please note that data collection for the PFS2 is limited to the date of progression and not specific read of the tumor assessments.

It is strongly recommended that the teams consult regulatory agencies for scientific advice given the limited experience with the use of this endpoint in regulatory setting in light of methodological issues wtih respect to censoring foreseen.

10.5.3.2.5 Time to treatment failure

This endpoint is often appropriate in studies of advanced disease where early discontinuation is typically related to intolerance of the investigational drug. In some protocols, time to treatment failure may be considered as a sensitivity analysis for time to progression. The list of discontinuation reasons to be considered or not as treatment failure may be adapted according to the specificities of the study or the disease.

Time to treatment failure (TTF) is the time from date of randomization/start of treatment to the earliest of date of progression, date of death due to any cause, or date of discontinuation due to reasons other than 'Protocol violation' or 'Administrative problems'. The time to treatment failure for participants who did not experience treatment failure will be censored at last adequate tumor assessment.

10.5.3.2.6 Duration of response

The analysis of the following variables should be performed with much caution when restricted to responders since treatment bias could have been introduced. There have been reports where a treatment with a significantly higher response rate had a significantly shorter duration of response but where this probably primarily reflected selection bias which is explained as follows: It is postulated that there are two groups of participants: a good risk group and a poor risk group. Good risk participants tend to get into response readily (and relatively quickly) and tend to remain in response after they have a response. Poor risk participants tend to be difficult to achieve a response, may have a longer time to respond, and tend to relapse quickly when they do respond. Potent agents induce a response in both good risk and poor risk participants. Less potent agents induce a response mainly in good risk participants only. This is described in more detail by Morgan 1988.

It is recommended that an analysis of all participants (both responders and non-responders) be performed whether or not a "responders only" descriptive analysis is presented. An analysis of responders should only be performed to provide descriptive statistics and even then interpreted with caution by evaluating the results in the context of the observed response rates. If an inferential comparison between treatments is required this should only be performed on all participants (i.e. not restricting to "responders" only) using appropriate statistical methods such as the techniques described in Ellis et al 2008. It should also be stated in the protocol if duration of response is to be calculated in addition for unconfirmed response.

For summary statistics on "responders" only the following definitions are appropriate. (Specific definitions for an all-participant analysis of these endpoints are not appropriate since the status of participants throughout the study is usually taken into account in the analysis).

Duration of overall response (CR or PR): For participants with a CR or PR (which may have to be confirmed the start date is the date of first documented response (CR or PR) and the end date and censoring is defined the same as that for time to progression.

The following two durations might be calculated in addition for a large Phase III study in which a reasonable number of responders is seen.

Duration of overall complete response (CR): For participants with a CR (which may have to be confirmed) the start date is the date of first documented CR and the end date and censoring is defined the same as that for time to progression.

Duration of stable disease (CR/PR/SD): For participants with a CR or PR (which may have to be confirmed) or SD the start and end date as well as censoring is defined the same as that for time to progression.

10.5.3.2.7 Time to response

Time to overall response (CR or PR) is the time between date of randomization/start of treatment until first documented response (CR or PR). The response may need to be confirmed depending on the type of study and its importance. Where the response needs to be confirmed then time to response is the time to the first CR or PR observed.

Although an analysis on the full population is preferred a descriptive analysis may be performed on the "responders" subset only, in which case the results should be interpreted with caution and in the context of the overall response rates, since the same kind of selection bias may be introduced as described for duration of response in Section 10.5.3.2.5. It is recommended that an analysis of all participants (both responders and non-responders) be performed whether or not a "responders only" descriptive analysis is presented. Where an inferential statistical comparison is required, then all participants should definitely be included in the analysis to ensure the statistical test is valid. For analysis including all participants, participants who did not achieve a response (which may have to be a confirmed response) will be censored using one of the following options.

- at maximum follow-up (i.e. FPFV to LPLV used for the analysis) for participants who had a PFS event (i.e. progressed or died due to any cause). In this case the PFS event is the worst possible outcome as it means the participant cannot subsequently respond. Since the statistical analysis usually makes use of the ranking of times to response it is sufficient to

assign the worst possible censoring time which could be observed in the study which is equal to the maximum follow-up time (i.e. time from FPFV to LPLV)

- at last adequate tumor assessment date otherwise. In this case participants have not yet progressed so they theoretically still have a chance of responding

Time to overall complete response (CR) is the time between dates of randomization/start of treatment until first documented CR. Similar analysis considerations including (if appropriate) censoring rules apply for this endpoint described for the time to overall response endpoint.

10.5.3.2.8 Definition of start and end dates for time to event variables

Assessment date

For each assessment (i.e. evaluation number), the **assessment date** is calculated as the latest of all measurement dates (e.g. X-ray, CT-scan) if the overall lesion response at that assessment is CR/PR/SD/NE. Otherwise - if overall lesion response is progression - the assessment date is calculated as the earliest date of all measurement dates at that evaluation number.

In the calculation of the assessment date for time to event variables, any unscheduled assessment should be treated similarly to other evaluations.

Start dates

For all "time to event" variables, other than duration of response, the randomization/ date of treatment start will be used as the start date.

For the calculation of duration of response the following start date should be used:

- Date of first documented response is the assessment date of the first overall lesion response of CR (for duration of overall complete response) or CR / PR (for duration of overall response) respectively, when this status is later confirmed.

End dates

The end dates which are used to calculate 'time to event' variables are defined as follows:

- Date of death (during treatment as recorded on the treatment completion page or during follow-up as recorded on the study evaluation completion page or the survival follow-up page).
- Date of progression is the first assessment date at which the overall lesion response was recorded as progressive disease.
- Date of last adequate tumor assessment is the date the last tumor assessment with overall lesion response of CR, PR or SD which was made before an event or a censoring reason occurred. In this case the last tumor evaluation date at that assessment is used. If no post-baseline assessments are available (before an event or a censoring reason occurred) the date of randomization/start of treatment is used.
- Date of next scheduled assessment is the date of the last adequate tumor assessment plus the protocol specified time interval for assessments. This date may be used if back-dating is considered when the event occurred beyond the acceptable time window for the next tumor assessment as per protocol (see Section 10.5.3.2.8).

Example (if protocol defined schedule of assessments is 3 months): tumor assessments at baseline - 3 months - 6 months - missing - missing - PD. Date of next scheduled assessment would then correspond to 9 months.

- Date of discontinuation is the date of the end of treatment visit.
- Date of last contact is defined as the last date the participant was known to be alive. This corresponds to the latest date for either the visit date, lab sample date or tumor assessment date. If available, the last known date participant alive from the survival follow-up page is used. If no survival follow-up is available, the date of discontinuation is used as last contact date.
- Date of secondary anti-cancer therapy is defined as the start date of any additional (secondary) antineoplastic therapy or surgery.

10.5.3.2.9 Handling of participants with non-measurable disease only at baseline

It is possible that participants with only non-measurable disease present at baseline are entered into the study, either because of a protocol violation or by design (e.g. in Phase III studies with PFS as the primary endpoint). In such cases the handling of the response data requires special consideration with respect to inclusion in any analysis of endpoints based on the overall response evaluations.

It is recommended that any participants with only non-measurable disease at baseline should be included in the main (ITT) analysis of each of these endpoints.

Although the text of the definitions described in the previous sections primarily relates to participants with measurable disease at baseline, participants without measurable disease should also be incorporated in an appropriate manner. The overall response for participants with non-measurable disease is derived slightly differently according to Table 10-10.

Table 10-10 Overall lesion response at each assessment: participants with non-target disease only

Non-target lesions	New Lesions	Overall lesion response
CR	No	CR
Non-CR/Non-PD[1]	No	Non-CR/non-PD
NE	No	NE
PD	Yes or No	PD
Any	Yes	PD

[1]As defined in Section 10.5.2.4.

In general, the **non-CR/non-PD response** for these participants is considered equivalent to an SD response in endpoint determination. In summary tables for best overall response participants with only non-measurable disease may be highlighted in an appropriate fashion e.g. in particular by displaying the specific numbers with the non-CR/non-PD category.

In considering how to incorporate data from these participants into the analysis the importance to each endpoint of being able to identify a PR and/or to determine the occurrence and timing of progression needs to be taken into account.

For ORR it is recommended that the main (ITT) analysis includes data from participants with only non-measurable disease at baseline, handling participants with a best response of CR as "responders" with respect to ORR and all other participants as "non-responders".

For PFS, it is again recommended that the main ITT analyses on these endpoints include all participants with only non-measurable disease at baseline, with possible sensitivity analyses which exclude these particular participants. Endpoints such as PFS which are reliant on the determination and/or timing of progression can incorporate data from participants with only non-measurable disease.

10.5.3.2.10 Sensitivity analyses

This section outlines the possible event and censoring dates for progression, as well as addresses the issues of missing tumor assessments during the study. For instance, if one or more assessment visits are missed prior to the progression event, to what date should the progression event be assigned? And should progression event be ignored if it occurred after a long period of a participant being lost to follow-up? It is important that the protocol and RAP specify the primary analysis in detail with respect to the definition of event and censoring dates and also include a description of one or more sensitivity analyses to be performed.

Based on definitions outlined in Section 10.5.3.2.7, and using the draft FDA guideline on endpoints (Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics, April 2005) as a reference, the following analyses can be considered:

Table 10-11 Options for event dates used in PFS, TTP, duration of response

	Situation	**Options for end-date (progression or censoring)[1]** **(1) = default unless specified differently in the protocol or RAP**	**Outcome**
A	No baseline assessment	(1) Date of randomization/start of treatment[3]	Censored
B	Progression at or before next scheduled assessment	(1) Date of progression (2) Date of next scheduled assessment[2]	Progressed Progressed
C1	Progression or death after **exactly one** missing assessment	(1) Date of progression (or death) (2) Date of next scheduled assessment[2]	Progressed Progressed
C2	Progression or death after **two or more** missing assessments	(1) Date of last adequate assessment[2] (2) Date of next scheduled assessment[2] (3) Date of progression (or death)	Censored Progressed Progressed
D	No progression	(1) Date of last adequate assessment	Censored

	Situation	**Options for end-date (progression or censoring)[1]** **(1) = default unless specified differently in the protocol or RAP**	**Outcome**
E	Treatment discontinuation due to 'Disease progression' without documented progression, i.e. clinical progression based on investigator claim	(1) Ignore clinical progression and follow situations above (2) Date of discontinuation (visit date at which clinical progression was determined)	As per above situations Progressed
F	New anticancer therapy given	• Ignore the new anticancer therapy and follow situations above (ITT approach) • Date of last adequate assessment prior to new anticancer therapy (3) Date of secondary anti-cancer therapy (4) Date of secondary anti-cancer therapy	As per above situations Censored Censored Event
G	Deaths due to reason other than deterioration of 'Study indication'	(1) Date of last adequate assessment	Censored (only TTP and duration of response)
[1]=Definitions can be found in Section 10.5.3.2.7. [2]=After the last adequate tumor assessment. "Date of next scheduled assessment" is defined in Section 10.5.3.2.7. [3]=The rare exception to this is if the participant dies no later than the time of the second scheduled assessment as defined in the protocol in which case this is a PFS event at the date of death.			

The primary analysis and the sensitivity analyses must be specified in the protocol. Clearly define if and why options (1) are not used for situations C, E and (if applicable) F.

Situations C (C1 and C2): Progression or death after one or more missing assessments: The primary analysis is usually using options (1) for situations C1 and C2, i.e.

1. (C1) taking the actual progression or death date, in the case of only one missing assessment.
2. (C2) censoring at the date of the last adequate assessment, in the case of two or more consecutive missing assessments.

In the case of two or missing assessments (situation C2), option (3) may be considered jointly with option (1) in situation C1 as sensitivity analysis. A variant of this sensitivity analysis consists of backdating the date of event to the next scheduled assessment as proposed with option (2) in situations C1 and C2.

Situation E: Treatment discontinuation due to 'Disease progression' without documented progression: By default, option (1) is used for situation E as participants without documented PD should be followed for progression after discontinuation of treatment. However, option (2)

may be used as sensitivity analysis. If progression is claimed based on clinical deterioration instead of tumor assessment by e.g. CT-scan, option (2) may be used for indications with high early progression rate or difficulties to assess the tumor due to clinical deterioration.

Situation F: New cancer therapy given: the handling of this situation must be specified in detail in the protocol. However, option (1) (ITT) is the recommended approach; events documented after the initiation of new cancer therapy will be considered for the primary analysis i.e. progressions and deaths documented after the initiation of new cancer therapy would be included as events. This will require continued follow-up for progression after the start of the new cancer therapy. In such cases, it is recommended that an additional sensitivity analysis be performed by censoring at last adequate assessment prior to initiation of new cancer therapy.

Option (2), i.e. censoring at last adequate assessment may be used as a sensitivity analysis. If a high censoring rate due to start of new cancer therapy is expected, a window of approximately 8 weeks performed after the start of new cancer therapy can be used to calculate the date of the event or censoring. This should be clearly specified in the analysis plan.

In some specific settings, local treatments (e.g. radiation/surgery) may not be considered as cancer therapies for assessment of event/censoring in PFS/TTP/DoR analysis. For example, palliative radiotherapy given in the trial for analgesic purposes or for lytic lesions at risk of fracture will not be considered as cancer therapy for the assessment of BOR and PFS analyses. The protocol should clearly state the local treatments which are not considered as antineoplastic therapies in the PFS/TTP/DoR analysis.

The protocol should state that tumor assessments will be performed every x weeks until radiological progression irrespective of initiation of new antineoplastic therapy. It is strongly recommended that a tumor assessment is performed before the participant is switched to a new cancer therapy.

Additional suggestions for sensitivity analyses

Other suggestions for additional sensitivity analyses may include analyses to check for potential bias in follow-up schedules for tumor assessments, e.g. by assigning the dates for censoring and events only at scheduled visit dates. The latter could be handled by replacing in Table 10-6 the "Date of last adequate assessment" by the "Date of previous scheduled assessment (from baseline)", with the following definition:

- **Date of previous scheduled assessment (from baseline)** is the date when a tumor assessment would have taken place, if the protocol assessment scheme was strictly followed from baseline, immediately before or on the date of the last adequate tumor assessment.

In addition, analyses could be repeated using the Investigators' assessments of response rather than the calculated response. The need for these types of sensitivity analyses will depend on the individual requirements for the specific study and disease area and have to be specified in the protocol or RAP documentation.

10.5.4 Data handling and programming rules

The following section should be used as guidance for development of the protocol, data handling procedures or programming requirements (e.g. on incomplete dates).

10.5.4.1 Study/project specific decisions

For each study (or project) various issues need to be addressed and specified in the protocol or RAP documentation. Any deviations from protocol must be discussed and defined at the latest in the RAP documentation.

The proposed primary analysis and potential sensitivity analyses should be discussed and agreed with the health authorities and documented in the protocol (or at the latest in the RAP documentation before database lock).

10.5.4.2 End of treatment phase completion

10.5.4.3 End of post-treatment follow-up (study phase completion)

End of post-treatment follow-up visit will be completed after discontinuation of study treatment and post-treatment evaluations but prior to collecting survival follow-up.

Participants may provide study phase completion information for one of the following reasons:

- Adverse event
- Lost to follow-up
- Physician decision
- Pregnancy
- Protocol deviation
- Technical problems
- Subject/guardian decision
- Death
- Progressive disease
- Study terminated by the sponsor

10.5.4.4 Medical validation of programmed overall lesion response

In order to be as objective as possible the RECIST programmed calculated response assessment is very strict regarding measurement methods (i.e. any assessment with more or less sensitive method than the one used to assess the lesion at baseline is considered NE) and not available evaluations (i.e. if any target or non-target lesion was not evaluated the whole overall lesion response is NE unless remaining lesions qualified for PD). This contrasts with the slightly more flexible guidance given to local investigators (and to the central reviewers) to use expert judgment in determining response in these type of situations, and therefore as a consequence discrepancies between the different sources of response assessment often arise. To ensure the quality of response assessments from the local site and/or the central reviewer, the responses may be re-evaluated by clinicians (based on local investigator data recorded in eCRF or based on central reviewer data entered in the database) at Novartis or external experts. In addition,

data review reports will be available to identify assessments for which the investigators' or central reader's opinion does not match the programmed calculated response based on RECIST criteria. This may be queried for clarification. However, the investigator or central reader's response assessment will never be overruled.

If Novartis elect to invalidate an overall lesion response as evaluated by the investigator or central reader upon internal or external review of the data, the calculated overall lesion response at that specific assessment is to be kept in a dataset. This must be clearly documented in the report and analysis plan (RAP) documentation and agreed before database lock. This dataset should be created and stored as part of the 'raw' data.

Any discontinuation due to 'Disease progression' without documentation of progression by RECIST criteria should be carefully reviewed. Only participants with documented deterioration of symptoms indicative of progression of disease should have this reason for discontinuation of treatment or study evaluation.

10.5.4.5 Programming rules

The following should be used for programming of efficacy results:

10.5.4.5.1 Calculation of time to event variables

Time to event = end date - start date + 1 (in days)

When no post-baseline tumor assessments are available, the date of randomization/start of treatment will be used as end date (duration = 1 day) when time is to be censored at last tumor assessment, i.e. time to event variables can never be negative.

10.5.4.5.2 Incomplete assessment dates

All investigation dates (e.g. X-ray, CT scan) must be completed with day, month and year.

If one or more investigation dates are incomplete but other investigation dates are available, this/these incomplete date(s) are not considered for calculation of the assessment date (and assessment date is calculated as outlined in Section 10.5.3.2.7). If all measurement dates have no day recorded, the 1st of the month is used.

If the month is not completed, for any of the investigations, the respective assessment will be considered to be at the date which is exactly between previous and following assessment. If a previous and following assessment is not available, this assessment will not be used for any calculation.

10.5.4.5.3 Incomplete dates for last known date participant alive or death

All dates must be completed with day, month and year. If the day is missing, the 15th of the month will be used for incomplete death dates or dates of last contact.

10.5.4.5.4 Non-target lesion response

If no non-target lesions are identified at baseline (and therefore not followed throughout the study), the non-target lesion response at each assessment will be considered 'not applicable (NA)'.

10.5.4.5.5 Study/project specific programming

The standard analysis programs need to be adapted for each study/project.

10.5.4.5.6 Censoring reason

In order to summarize the various reasons for censoring, the following categories will be calculated for each time to event variable based on the treatment completion page, the study evaluation completion page and the survival page.

For survival the following censoring reasons are possible:

- Alive
- Lost to follow-up

For PFS and TTP (and therefore duration of responses) the following censoring reasons are possible:

- Ongoing without event
- Lost to follow-up
- Withdrew consent
- Adequate assessment no longer available*
- Event documented after two or more missing tumor assessments (optional, see Table 10-6)
- Death due to reason other than underlying cancer *(only used for TTP and duration of response)*
- Initiation of new anti-cancer therapy

* Adequate assessment is defined in Section 10.5.3.2.7. This reason is applicable when adequate evaluations are missing for a specified period prior to data cut-off (or prior to any other censoring reason) corresponding to the unavailability of two or more planned tumor assessments prior to the cut-off date. The following clarifications concerning this reason should also be noted:

- This may be when there has been a definite decision to stop evaluation (e.g. reason="Sponsor decision" on study evaluation completion page), when participants are not followed for progression after treatment completion or when only NE assessments are available just prior to data cut-off).
- The reason "Adequate assessment no longer available" also prevails in situations when another censoring reason (e.g. withdrawal of consent, loss to follow-up or alternative anti-cancer therapy) has occurred more than the specified period following the last adequate assessment.
- This reason will also be used to censor in case of no baseline assessment.

10.6 Appendix 6: Patient Engagement

The following patient engagement initiatives are included in this study and will be provided, as available, for distribution to study participants at the time points indicated. If compliance is impacted by cultural norms or local laws and regulations, sites may discuss modifications to these requirements with Novartis.

- Thank You letter
- Plain language trial summary - after CSR publication

11 References

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